

FIG. 1

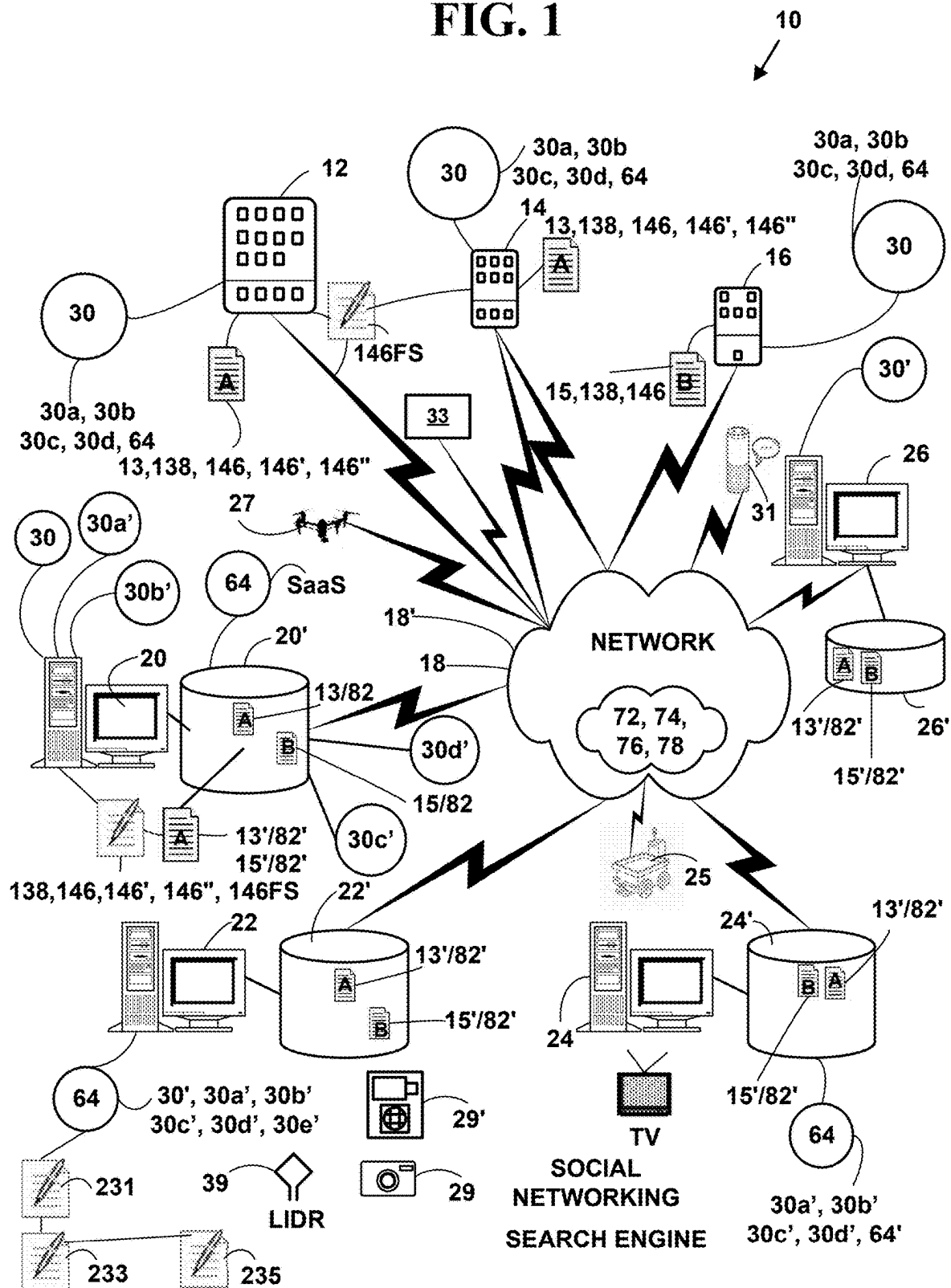
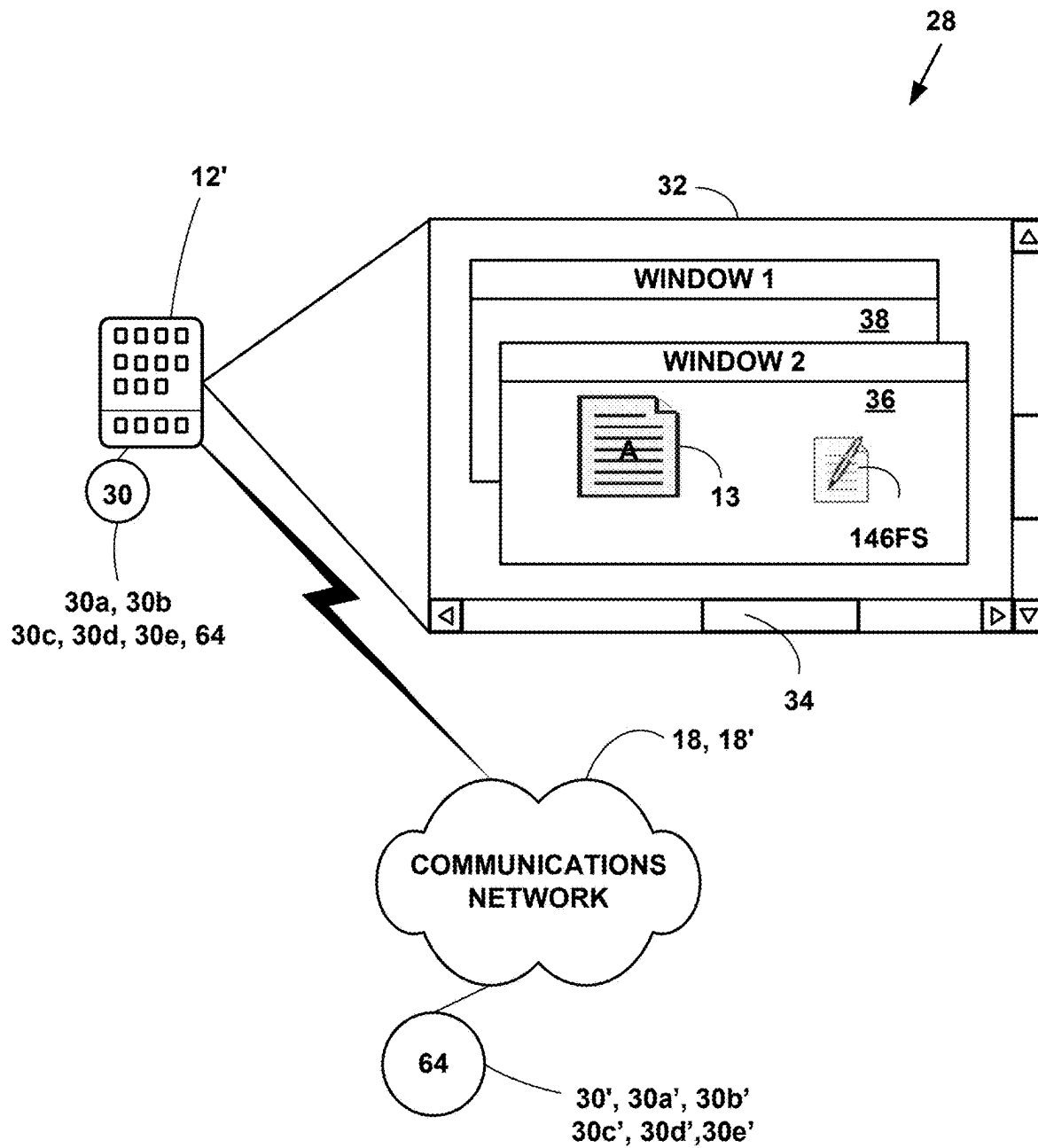


FIG. 2



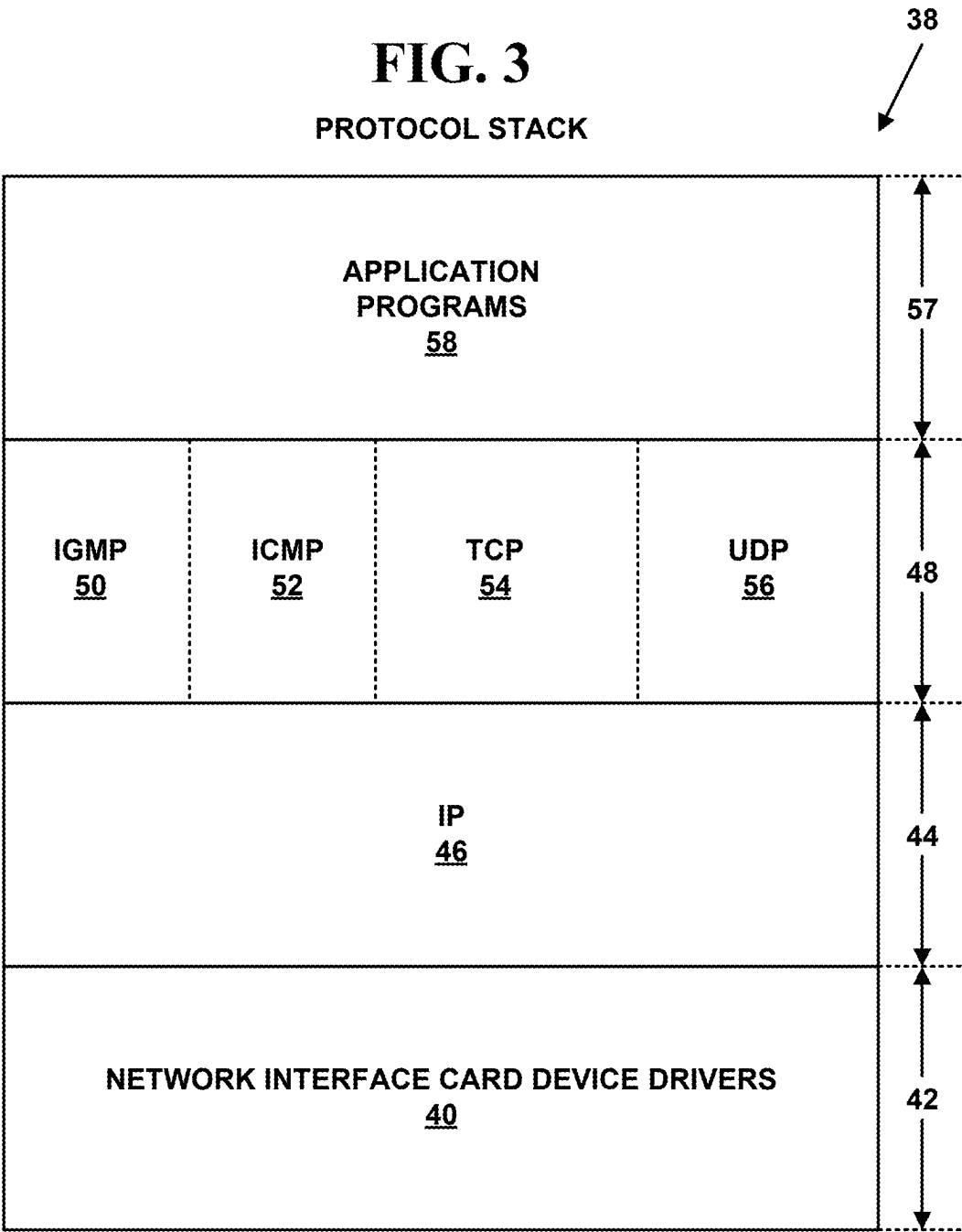


FIG. 4

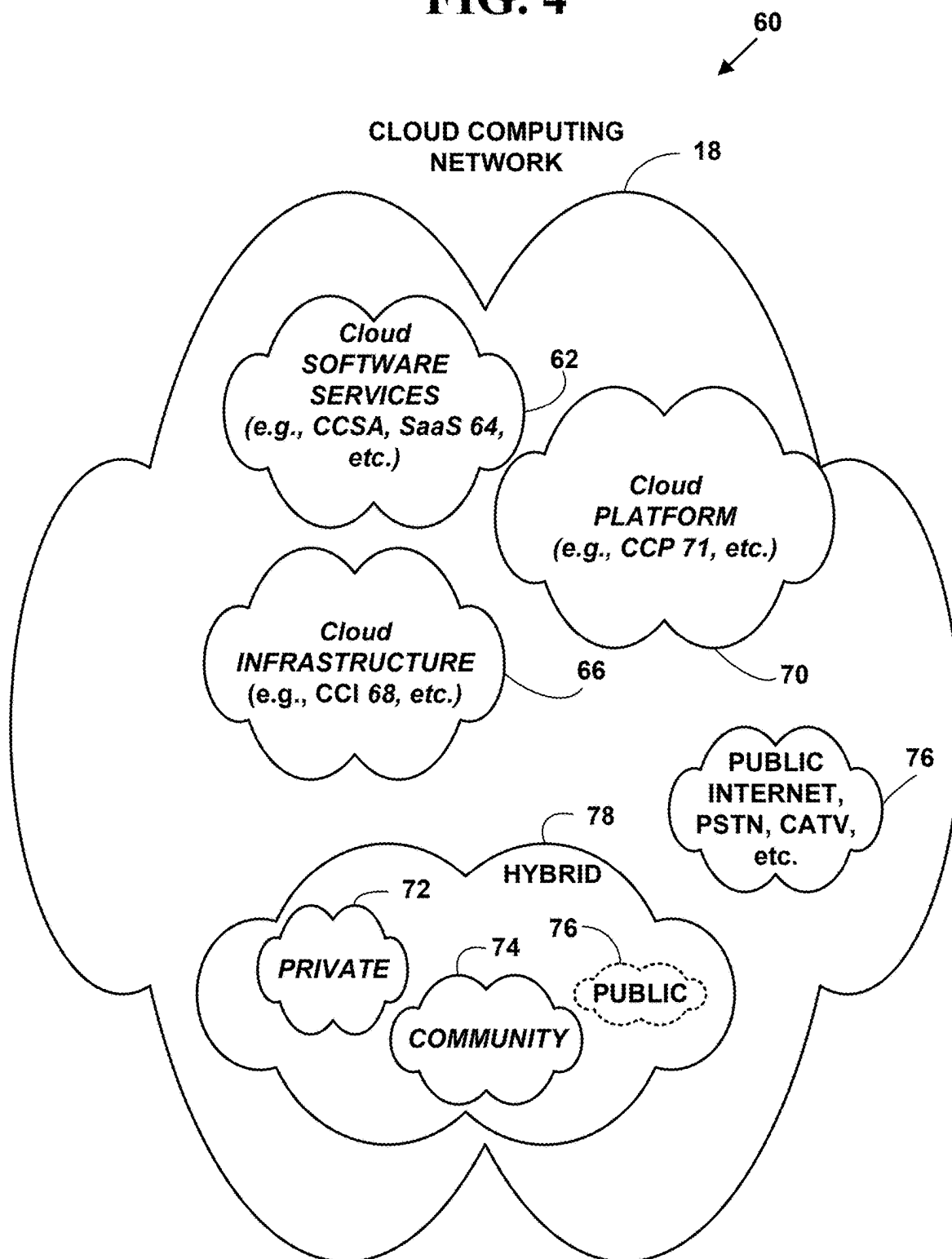


FIG. 5

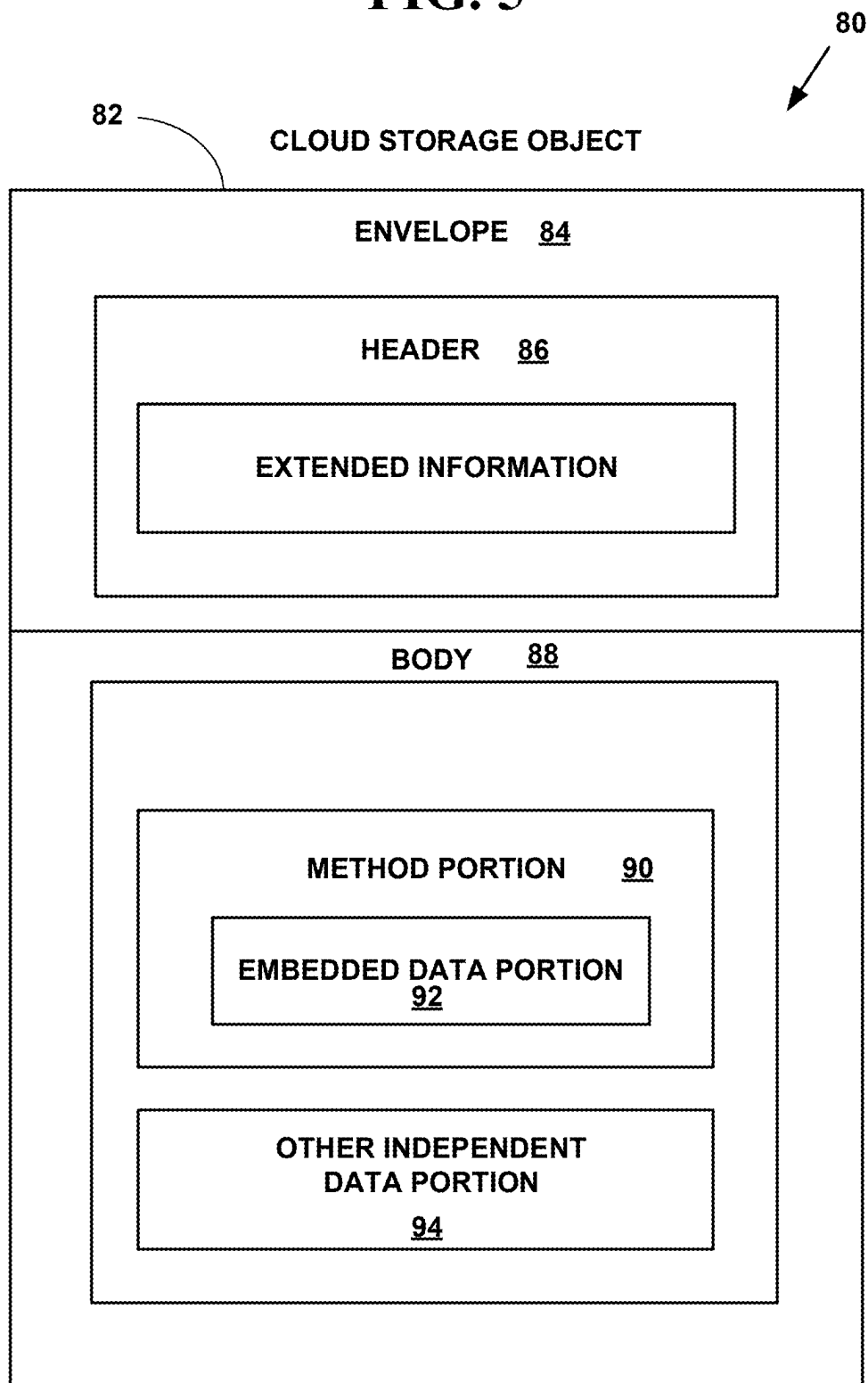


FIG. 6

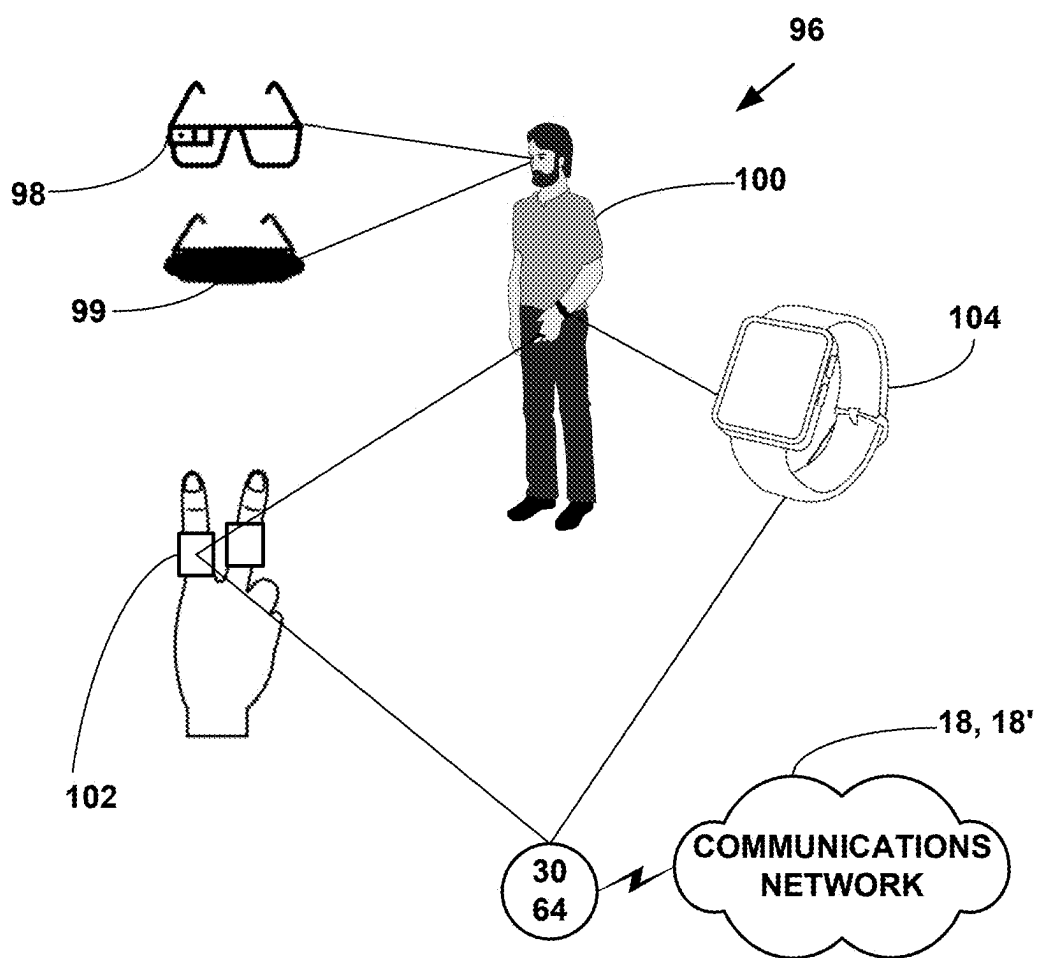


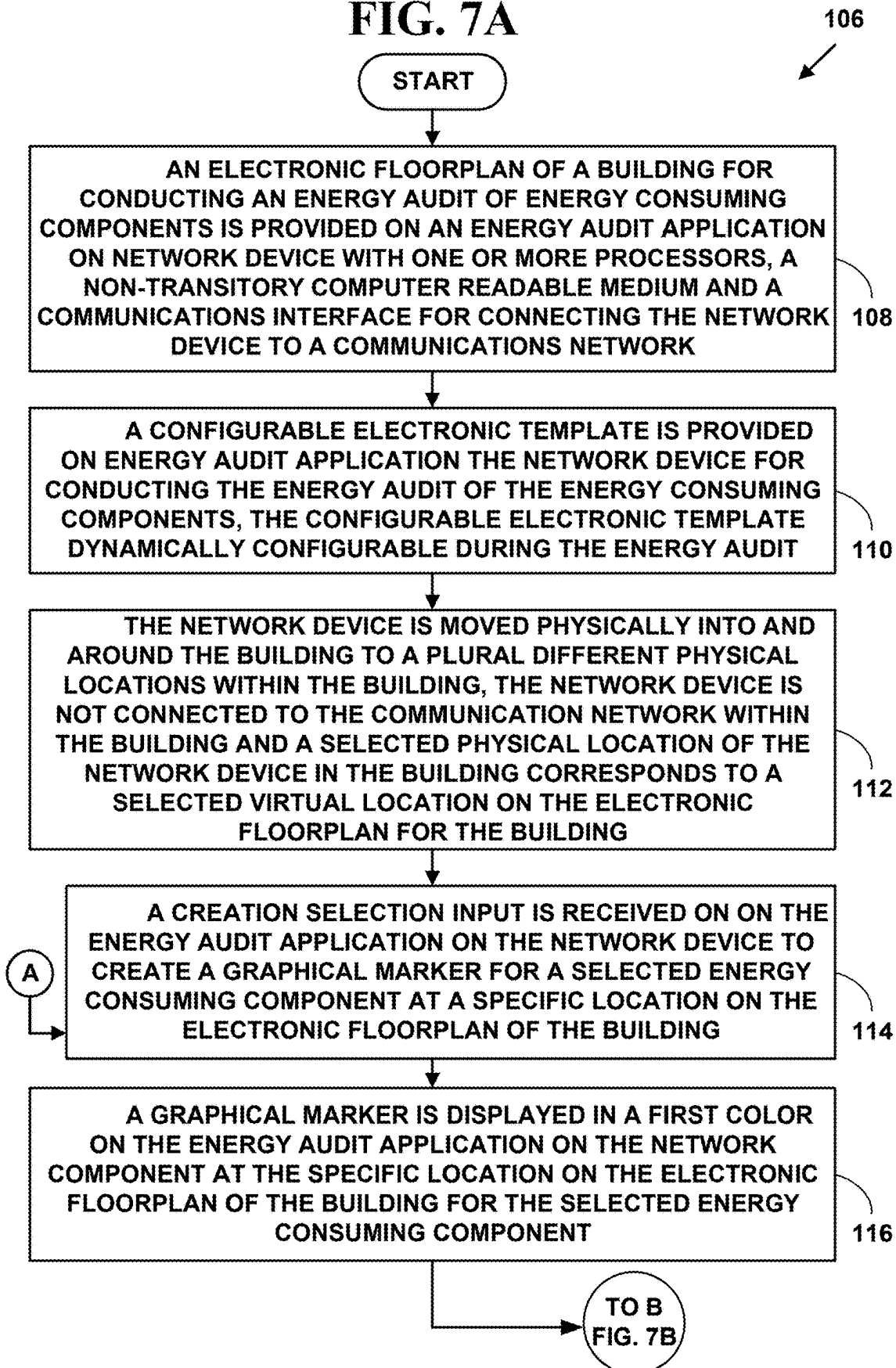
FIG. 7A

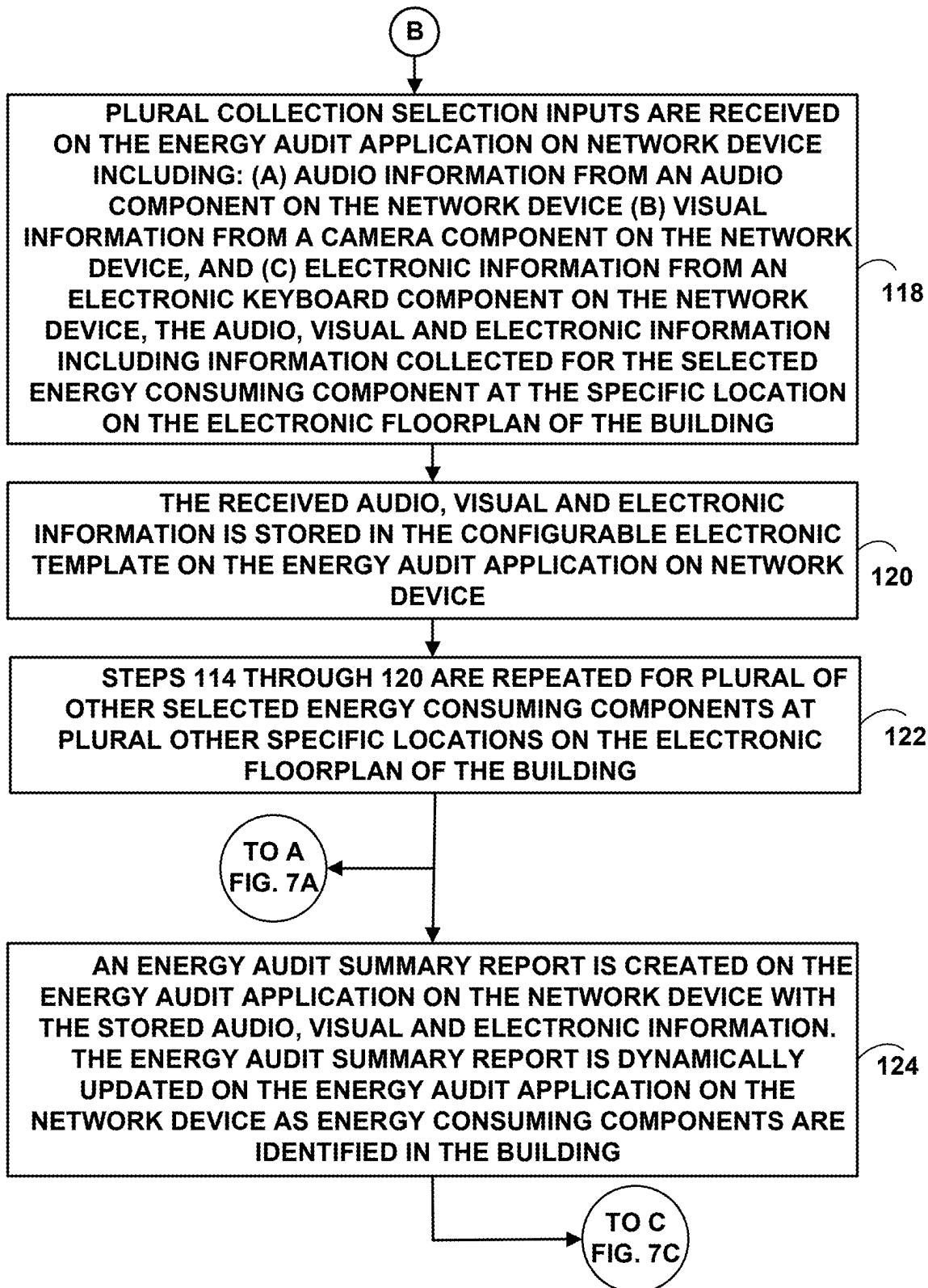
FIG. 7B

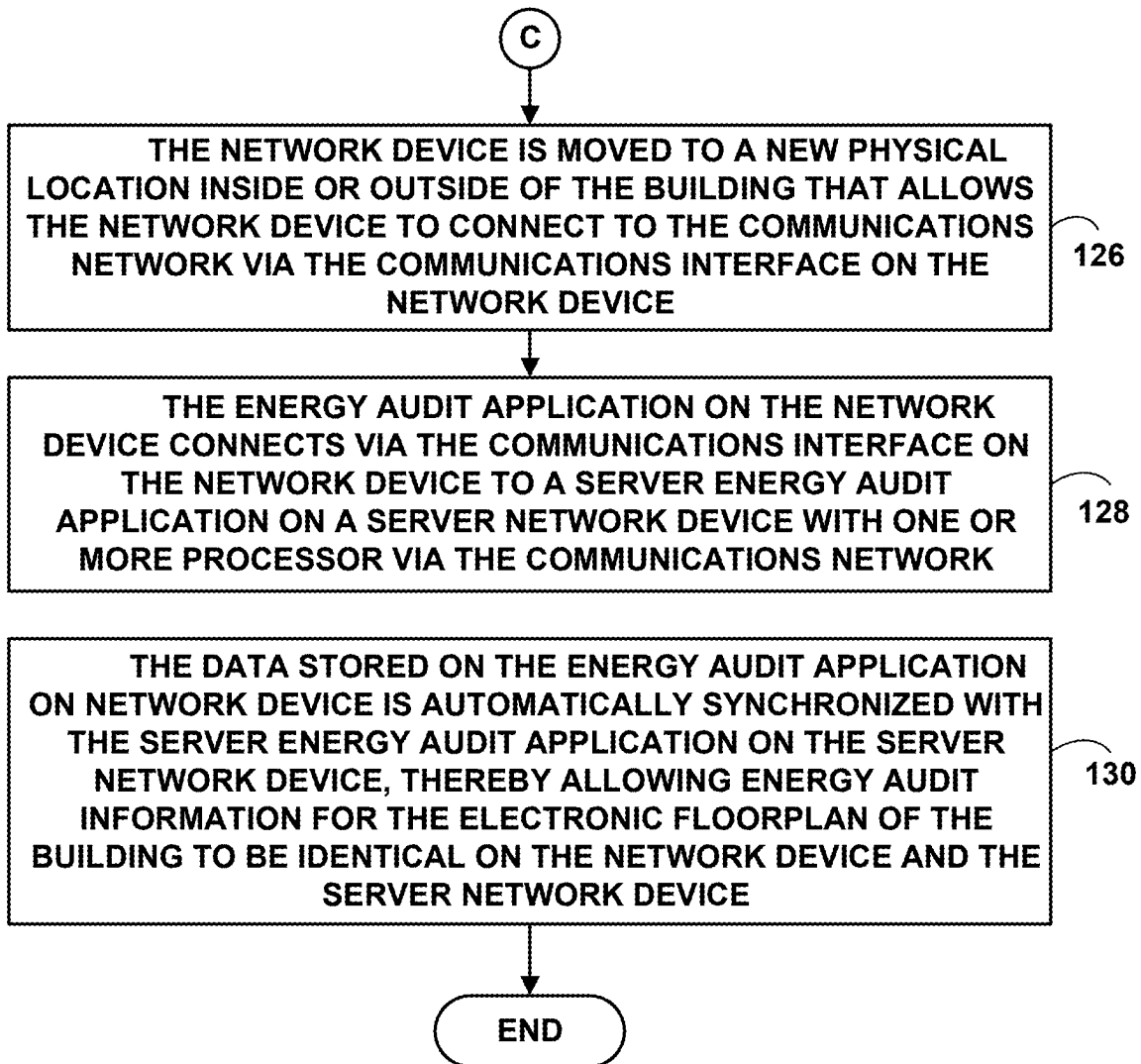
FIG. 7C

Figure 1 is a schematic diagram of a smart building system. The diagram illustrates a 3D perspective of a building structure with various components labeled. At the top left, a control unit 12 is connected to two document-like icons 138 and 146. A large arrow 132 points towards the building. On the right, a power source 136, 136' is connected to the building. Inside the building, several light bulbs are shown, some labeled 140, 140a, 140b, 140c, 140d, 140e, 135, 135', 133, 137, 144, 144', 146, 141. A battery icon 139 is shown with the text "LEVEL OF COMPLETION" below it. At the bottom, two arrows 142 and 145 point left and right respectively, labeled "ENERGY CONSUMING DEVICES" and "ENERGY PRODUCING DEVICE".

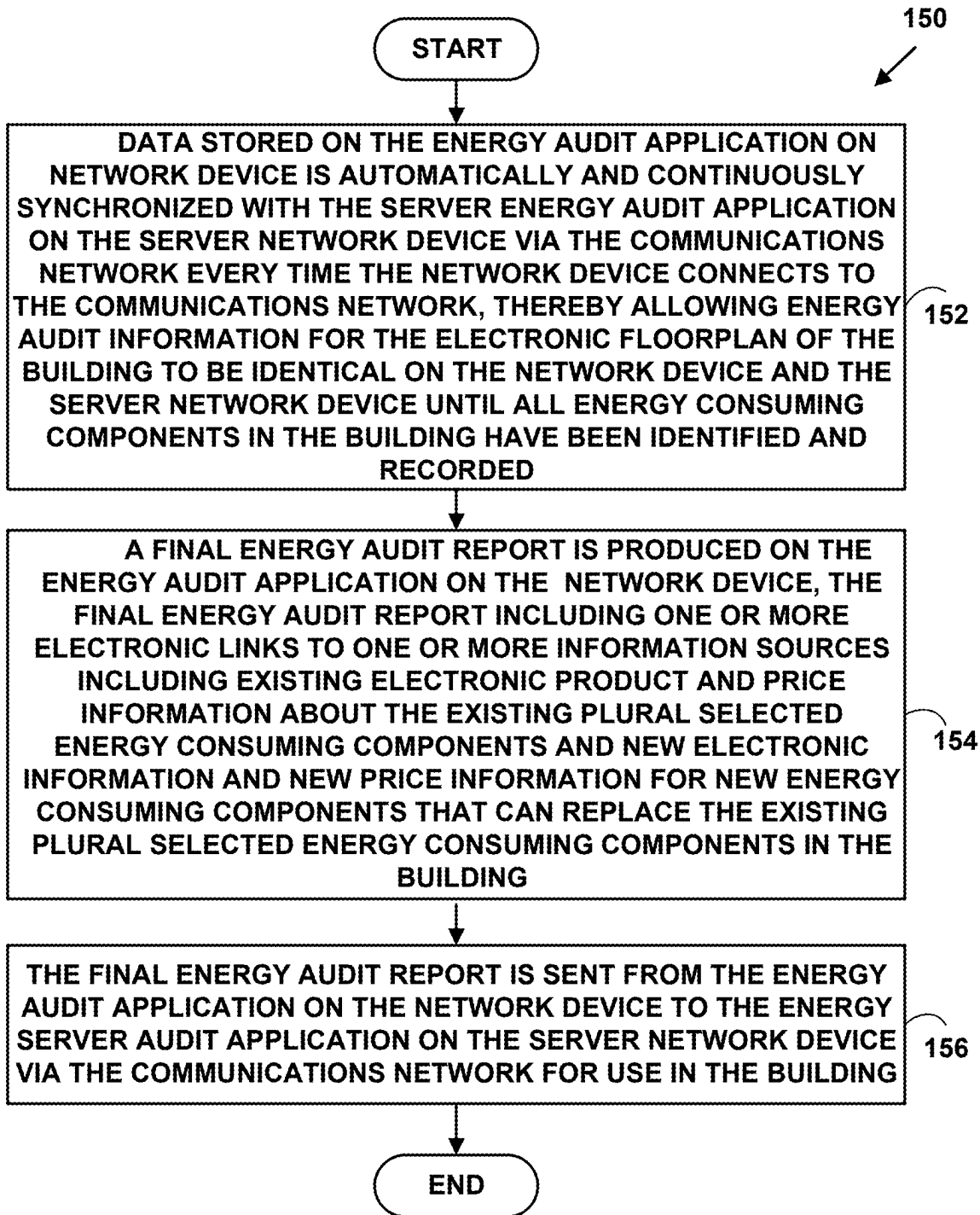
FIG. 9

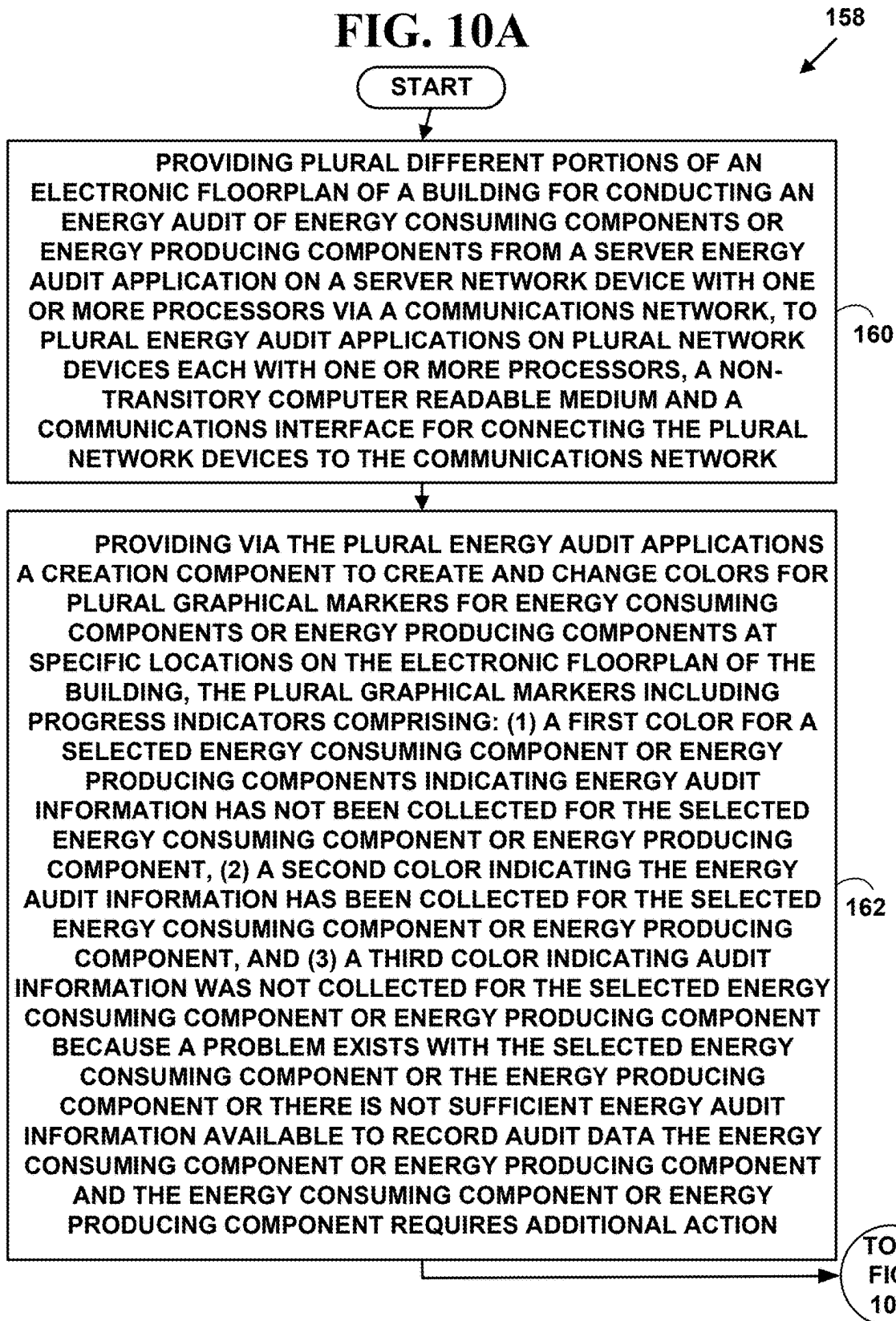
FIG. 10A

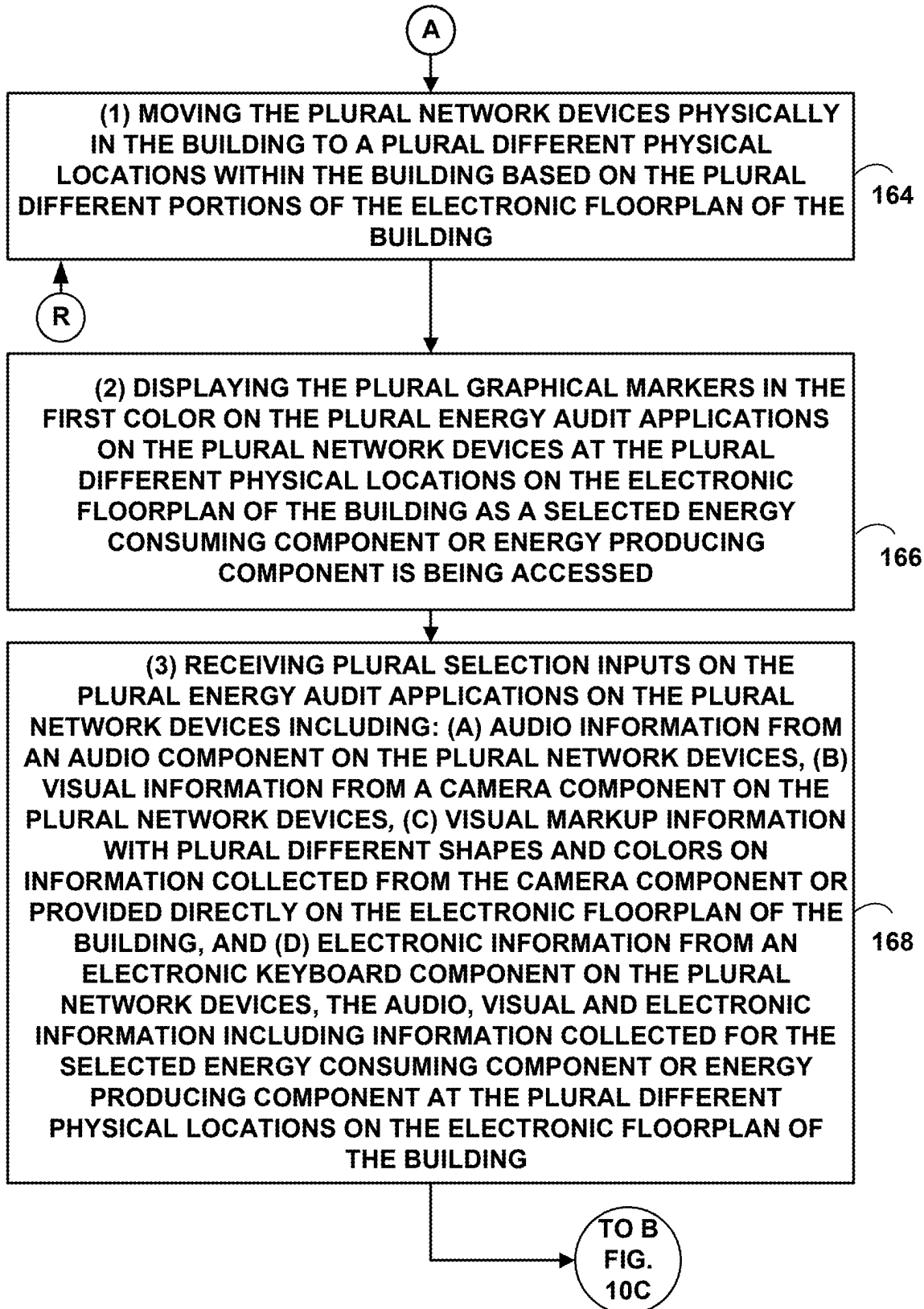
FIG. 10B

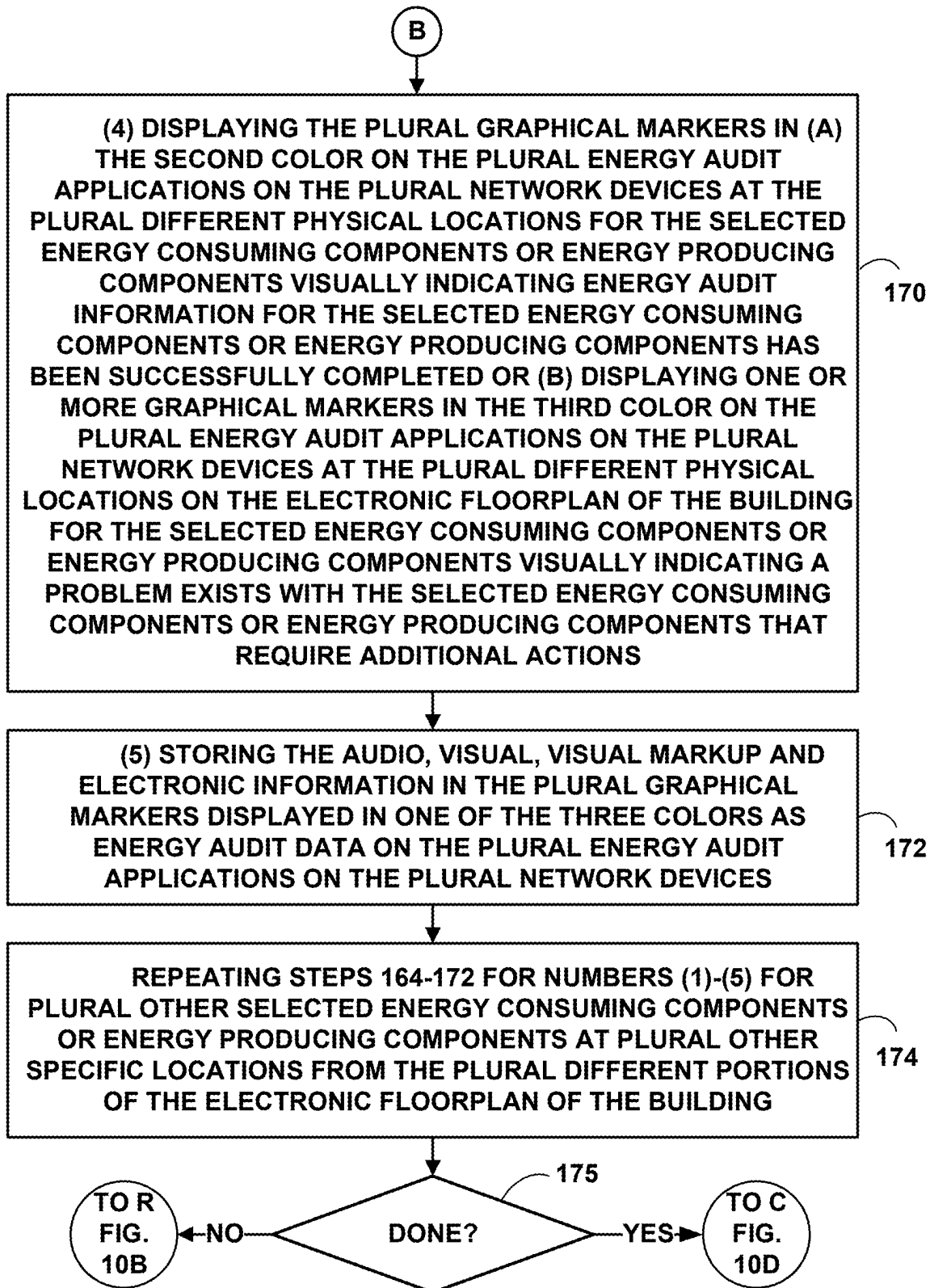
FIG. 10C

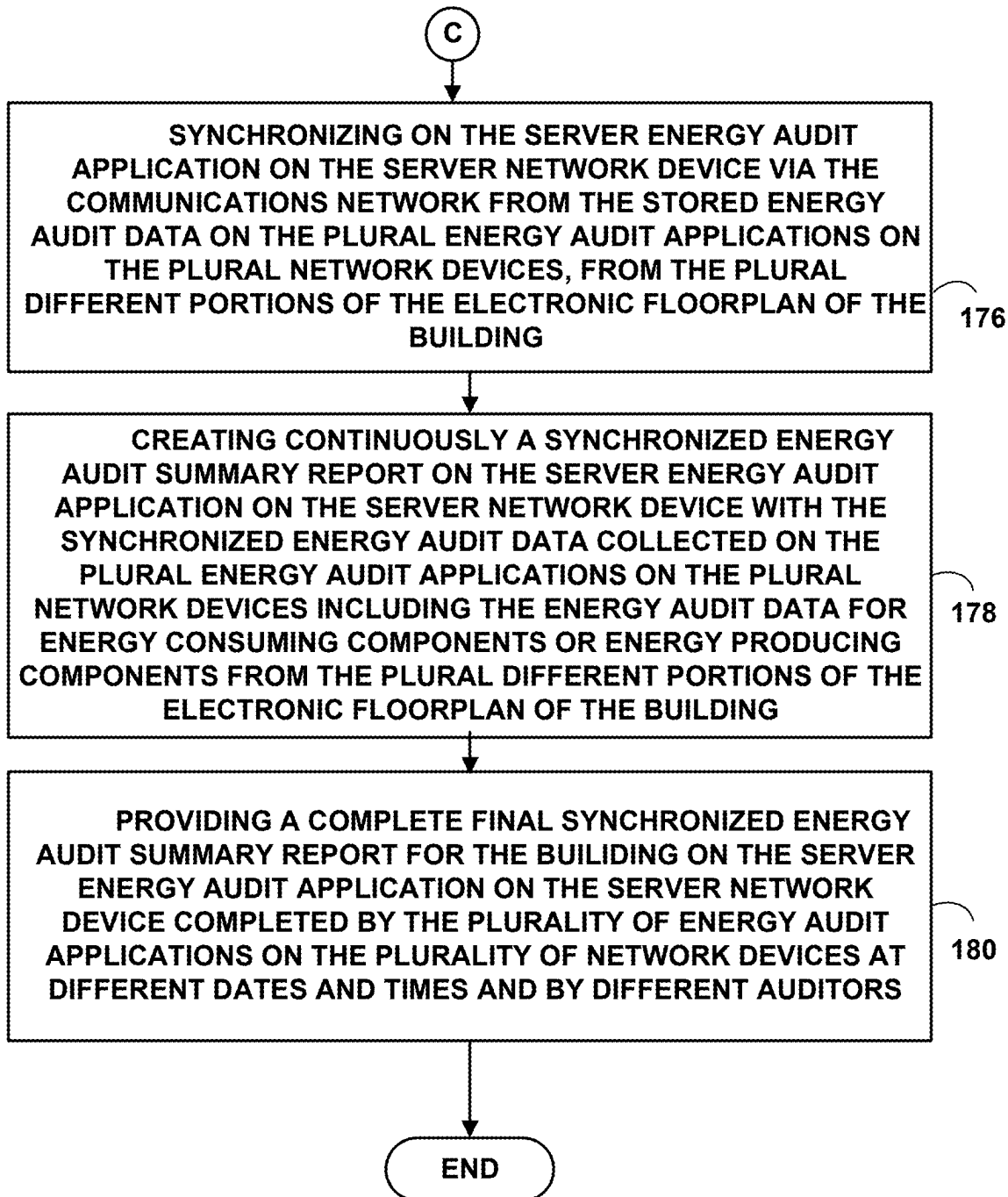
FIG. 10D

FIG. 11

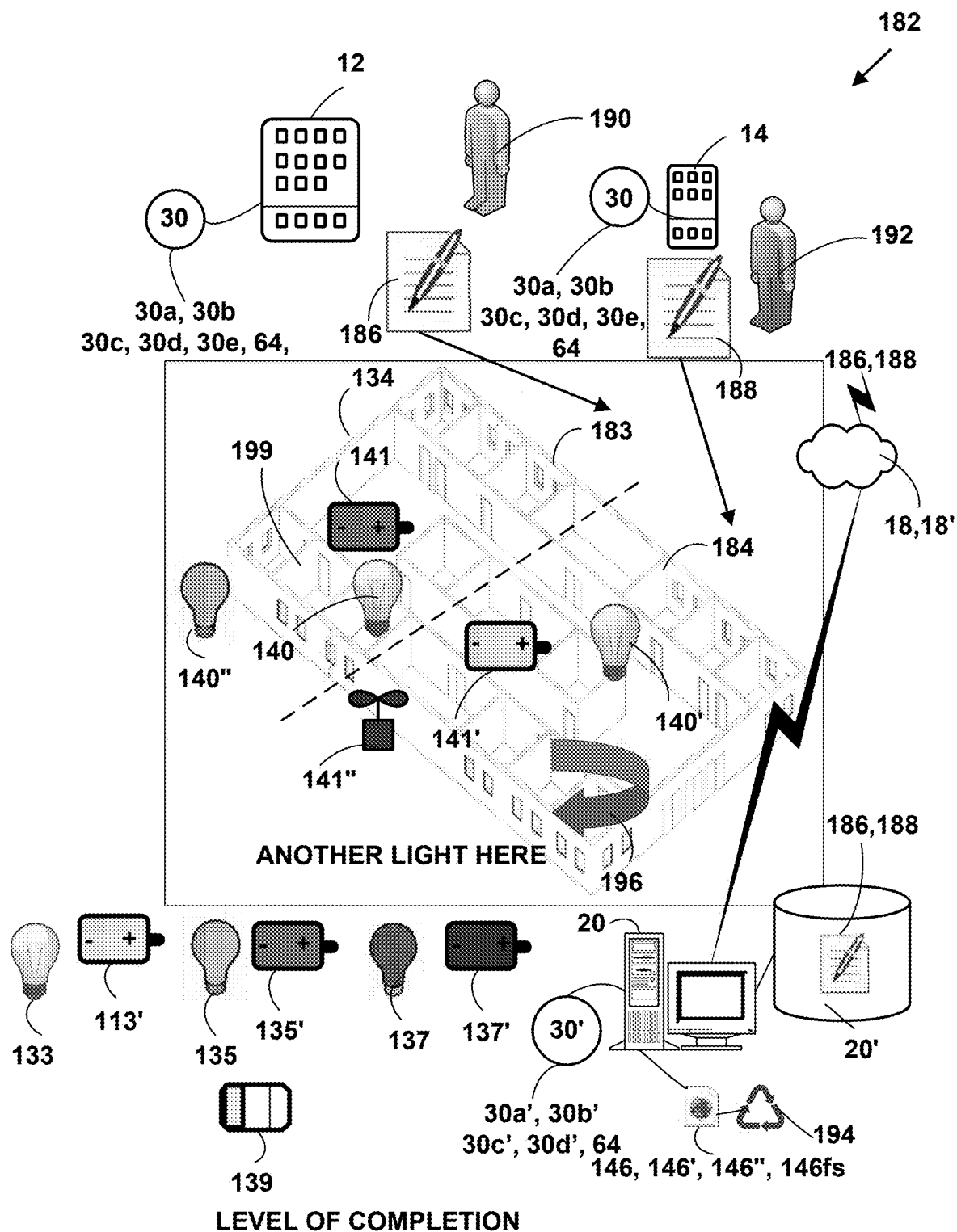


FIG. 12

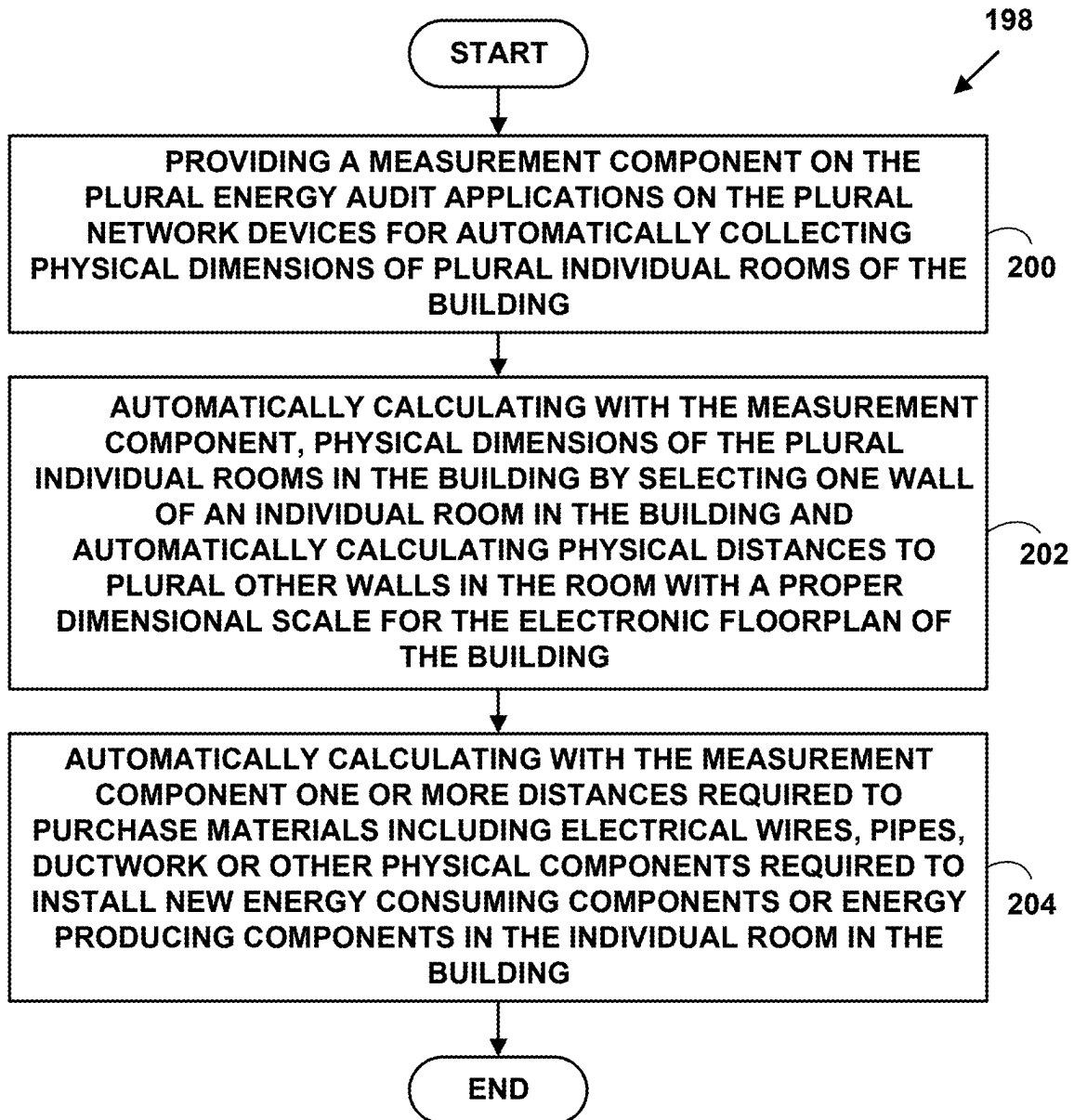


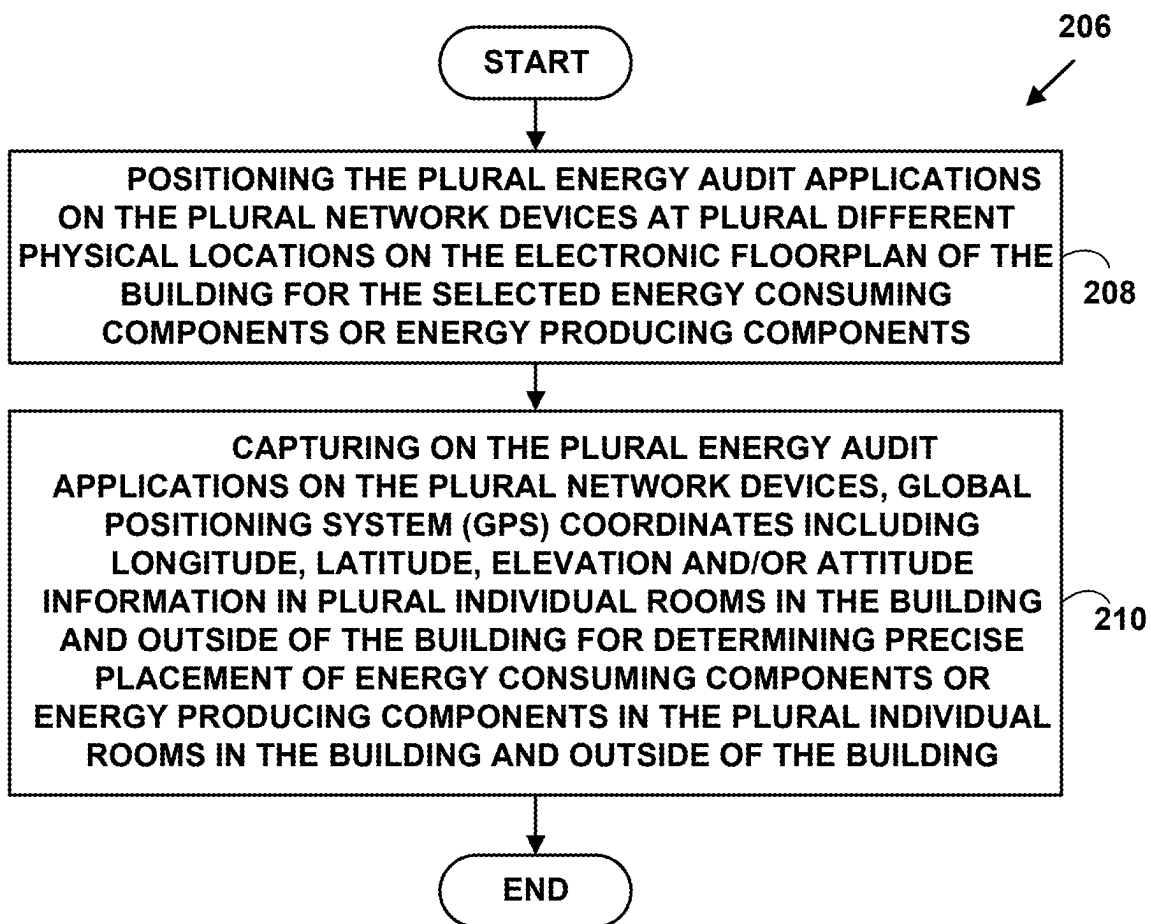
FIG. 13

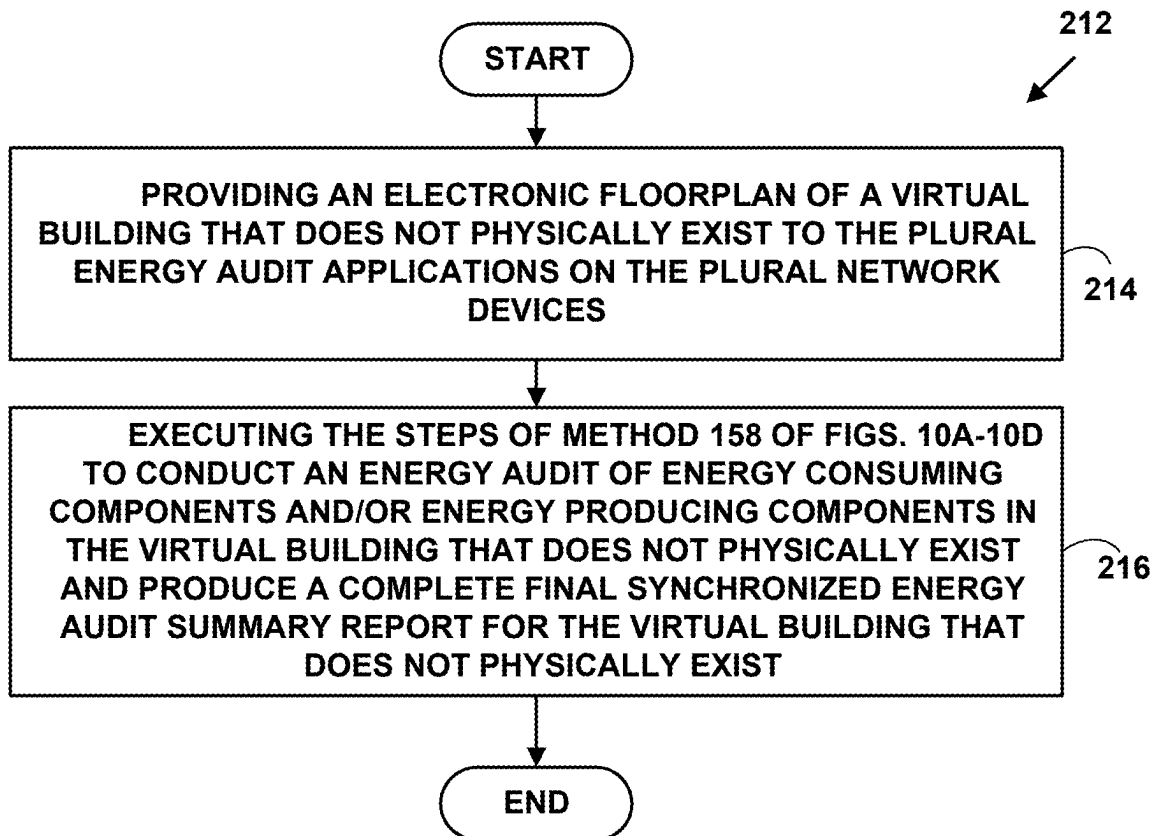
FIG. 14

FIG. 15

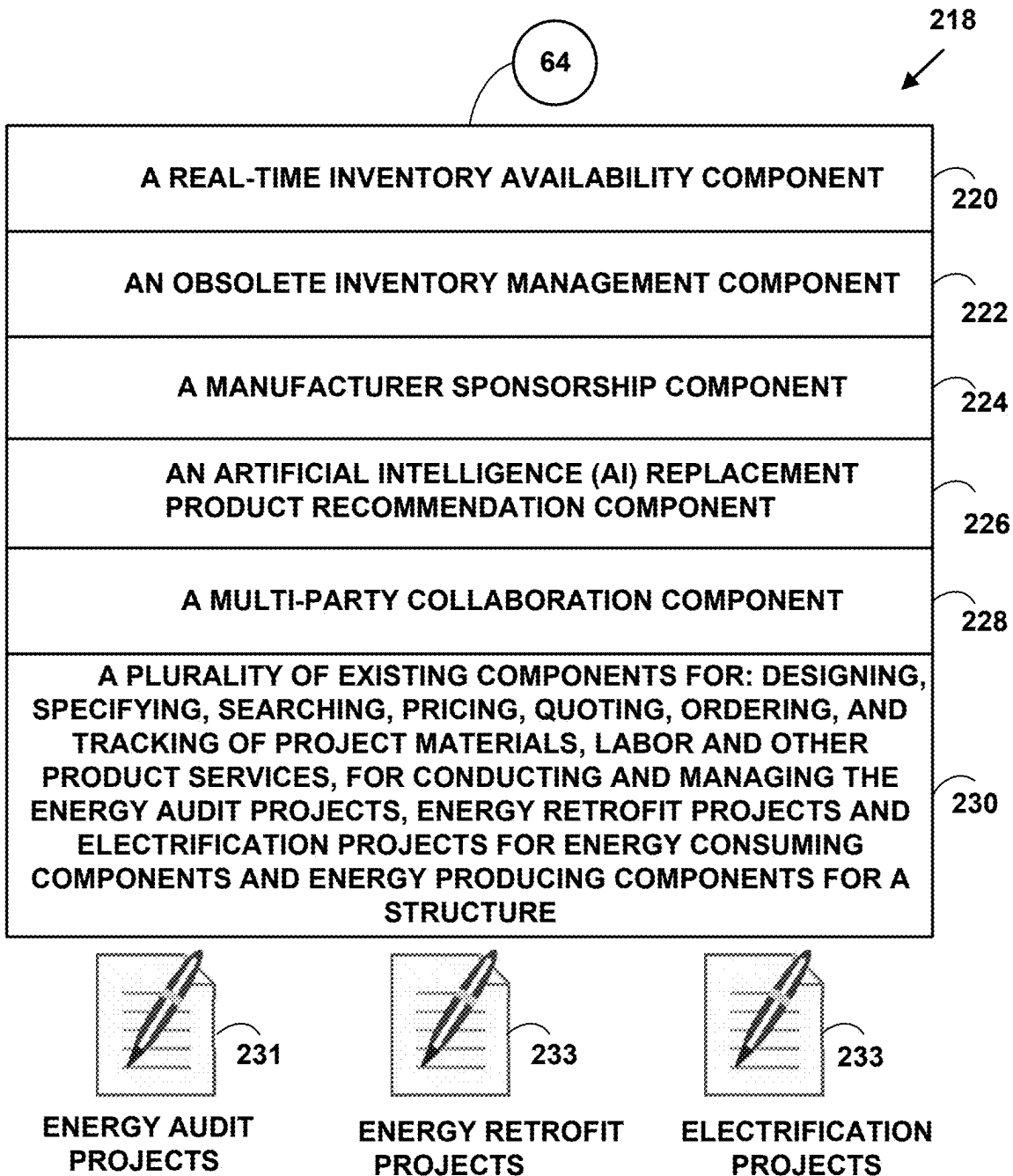


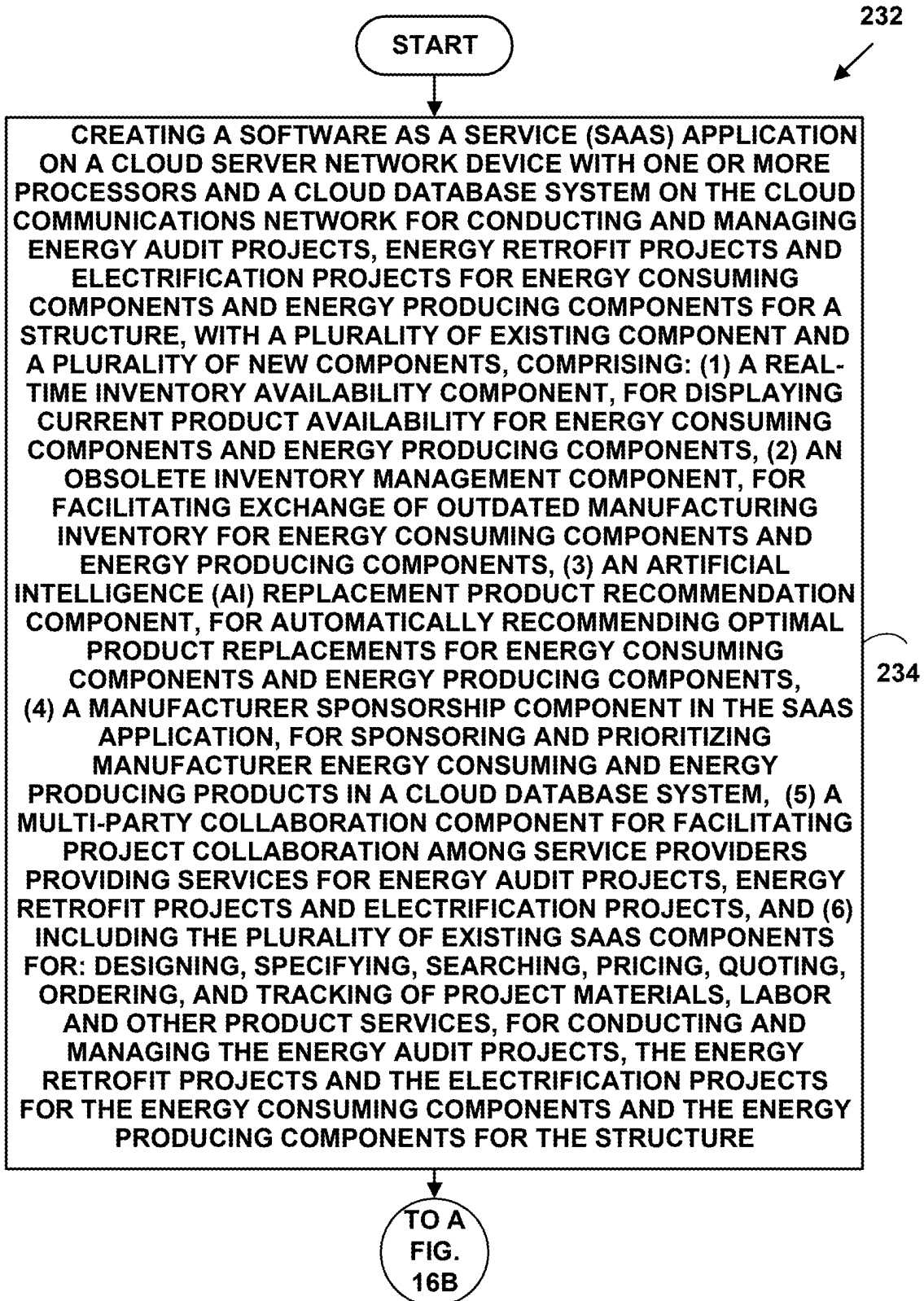
FIG. 16A

FIG. 16B

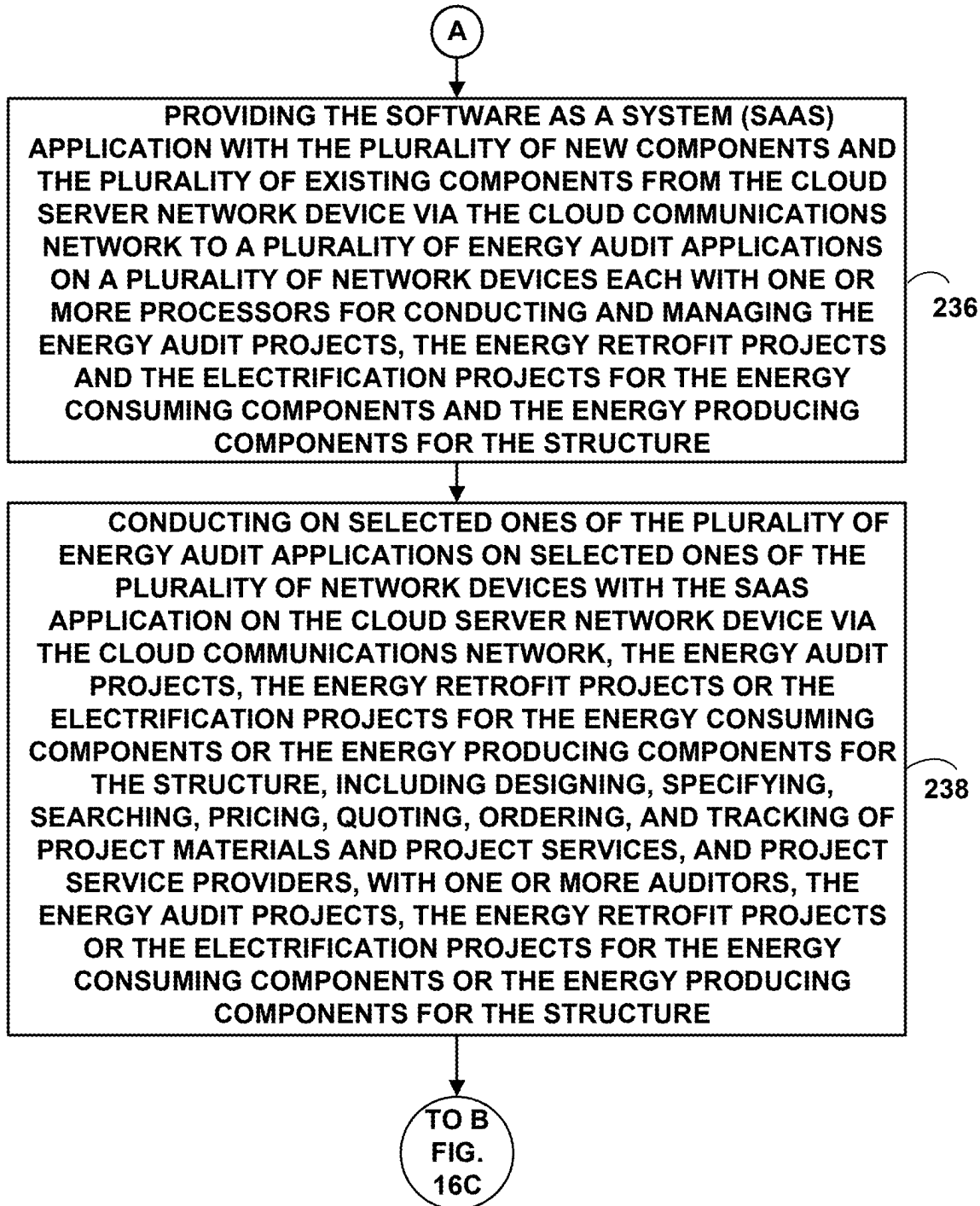
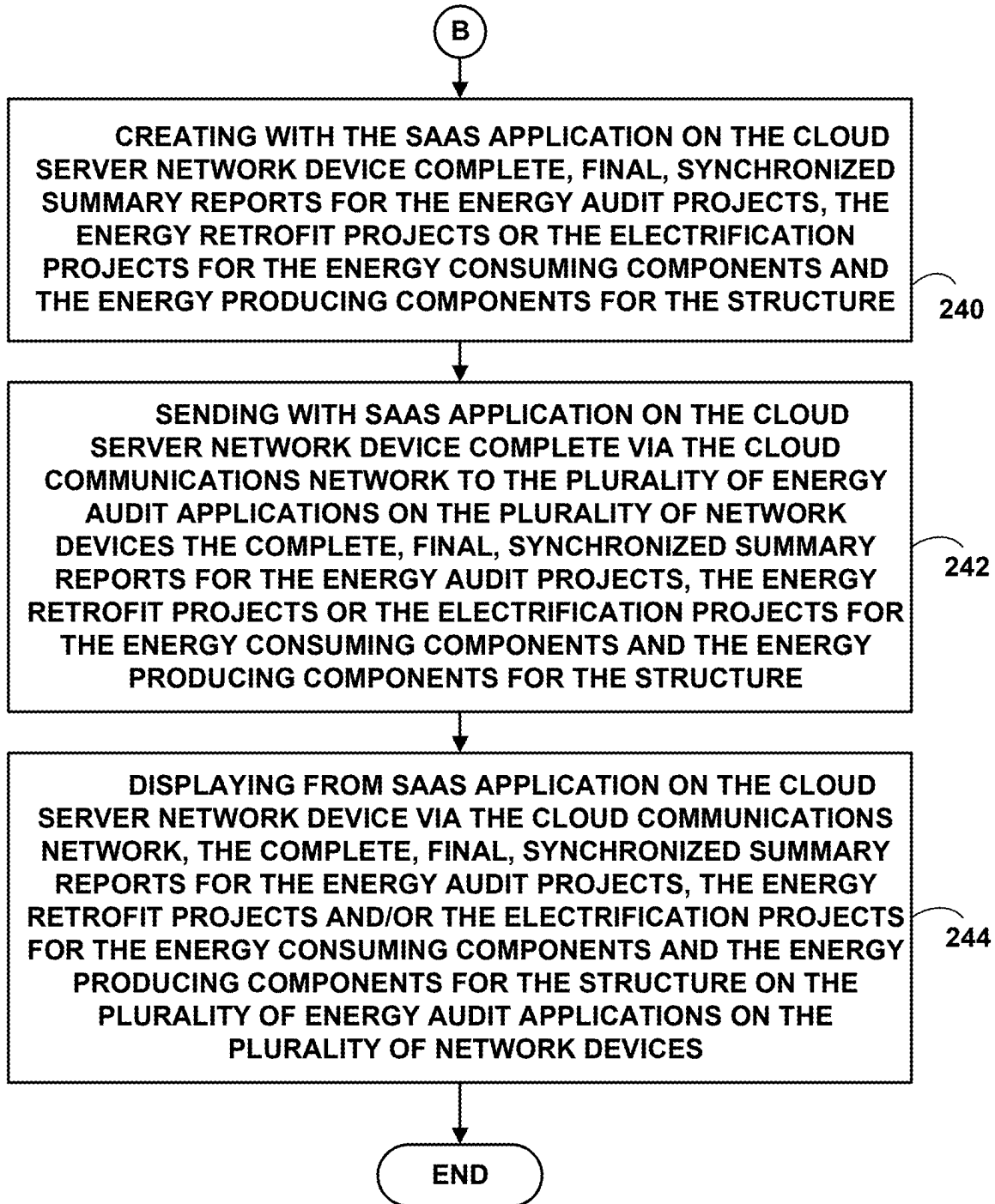


FIG. 16C



**METHOD AND SYSTEM FOR PROVIDING
ENERGY AUDIT, ENERGY RETOFIT AND
ENERGY ELECTRIFICATION PROJECTS
WITH MULTIPLE PARTICIPANTS**

**CROSS REFERENCES TO RELATED
APPLICATIONS**

[0001] The U.S. Utility patent application is a Continuation-In-Part (CIP) of U.S. utility patent application Ser. No. 18/208,763, filed Jun. 12, 2023, which issued as U.S. Pat. No. 12,307,398, on May 20, 2025, which is a CIP of U.S. utility patent application Ser. No. 17/172,957, filed on Feb. 10, 2021, which issued as U.S. Pat. No. 11,676,106, on Jun. 13, 2023, which is a CIP of U.S. utility patent application number, 17/070,720, filed Oct. 14, 2020, which issued as U.S. Pat. No. 11,402,405, on Aug. 2, 2022, the contents of all of which are incorporated by reference.

FIELD OF INVENTION

[0002] This invention relates to energy consuming and energy producing devices. More specifically, it relates to a method and system for providing energy audit, energy retrofit and energy electrification projects with multiple participants.

BACKGROUND OF THE INVENTION

[0003] In commercial and industrial buildings, there are a large number of energy consuming devices such as lighting, heating, ventilation, air conditioning, etc. These energy consuming devices often include devices manufactured by multiple different companies and include devices installed and replaced during many different time periods.

[0004] A commercial or industrial building (e.g., factory, etc.) typically includes a variety of energy consuming devices that may be using older technologies, be inefficient, include devices that are no longer available, etc. For example, a building may include a mix of old inefficient incandescent lighting or fluorescent lighting and more efficient Light Emitting Diode (LED) lighting. The building may include transformers or starters for their fluorescent lighting that are energy inefficient or are no longer manufactured and available for purchase.

[0005] A commercial or industrial building may also include, or be associated with energy producing devices, such as solar panels, wind turbines, hydroelectric generators, biomass generators or nuclear reactors. These energy producing devices also include components manufactured by multiple different companies and include components installed and replaced during many different time periods.

[0006] There are many problems associated with such a mix of energy consuming and energy producing components. One problem is that companies performing energy consumption audits and retrofit projects on commercial and industrial buildings must first gather the “as-is” environment of a building to create an existing energy footprint, and to understand critical space details required to estimate and carryout an energy audit or energy upgrade project.

[0007] To complete an energy audit or energy retrofit project, a building engineer must physically walk through the building collecting counts, space details and pictures of energy consuming devices and energy produces devices that reside in the building. This data collection process is referred

to as an Energy “Audit,” “Survey,” “Walk-through,” “Assessment,” or “Pre-view.”

[0008] Historically, this data collection process has been done on paper and then transcribed into a spreadsheet, a process that is time-consuming and error prone.

[0009] Another problem is that commercial buildings and industrial buildings are large and include a huge number of energy consumption or energy producing components. A commercial building may include many different floors and an industrial building may include many hundreds of thousands of square feet. Any energy audit takes a large amount of time is subject to many different types of collection errors.

[0010] Another problem is that when building engineers conduct an energy audit, they use their own data collection process and notations. This introduces inconsistencies in the data collection process and is less application than if a standard data collection process with standard terminology is used.

[0011] Another problem is that building engineers may take multiple individual pictures of energy consumption components or energy producing components during a data collection process. Managing such multiple individual pictures is difficult.

[0012] Another problem is that in many commercial and industrial buildings electronic devices cannot connect to communications networks, such as the Internet, a telecommunications network, a WiFi network, etc. because such buildings typically include concrete, steel and other building materials that block receiving electronic signals from such communications networks. Thus, a building engineer who is conducting a data collection process for an energy audit or energy retrofit is not able to use any electronic device such as a mobile phone, tablet, etc. to complete the data collection process being online and connected to a communications network, cloud communications network, etc.

[0013] Another problem is that if an electronic device cannot connect to communications networks, such as the Internet, a telecommunications network, a WiFi network, etc. because such buildings typically include concrete, steel and other building materials that block receiving electronic signals from such communications networks, a device such as a smart phone, tablet, etc. will continuously try to connect to the communications network and quickly drain its battery and then can no longer be used within the building.

[0014] Another problem is that when data collection for an energy audit or energy retrofit is being conducted, exact locations of the energy consumption components or energy producing components within the buildings cannot be determined or properly recorded because drawings for the building are no longer available, are not up to date or are not available electronically.

[0015] Another problem is that exact or estimated measurements of the energy consumption components or energy producing components because documentation for the components may be lost or no longer be available to the building engineer because the component is no longer manufactured.

[0016] Another problem is that when data collection for an energy audit or energy retrofit is being conducted the data collection is done in an ad-hoc manner and the data collected is not consistent and may not be reproducible.

[0017] Another problem is that data collection for an energy audit or energy retrofit is being conducted, the audit cannot be completed at different times and dates by different people and synchronized in one place at a later time.

[0018] Another problem is that when data collection for an energy audit or energy retrofit is being conducted, the data collection information is not synchronized among devices or parties and is therefore prone to errors.

[0019] Another problem is that energy audit or energy retrofit project systems are not available for sponsorships by different manufactures of energy consuming and energy consuming products.

[0020] Another problem is that energy audit or energy retrofit project systems do not include a real-time energy consuming and energy consuming products inventory management system.

[0021] Another problem is that energy audit or energy retrofit project system do not include a management system for obsolete energy consuming and energy consuming products.

[0022] Another problem is that energy audit or energy retrofit project systems do not include a management system for facilitating project collaboration among service multiple service providers for energy audit or energy retrofit projects.

[0023] Another problem is that energy audit or energy retrofit project systems do not include an Artificial Intelligence (AI) system with Big Data for automatically recommending optimal product replacements for energy audit or energy retrofit projects.

[0024] There have been some solutions solving some of the problems associated with energy audit, energy retrofit and electrification systems.

[0025] For example, U.S. Pat. No. 11,402,415, that issued to Seifert, et al., teaches “A method and system for providing energy audits including installation, verification and inspection of energy consuming components and energy producing components. An energy audit is conducted offline on an energy audit application on a network device for energy consuming components (e.g., lighting, heating, ventilation, air condition, etc.) and energy producing components (e.g., solar, wind, water, etc.) and automatically synchronized to a server network device when the network device is able to connect to a communications network and go on-line. Energy audit templates are provided to ensure all energy consuming components and energy producing components are fully defined and consistently identified on an electronic floor plan for a commercial or industrial building.”

[0026] U.S. Pat. No. 11,676,106, that issued to Seifert, et al., teaches “A method and system for managing and so materials and services for energy retrofit projects. An electronic energy retrofit project portal provides real-time designing, specifying, searching, pricing, quoting, ordering, and tracking of energy audit project materials and services required for a desired energy retrofit project. The functionality of the electronic energy retrofit project portal is available in real-time directly and from within existing design programs that create and provide energy audits and available as a standalone portal.”

[0027] U.S. Pat. No. 12,265,937, that issued to Seifert, et al., teaches “A method and system for providing energy audits or energy retrofits with plural auditors. An energy audit or energy retrofit is conducted in plural parts by plural different auditors on plural different energy audit applications on plural different network devices for energy consuming components (e.g., lighting, heating, ventilation, air condition, etc.) and energy producing components (e.g., solar, wind, water, etc.). The plural audit parts are automatically synchronized to a server network device allowing

energy audits or energy retrofits to be completed at different dates and times by different auditors.”

[0028] U.S. Pat. No. 12,307,398, that issued to Seifert, et al., teaches “A method and system for managing and sourcing materials and services for electrification projects. An electrification project portal provides real-time designing, specifying, searching, pricing, quoting, ordering, and tracking of energy audit project materials and services required for a desired electrification project. An electrification project includes replacing fossil-fuel powered devices with devices powered by electricity generated from stored power devices and other electricity sources. The functionality of the electrification project portal is available in real-time directly and from within existing design programs that create and provide electrification projects and available as a standalone portal.”

[0029] However, none of these solutions solve all of the problems associated with energy audit, energy retrofit or electrification systems.

[0030] Thus, it is desirable to solve some of the problems associated with energy audit or energy retrofit project systems conducting energy audits, energy retrofits and electrification of energy consuming and energy producing devices in a commercial or industrial building with multiple participants.

SUMMARY OF THE INVENTION

[0031] In accordance with preferred embodiments of the present invention, some of the problems associated with energy audit, energy retrofit and electrification systems conducting energy audits, energy retrofits and electrification of energy consuming and energy producing devices in a structure with multiple participants are overcome. A method and system for providing energy audit, energy retrofit and energy electrification projects with multiple participants is presented.

[0032] A cloud Software as a Service (SaaS) application with components for designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services and additional components including a real-time inventory availability component, an obsolete inventory management component, an artificial intelligence (AI) replacement product recommendation component, a manufacturer sponsorship component and a multi-party collaboration component, for conducting and managing energy audit projects, energy retrofit projects and/or electrification projects for energy consuming components and energy producing components for a building or other structures.

[0033] The foregoing and other features and advantages of preferred embodiments of the present invention will be more readily apparent from the following detailed description. The detailed description proceeds with references to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] Preferred embodiments of the present invention are described with reference to the following drawings, wherein:

[0035] FIG. 1 is a block diagram illustrating an exemplary energy audit processing and display system;

[0036] FIG. 2 is a block diagram illustrating an exemplary energy audit display system;

[0037] FIG. 3 is a block diagram illustrating an exemplary networking protocol stack;

[0038] FIG. 4 is a block diagram illustrating an exemplary cloud communications network;

[0039] FIG. 5 is a block diagram illustrating an exemplary cloud storage object;

[0040] FIG. 6 is a block diagram illustrating wearable network devices;

[0041] FIGS. 7A, 7B and 7C are a flow diagram illustrating a method for providing energy audits;

[0042] FIG. 8 is a block diagram illustrating components of energy audits;

[0043] FIG. 9 is a flow diagram illustrating a method for providing energy audits;

[0044] FIG. 10A-10D are a flow diagram illustrating a method for providing energy audits with multiple auditors;

[0045] FIG. 11 is a block diagram illustrating providing energy audits with multiple auditors;

[0046] FIG. 12 is a flow diagram illustrating a method for providing energy audits with multiple auditors;

[0047] FIG. 13 is a flow diagram illustrating a method for providing energy audits with multiple auditors;

[0048] FIG. 14 is a flow diagram illustrating a method for providing energy audits with multiple auditors;

[0049] FIG. 15 is a block diagram illustrating additional components of a cloud Software as a Service (SaaS); and

[0050] FIGS. 16A, 16B and 16C are a flow diagram illustrating a method for conducting and managing energy audits, energy retrofits and electrification projects.

DETAILED DESCRIPTION OF THE INVENTION

Exemplary Energy Audit Processing and Display System

[0051] FIG. 1 is a block diagram illustrating an exemplary electronic energy audit processing and display system 10. The exemplary electronic system 10 includes, but is not limited to, one or more target network devices 12, 14, 16, etc. each with one or more processors and each with a non-transitory computer readable medium.

[0052] The one or more target network devices 12, 14, 16 (illustrated in FIG. 1 only as a tablet and two smart phones for simplicity) include, but are not limited to, desktop and laptop computers, tablet computers, mobile phones, non-mobile phones with displays, three-dimensional (3D) printers, robots, smart phones, Internet phones, Internet appliances, personal digital/data assistants (PDA), portable, handheld and desktop video game devices, driverless vehicles 25, unmanned aerial vehicles (UAVs) 27, cameras 29, smart speakers 31, Internet of Things (IoT) devices 33, cable television (CATV), satellite television (SATV) and Internet television set-top boxes, digital televisions including high definition television (HDTV), three-dimensional (3DTV) televisions, wearable network devices 98-102 (FIG. 6), including Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR) network devices, 99, and/or other types of network devices.

[0053] The Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR) network devices, 99 includes VR, AR and/or MR artificial realities.

[0054] Virtual reality (VR) is a simulated experience that can be similar to or completely different from the real world. Current applications of virtual reality include entertainment (e.g. video games, etc.), education (e.g., technical training,

medical training, military training, etc.) and business (e.g. virtual meetings, etc.). Other distinct types of VR-style technology include augmented reality and mixed reality, sometimes referred to as extended reality or XR.

[0055] There are two types of VR; immersive VR and text-based networked VR (also known as “Cyberspace VR”). The immersive VR changes your view, when you move your head. While both VRs are appropriate for training, Cyberspace VR is used for distance learning. In some cases, these two types are even complementary to each other.

[0056] Currently, standard virtual reality systems use either virtual reality headsets or multi-projected environments to generate realistic images, sounds and other sensations that simulate a user’s physical presence in a virtual environment. A person using virtual reality equipment is able to look around the artificial world, move around in it, and interact with virtual features or items. The effect is commonly created by VR headsets consisting of a head-mounted display with a small screen in front of the eyes, but can also be created through specially designed rooms with multiple large screens. Virtual reality typically incorporates auditory and video feedback, but may also allow other types of sensory and force feedback through haptic technology.

[0057] Augmented Reality (AR) is a type of virtual reality technology that blends what the user sees in their real surroundings with digital content generated by computer software. The additional software-generated images with the virtual scene typically enhance how the real surroundings look in some way. AR systems layer virtual information over a camera live feed into a headset or smart glasses or through mobile devices giving the user the ability to view three-dimensional images.

[0058] The terms augmented reality (AR) and virtual reality (VR) are often confused. AR is any hardware and/or software that overlays text or graphics on real-world images or video. VR is any hardware and/or software that places the user in a completely artificial environment.

[0059] For example, the digital application POKEMON GO, is an AR mobile game developed and published by Niantic in collaboration with NINTENDO and The POKEMON COMPANY for APPLE iOS and ANDROID devices. POKEMON GO uses mobile devices with GPS to locate, capture, train, and battle virtual creatures, called POKEMON, which appear as if they are in the player’s real-world physical location.

[0060] Mixed reality (MR) is the merging of the real world and virtual worlds to produce new environments and visualizations where physical and digital objects co-exist and interact in real-time.

[0061] A “smart phone” is a mobile phone 14 that offers more advanced computing ability and connectivity than a contemporary basic feature phone. Smart phones and feature phones may be thought of as handheld computers integrated with a mobile telephone, but while most feature phones are able to run applications based on platforms such as JAVA ME, a smart phone usually allows the user to install and run more advanced applications. Smart phones and/or tablet computers run complete operating system software providing a platform for application developers.

[0062] The tablet computers 12 include, but are not limited to, tablet computers such as the IPAD, by APPLE, Inc., the HP Tablet, by HEWLETT PACKARD, Inc., the PLAYBOOK, by RIM, Inc., the TABLET, by SONY, Inc., etc.

[0063] A “smart speaker” **31** is a type of wireless speaker and voice command device with an integrated virtual assistant that offers interactive actions and hands-free activation with the help of one “hot word” (or several “hot words”). Some smart speakers can also act as a smart device that utilizes Wi-Fi, Bluetooth and other wireless protocol standards to extend usage beyond audio playback, such as to control home automation devices. This can include, but is not be limited to, features such as compatibility across a number of services and platforms, peer-to-peer connection through mesh networking, virtual assistants, and others. Each can have its own designated interface and features in-house, usually launched or controlled via application or home automation software. Some smart speakers also include a screen to show the user a visual response.

[0064] The IoT network devices, include but are not limited to, security cameras, doorbells with real-time video cameras, baby monitors, televisions, set-top boxes, lighting, heating (e.g., smart thermostats, etc.), ventilation, air conditioning (HVAC) systems, and appliances such as washers, dryers, robotic vacuums, air purifiers, ovens, refrigerators, freezers, toys, game platform controllers, game platform attachments (e.g., guns, googles, sports equipment, etc.), and/or other IoT network devices.

[0065] The target network devices **12, 14, 16, 25, 27, 29, 31, 33, 98-104** are in communications with a cloud communications network **18** or a non-cloud computing network **18'** via one or more wired and/or wireless communications interfaces. The cloud communications network **18**, is also called a “cloud computing network” herein and the terms may be used interchangeably.

[0066] The plural target network devices **12, 14, 16, 25, 27, 29, 31, 33, 98-104** make requests for electronic content **13, 15, 13'/82, 15'/82, 138, 146, 146'** via the cloud communications network **18** and/or non-cloud communications network **18'**.

[0067] The cloud communications network **18** and non-cloud communications network **18'** includes, but is not limited to, communications over a wire connected to the target network devices, wireless communications, and other types of communications using one or more communications and/or networking protocols.

[0068] Plural server network devices **20, 22, 24, 26** (only four of which are illustrated for simplicity) each with one or more processors and a non-transitory computer readable medium include one or more associated databases **20', 22', 24', 26'** including server applications **30a, 30b, 30c, 30d, 30e** and/or cloud Software as a Service **64** (SaaS) applications. The plural network devices **20, 22, 24, 26** are in communications with the one or more target devices **12, 14, 16, 25, 27, 29, 31, 33, 98-104** via the cloud communications network **18** and non-cloud communications network **18'**.

[0069] Plural server network devices **20, 22, 24, 26** are physically located on one more public networks **76** (See FIG. 4), private networks **72**, community networks **74** and/or hybrid networks **78** comprising the cloud network **18**.

[0070] One or more server network devices (e.g., **20, 22, 24, 26**, etc.) store portions **13', 15'** of the electronic content **13, 15, 138, 146, 146'** (e.g., energy audit data, etc.) as cloud storage objects **82** (FIG. 5) as is described herein.

[0071] The plural server network devices **20, 22, 24, 26**, may be connected to, but are not limited to, World Wide Web servers, Internet servers, search engine servers, vertical search engine servers, social networking site servers, file

servers, other types of electronic information servers, and other types of server network devices (e.g., edge servers, firewalls, routers, gateways, etc.).

[0072] The plural server network devices **20, 22, 24, 26** also include, but are not limited to, network servers used for cloud computing providers, etc.

[0073] The cloud communications network **18** and non-cloud communications network **18'** includes, but is not limited to, a wired and/or wireless communications network comprising one or more portions of: the Internet, an intranet, a Local Area Network (LAN), a wireless LAN (WiLAN), a Wide Area Network (WAN), a Metropolitan Area Network (MAN), a Public Switched Telephone Network (PSTN), a Wireless Personal Area Network (WPAN) and other types of wired and/or wireless communications networks **18**.

[0074] The cloud communications network **18** and non-cloud communications network **18'** includes one or more gateways, routers, bridges and/or switches. A gateway connects computer networks using different network protocols and/or operating at different transmission capacities. A router receives transmitted messages and forwards them to their correct destinations over the most efficient available route. A bridge is a device that connects networks using the same communications protocols so that information can be passed from one network device to another. A switch is a device that filters and forwards packets between network segments based on some pre-determined sequence (e.g., timing, sequence number, etc.).

[0075] An operating environment for the network devices of the exemplary electronic information display system **10** include a processing system with one or more high speed Central Processing Unit(s) (CPU), processors, one or more memories and/or other types of non-transitory computer readable mediums. In accordance with the practices of persons skilled in the art of computer programming, the present invention is described below with reference to acts and symbolic representations of operations or instructions that are performed by the processing system, unless indicated otherwise. Such acts and operations or instructions are referred to as being “computer-executed,” “CPU-executed,” or “processor-executed.”

[0076] It will be appreciated that acts and symbolically represented operations or instructions include the manipulation of electrical information by the CPU or processor. An electrical system represents data bits which cause a resulting transformation or reduction of the electrical information or biological information, and the maintenance of data bits at memory locations in a memory system to thereby reconfigure or otherwise alter the CPU's or processor's operation, as well as other processing of information. The memory locations where data bits are maintained are physical locations that have particular electrical, magnetic, optical, or organic properties corresponding to the data bits.

[0077] The data bits may also be maintained on a non-transitory computer readable medium including magnetic disks, optical disks, organic memory, and any other volatile (e.g., Random Access Memory (RAM)) or non-volatile (e.g., Read-Only Memory (ROM), flash memory, etc.) mass storage system readable by the CPU. The non-transitory computer readable medium includes cooperating or interconnected computer readable medium, which exist exclusively on the processing system or can be distributed among multiple interconnected processing systems that may be local or remote to the processing system.

Exemplary Electronic Energy Audit Content Display System

[00778] FIG. 2 is a block diagram illustrating an exemplary electronic energy audit information display system 28. The exemplary electronic message information display system 12' includes, but is not limited to a target network device (e.g., 12, etc.) with an application 30 and a display component 32. The application 30 presents a graphical user interface (GUI) 34 on the display 32 component. The GUI 32 presents a multi-window 36, 38, etc. (only two of which are illustrated) interface to a user.

[0079] In one embodiment of the invention, the application 30 is a software application. However, the present invention is not limited to this embodiment and the application 30 can be hardware, firmware, hardware and/or any combination thereof. In one embodiment, the application 30 includes a mobile application for a smart phone, electronic tablet and/or other network device. In one embodiment, the application 30 includes web-browser based application. In one embodiment, the application 30 includes a web-chat client application. In another embodiment, the server application 30a, 30b, 30c, 30d, 30e, SaaS 64 includes a cloud application used on a cloud communications network 18 provided by a cloud service 64. However, the present invention is not limited these embodiments and other embodiments can be used to practice the invention

[0080] In another embodiment, a portion of the application 30 is executing on the target network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 and another portion of the application 30a, 30b, 30c, 30d, 30e, SaaS 64 is executing on the server network devices 20, 22, 24, 26. The applications also include one or more library applications. However, the present invention is not limited these embodiments and other embodiments can be used to practice the invention.

Exemplary Networking Protocol Stack

[0081] FIG. 3 is a block diagram illustrating a layered protocol stack 38 for network devices in the electronic message information display system 10. The layered protocol stack 38 is described with respect to Internet Protocol (IP) suites comprising in general from lowest-to-highest, a link 42, network 44, transport 48 and application 56 layers. However, more or fewer layers could also be used, and different layer designations could also be used for the layers in the protocol stack 38 (e.g., layering based on the Open Systems Interconnection (OSI) model including from lowest-to-highest, a physical, data-link, network, transport, session, presentation and application layer.).

[0082] The network devices 12, 14, 16, 20, 22, 24, 26, 31, 98-104 are connected to the communication network 18 with Network Interface Card (NIC) cards including device drivers 40 in a link layer 42 for the actual hardware connecting the network devices 12, 14, 16, 20, 22, 24, 26, 31, 98-104 to the cloud communications network 18. For example, the NIC device drivers 40 may include a serial port device driver, a digital subscriber line (DSL) device driver, an Ethernet device driver, a wireless device driver, a wired device driver, etc. The device drivers interface with the actual hardware being used to connect the network devices to the cloud communications network 18. The NIC cards have a medium access control (MAC) address that is unique to each NIC and unique across the whole cloud network 18.

The Medium Access Control (MAC) protocol is used to provide a data link layer of an Ethernet LAN system and for other network systems.

[0083] Above the link layer 42 is a network layer 44 (also called the Internet Layer for Internet Protocol (IP) suites). The network layer 44 includes, but is not limited to, an IP layer 46.

[0084] IP 46 is an addressing protocol designed to route traffic within a network or between networks. However, more, fewer or other protocols can also be used in the network layer 44, and the present invention is not limited to IP 46. For more information on IP 46 see IETF RFC-791, incorporated herein by reference.

[0085] Above network layer 44 is a transport layer 48. The transport layer 48 includes, but is not limited to, an optional Internet Group Management Protocol (IGMP) layer 50, a Internet Control Message Protocol (ICMP) layer 52, a Transmission Control Protocol (TCP) layer 54 and a User Datagram Protocol (UDP) layer 56. However, more, fewer or other protocols could also be used in the transport layer 48.

[0086] Optional IGMP layer 50, hereinafter IGMP 50, is responsible for multicasting. For more information on IGMP 50 see RFC-1112, incorporated herein by reference. ICMP layer 52, hereinafter ICMP 52 is used for IP 46 control. The main functions of ICMP 52 include error reporting, reachability testing (e.g., ping, etc.), route-change notification, performance, subnet addressing and other maintenance. For more information on ICMP 52 see RFC-792, incorporated herein by reference. Both IGMP 50 and ICMP 52 are not required in the protocol stack 38. ICMP 52 can be used alone without optional IGMP layer 50.

[0087] TCP layer 54, hereinafter TCP 54, provides a connection-oriented, end-to-end reliable protocol designed to fit into a layered hierarchy of protocols which support multi-network applications. TCP 54 provides for reliable inter-process communication between pairs of processes in network devices attached to distinct but interconnected networks. For more information on TCP 54 see RFC-793, incorporated herein by reference.

[0088] UDP layer 56, hereinafter UDP 56, provides a connectionless mode of communications with datagrams in an interconnected set of computer networks. UDP 56 provides a transaction-oriented datagram protocol, where delivery and duplicate packet protection are not guaranteed. For more information on UDP 56 see RFC-768, incorporated herein by reference. Both TCP 54 and UDP 56 are not required in protocol stack 38. Either TCP 54 or UDP 56 can be used without the other.

[0089] Above transport layer 48 is an application layer 57 where application programs 58 (e.g., 30, 30a, 30b, 30c, 30d, 30e, SaaS 64 etc.) to carry out desired functionality for a network device reside. For example, the application programs 58 for the client network devices 12, 14, 16, 27, 29, 31, 98-104 may include web-browsers or other application programs, application program 30, while application programs for the server network devices 20, 22, 24, 26 may include other application programs (e.g., 30a, 30b, 30c, 30d, 30e, SaaS 64 etc.).

[0090] In one embodiment, application program 30 includes an Energy Audit, application 30a, an Energy Audit offline functionality application 30b, an Artificial Intelligence (AI) application 30c, a creation component application, 30d, 30e and/or a measurement component application

30e. The AI application **30c** is used with and/or without Big Data. However, the present invention is not limited to such an embodiment and more, fewer and/or other types of applications can be used to practice the invention.

[0091] However, the protocol stack **38** is not limited to the protocol layers illustrated and more, fewer or other layers and protocols can also be used in protocol stack **38**. In addition, other protocols from the Internet Protocol suites (e.g., Simple Mail Transfer Protocol, (SMTP), Hyper Text Transfer Protocol (HTTP), File Transfer Protocol (FTP), Dynamic Host Configuration Protocol (DHCP), DNS, etc.) and/or other protocols from other protocol suites may also be used in protocol stack **38**.

[0092] In addition, markup languages such as HyperText Markup Language (HTML), Extensible Markup Language (XML) and/or others are used. HyperText Markup Language (HTML) is a markup language for creating web pages and other information that can be displayed in a web browser.

[0093] HTML is written in the form of HTML elements consisting of tags enclosed in angle brackets within the web page content. HTML tags most commonly come in pairs although some tags represent empty elements and so are unpaired. The first tag in a pair is the start tag, and the second tag is the end tag (they are also called opening tags and closing tags). In between these tags web designers can add text, further tags, comments and other types of text-based content.

[0094] The purpose of a web browser is to read HTML documents and compose them into visible or audible web pages. The browser does not display the HTML tags, but uses the tags to interpret the content of the page.

[0095] HTML elements form the building blocks of all websites. HTML allows images and objects to be embedded and can be used to create interactive forms. It provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items. It can embed scripts written in languages such as JavaScript which affect the behavior of HTML web pages.

[0096] Extensible Markup Language (XML) is another markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable. It is defined in the XML 1.0 Specification produced by the W3C, the contents of which are incorporated by reference and several other related specifications, all free open standards.

[0097] XML a textual data format with strong support via Unicode for the languages of the world. Although the design of XML focuses on documents, it is widely used for the representation of arbitrary data structures, for example in web services. The oldest schema language for XML is the Document Type Definition (DTD). DTDs within XML documents define entities, which are arbitrary fragments of text and/or markup tags that the XML processor inserts in the DTD itself and in the XML document wherever they are referenced, like character escapes.

[0098] Preferred embodiments of the present invention include network devices and wired and wireless interfaces that are compliant with all or part of standards proposed by the Institute of Electrical and Electronic Engineers (IEEE), International Telecommunications Union-Telecommunication Standardization Sector (ITU), European Telecommunications Standards Institute (ETSI), Internet Engineering Task Force (IETF), U.S. National Institute of Security

Technology (NIST), American National Standard Institute (ANSI), Wireless Application Protocol (WAP) Forum, Bluetooth Forum, or the ADSL Forum.

Wireless Interfaces

[0099] In one embodiment of the present invention, the wireless interfaces on network devices **12, 14, 16, 20, 22, 24, 26, 31, 98-104** include but are not limited to, IEEE 802.11a, 802.11b, 802.11g, 802.11n, 802.15.4 (ZigBee), “Wireless Fidelity” (Wi-Fi), “Worldwide Interoperability for Microwave Access” (WiMAX), ETSI High Performance Radio Metropolitan Area Network (HIPERMAN) or “RF Home” wireless interfaces. In another embodiment of the present invention, the wireless sensor device may include an integral or separate Bluetooth and/or infra data association (IrDA) module for wireless Bluetooth or wireless infrared communications. However, the present invention is not limited to such an embodiment and other 802.11xx and other types of wireless interfaces can also be used.

[0100] 802.11b is a short-range wireless network standard. The IEEE 802.11b standard defines wireless interfaces that provide up to 11 Mbps wireless data transmission to and from wireless devices over short ranges. 802.11a is an extension of the 802.11b and can deliver speeds up to 54Mbps. 802.11g deliver speeds on par with 802.11a. However, other 802.11XX interfaces can also be used and the present invention is not limited to the 802.11 protocols defined. The IEEE 802.11a, 802.11b and 802.11g standards are incorporated herein by reference.

[0101] Wi-Fi is a type of 802.11xx interface, whether 802.11b, 802.11a, dual-band, etc. Wi-Fi devices include RF interfaces such as 2.4GHz for 802.11b or 802.11g and 5 GHz for 802.11a.

[0102] 802.15.4 (Zigbee) is low data rate network standard used for mesh network devices such as sensors, interactive toys, smart badges, remote controls, and home automation. The 802.15.4 standard provides data rates of 250 kbps, 40 kbps, and 20 kbps., two addressing modes; 16-bit short and 64-bit IEEE addressing, support for critical latency devices, such as joysticks, Carrier Sense Multiple Access/Collision Avoidance, (CSMA-CA) channel access, automatic network establishment by a coordinator, a full handshake protocol for transfer reliability, power management to ensure low power consumption for multi-month to multi-year battery usage and up to 16 channels in the 2.4 GHz Industrial, Scientific and Medical (ISM) band (Worldwide), 10 channels in the 915 MHz (US) and one channel in the 868 MHz band (Europe). The IEEE 802.15.4-2003 standard is incorporated herein by reference.

[0103] WiMAX is an industry trade organization formed by leading communications component and equipment companies to promote and certify compatibility and interoperability of broadband wireless access equipment that conforms to the IEEE 802.16XX and ETSI HIPERMAN. HIPERMAN is the European standard for metropolitan area networks (MAN).

[0104] The IEEE The 802.16a and 802.16g standards are wireless MAN technology standard that provides a wireless alternative to cable, DSL and T1/E1 for last mile broadband access. It is also used as complementary technology to connect IEEE 802.11XX hot spots to the Internet.

[0105] The IEEE 802.16a standard for 2-11 GHz is a wireless MAN technology that provides broadband wireless connectivity to fixed, portable and nomadic devices. It

provides up to 50-kilometers of service area range, allows users to get broadband connectivity without needing direct line of sight with the base station, and provides total data rates of up to 280 Mbps per base station, which is enough bandwidth to simultaneously support hundreds of businesses with T1/E1-type connectivity and thousands of homes with DSL-type connectivity with a single base station. The IEEE 802.16g provides up to 100 Mbps.

[0106] The IEEE 802.16e standard is an extension to the approved IEEE 802.16a/16g standard. The purpose of 802.16e is to add limited mobility to the current standard which is designed for fixed operation.

[0107] The ETSI HIPERMAN standard is an interoperable broadband fixed wireless access standard for systems operating at radio frequencies between 2 GHz and 11 GHz.

[0108] The IEEE 802.16a, 802.16e and 802.16g standards are incorporated herein by reference. WiMAX can be used to provide a WLP.

[0109] The ETSI HIPERMAN standards TR 101 031, TR 101 475, TR 101 493-1 through TR 101 493-3, TR 101 761-1 through TR 101 761-4, TR 101 762, TR 101 763-1 through TR 101 763-3 and TR 101 957 are incorporated herein by reference. ETSI HIPERMAN can be used to provide a WLP.

[0110] In one embodiment, the plural server network devices **20, 22, 24, 26** include a connection to plural network interface cards (NICs) in a backplane connected to a communications bus. The NIC cards provide gigabit/second (1×10^9 bits/second) communications speed of electronic information. This allows “scaling out” for fast electronic content retrieval. The NICs are connected to the plural server network devices **20, 22, 24, 26** and the cloud communications network **18**. However, the present invention is not limited to the NICs described and other types of NICs in other configurations and connections with and/or without buses can also be used to practice the invention.

[0111] In one embodiment, of the invention, the wireless interfaces also include wireless personal area network (WPAN) interfaces. As is known in the art, a WPAN is a personal area network for interconnecting devices centered around an individual person’s devices in which the connections are wireless. A WPAN interconnects all the ordinary computing and communicating devices that a person has on their desk (e.g. computer, etc.) or carry with them (e.g., PDA, mobile phone, smart phone, table computer two-way pager, etc.)

[0112] A key concept in WPAN technology is known as “plugging in.” In the ideal scenario, when any two WPAN-equipped devices come into close proximity (within several meters and/or feet of each other) or within a few miles and/or kilometers of a central server (not illustrated), they can communicate via wireless communications as if connected by a cable. WPAN devices can also lock out other devices selectively, preventing needless interference or unauthorized access to secure information. Zigbee is one wireless protocol used on WPAN networks such as cloud communications network **18** or non-cloud communications network **18’**.

[0113] The one or more target network devices **12, 14, 16, 20, 22, 24, 26, 31, 98-104** and one or more server network devices **20, 22, 24, 26** communicate with each other and other network devices with near field communications (NFC) and/or machine-to-machine (M2M) communications.

[0114] “Near field communication (NFC)” is a set of standards for smartphones and similar network devices to establish radio communication with each other by touching them together or bringing them into close proximity, usually no more than a few centimeters. Present applications include contactless transactions, data exchange, and simplified setup of more complex communications such as Wi-Fi. Communication is also possible between an NFC device and an unpowered NFC chip, called a “tag” including radio frequency identifier (RFID) tags and/or sensor.

[0115] NFC standards cover communications protocols and data exchange formats, and are based on existing radio-frequency identification (RFID) standards including ISO/IEC 14443 and FeliCa. These standards include ISO/IEC 1809 and those defined by the NFC Forum, all of which are incorporated by reference.

[0116] An “RFID tag” is an object that can be applied to or incorporated into a product, animal, or person for the purpose of identification and/or tracking using RF signals.

[0117] An “RFID sensor” is a device that measures a physical quantity and converts it into an RF signal which can be read by an observer or by an instrument (e.g., target network devices **12, 14, 16, 20, 22, 24, 26, 31, 98-104**, server network devices **20, 22, 24, 26**, etc.).

[0118] “Machine to machine (M2M)” refers to technologies that allow both wireless and wired systems to communicate with other devices of the same ability. M2M uses a device to capture an event (such as option purchase, etc.), which is relayed through a network (wireless, wired cloud, etc.) to an application (software program), that translates the captured event into meaningful information. Such communication was originally accomplished by having a remote network of machines relay information back to a central hub for analysis, which would then be rerouted into a system like a personal computer.

[0119] However, modern M2M communication has expanded beyond a one-to-one connection and changed into a system of networks that transmits data many-to-one and many-to-many to plural different types of devices and appliances. The expansion of IP networks across the world has made it far easier for M2M communication to take place and has lessened the amount of power and time necessary for information to be communicated between machines.

[0120] However, the present invention is not limited to such wireless interfaces and wireless networks and more, fewer and/or other wireless interfaces can be used to practice the invention.

Wired Interfaces

[0121] In one embodiment of the present invention, the wired interfaces include wired interfaces and corresponding networking protocols for wired connections to the Public Switched Telephone Network (PSTN) and/or a cable television network (CATV) and/or satellite television networks (SATV) and/or three-dimensional television (3DTV), including HDTV that connect the network devices **12, 14, 16, 20, 22, 24, 26, 31, 98-104** via one or more twisted pairs of copper wires, digital subscriber lines (e.g. DSL, ADSL, VDSL, etc.) coaxial cable, fiber optic cable, other connection media or other connection interfaces. The PSTN is any public switched telephone network provided by AT&T, GTE, Sprint, MCI, SBC, Verizon and others. The CATV is any cable television network provided by the Comcast, Time Warner, etc. However, the present invention is not limited to

such wired interfaces and more, fewer and/or other wired interfaces can be used to practice the invention.

Television Services

[0122] In one embodiment, the cloud applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** provide cloud Software as a Service **64** (SaaS) **64**, **64'** services and/or non-cloud application services from television services over the cloud communications network **18** or application services over the non-cloud communications network **18'**. The television services include digital television services, including, but not limited to, cable television, satellite television, high-definition television, three-dimensional, televisions and other types of network devices.

[0123] However, the present invention is not limited to such television services and more, fewer and/or other television services can be used to practice the invention.

Internet Television Services

[0124] In one embodiment, the cloud applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** provide cloud SaaS **64**, **64'** services and/or non-cloud application services from Internet television services over the cloud communications network **18** or non-cloud communications network **18'**. The television services include Internet television, Web-TV, and/or Internet Protocol Television (IPTv) and/or other broadcast television services.

[0125] "Internet television" allows users to choose a program or the television show they want to watch from an archive of programs or from a channel directory. The two forms of viewing Internet television are streaming content directly to a media player or simply downloading a program to a viewer's set-top box, game console, computer, or other network device.

[0126] "Web-TV" delivers digital content via broadband and mobile networks. The digital content is streamed to a viewer's set-top box, game console, computer, or other network device.

[0127] "Internet Protocol television (IPTv)" is a system through which Internet television services are delivered using the architecture and networking methods of the Internet Protocol Suite over a packet-switched network infrastructure, e.g., the Internet and broadband Internet access networks, instead of being delivered through traditional radio frequency broadcast, satellite signal, and cable television formats.

[0128] However, the present invention is not limited to such Internet Television services and more, fewer and/or other Internet Television services can be used to practice the invention.

General Search Engine Services

[0129] In one embodiment, the cloud applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** provide cloud SaaS **64**, **64'** services and/or non-cloud application services from general search engine services. A search engine is designed to search for information on a cloud communications network **18** or non-cloud communications network **18'** such as the Internet including World Wide Web servers, HTTP, FTP servers etc. The search results are generally presented in a list of electronic results. The information may consist of web pages, images, electronic information, multimedia information, and other types of files. Some search engines also mine

data available in databases or open directories. Unlike web directories, which are maintained by human editors, search engines typically operate algorithmically and/or are a mixture of algorithmic and human input.

[0130] In one embodiment, the cloud applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** provide cloud SaaS **64**, **64'** services and/or non-cloud application services from general search engine services. In another embodiment, the cloud applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** provide general search engine services by interacting with one or more other public search engines (e.g., GOOGLE, BING, YAHOO, etc.) and/or private search engine services.

[0131] In another embodiment, the cloud applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** provide cloud SaaS **64**, **64'** services and/or non-cloud application services from specialized search engine services, such as vertical search engine services by interacting with one or more other public vertical search engines and/or private search engine services.

[0132] However, the present invention is not limited to such general and/or vertical search engine services and more, fewer and/or other general search engine services can be used to practice the invention.

Social Networking Services

[0133] In one embodiment, the cloud applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** provide cloud SaaS **64**, **64'** services and/or non-cloud application services from one more social networking services including to/from one or more social networking web-sites (e.g., FACEBOOK, YOUTUBE, TWITTER, INSTAGRAM, etc.). The social networking web-sites also include, but are not limited to, social couponing sites, dating web-sites, blogs, RSS feeds, and other types of information web-sites in which messages can be left or posted for a variety of social activities.

[0134] However, the present invention is not limited to the social networking services described and other public and private social networking services can also be used to practice the invention.

Security And Encryption

[0135] Network devices **12**, **14**, **16**, **20**, **22**, **24**, **26**, **31**, **98-104** with wired and/or wireless interfaces of the present invention include one or more of the security and encryption techniques discussed herein for secure communications on the cloud communications network **18** or non-cloud communications network **18'**.

[0136] Application programs **58** (FIG. 2) include security and/or encryption application programs integral to and/or separate from the applications **30**, **30a**, **30b**, **30c**, **30d**, **30e**. Security and/or encryption programs may also exist in hardware components on the network devices (**12**, **14**, **16**, **20**, **22**, **24**, **26**, **31**, **98-104**) described herein and/or exist in a combination of hardware, software and/or firmware.

[0137] Wireless Encryption Protocol (WEP) (also called "Wired Equivalent Privacy") is a security protocol for WiLANs defined in the IEEE 802.11b standard. WEP is cryptographic privacy algorithm, based on the Rivest Cipher 4 (RC4) encryption engine, used to provide confidentiality for 802.11b wireless data.

[0138] RC4 is cipher designed by RSA Data Security, Inc. of Bedford, Massachusetts, which can accept encryption keys of arbitrary length, and is essentially a pseudo random

number generator with an output of the generator being XORed with a data stream to produce encrypted data.

[0139] WEP is used at the two lowest layers of the OSI model, the physical layer and the data link layer, therefore, it typically does not offer end-to-end security. One another problem with WEP is that its encryption keys are static rather than dynamic. To update WEP encryption keys, an individual has to manually update a WEP key. WEP also typically uses 40-bit static keys for encryption and thus provides “weak encryption,” making a WEP device a target of hackers.

[0140] The IEEE 802.11 Working Group is working on a security upgrade for the 802.11 standard called “802.11i.” This supplemental draft standard is intended to improve WiLAN security. It describes the encrypted transmission of data between systems 802.11X WiLANs. It also defines new encryption key protocols including the Temporal Key Integrity Protocol (TKIP). The IEEE 802.11i draft standard, version 4, completed Jun. 6, 2003, is incorporated herein by reference.

[0141] The 802.11i standard is based on 802.1x port-based authentication for user and device authentication. The 802.11i standard includes two main developments: Wi-Fi Protected Access (WPA) and Robust Security Network (RSN). WPA uses the same RC4 underlying encryption algorithm as WEP. However, WPA uses TKIP to improve security of keys used with WEP. WPA keys are derived and rotated more often than WEP keys and thus provide additional security. WPA also adds a message-integrity-check function to prevent packet forgeries.

[0142] RSN uses dynamic negotiation of authentication and selectable encryption algorithms between wireless access points and wireless devices. The authentication schemes proposed in the draft standard include Extensible Authentication Protocol (EAP). One proposed encryption algorithm is an Advanced Encryption Standard (AES) encryption algorithm.

[0143] Dynamic negotiation of authentication and encryption algorithms lets RSN evolve with the state of the art in security, adding algorithms to address new threats and continuing to provide the security necessary to protect information that WiLANs carry.

[0144] The NIST developed a new encryption standard, the Advanced Encryption Standard (AES) to keep government information secure. AES is intended to be a stronger, more efficient successor to Triple Data Encryption Standard (3DES).

[0145] DES is a popular symmetric-key encryption method developed in 1975 and standardized by ANSI in 1981 as ANSI X.3.92, the contents of which are incorporated herein by reference. As is known in the art, 3DES is the encrypt-decrypt-encrypt (EDE) mode of the DES cipher algorithm. 3DES is defined in the ANSI standard, ANSI X9.52-1998, the contents of which are incorporated herein by reference. DES modes of operation are used in conjunction with the NIST Federal Information Processing Standard (FIPS) for data encryption (FIPS 46-3, October 1999), the contents of which are incorporated herein by reference.

[0146] The NIST approved a FIPS for the AES, FIPS-197. This standard specified “Rijndael” encryption as a FIPS-approved symmetric encryption algorithm that may be used by U.S. Government organizations (and others) to protect

sensitive information. The NIST FIPS-197 standard (AES FIPS PUB 197, November 2001) is incorporated herein by reference.

[0147] The NIST approved a FIPS for U.S. Federal Government requirements for information technology products for sensitive but unclassified (SBU) communications. The NIST FIPS Security Requirements for Cryptographic Modules (FIPS PUB 140-2, May 2001) is incorporated herein by reference.

[0148] RSA is a public key encryption system which can be used both for encrypting messages and making digital signatures. The letters RSA stand for the names of the inventors: Rivest, Shamir and Adleman. For more information on RSA, see U.S. Pat. No. 4,405,829, now expired and incorporated herein by reference.

[0149] “Hashing” is the transformation of a string of characters into a usually shorter fixed-length value or key that represents the original string. Hashing is used to index and retrieve items in a database because it is faster to find the item using the shorter hashed key than to find it using the original value. It is also used in many encryption algorithms.

[0150] Secure Hash Algorithm (SHA), is used for computing a secure condensed representation of a data message or a data file. When a message of any length $<2^{64}$ bits is input, the SHA-1 produces a 160-bit output called a “message digest.” The message digest can then be input to other security techniques such as encryption, a Digital Signature Algorithm (DSA) and others which generates or verifies a security mechanism for the message. SHA-512 outputs a 512-bit message digest. The Secure Hash Standard, FIPS PUB 180-1, Apr. 17, 1995, is incorporated herein by reference.

[0151] Message Digest-5 (MD-5) takes as input a message of arbitrary length and produces as output a 128-bit “message digest” of the input. The MD5 algorithm is intended for digital signature applications, where a large file must be “compressed” in a secure manner before being encrypted with a private (secret) key under a public-key cryptosystem such as RSA. The IETF RFC-1321, entitled “The MD5 Message-Digest Algorithm” is incorporated here by reference.

[0152] Providing a way to check the integrity of information transmitted over or stored in an unreliable medium such as a wireless network is a prime necessity in the world of open computing and communications. Mechanisms that provide such integrity check based on a secret key are called “message authentication codes” (MAC). Typically, message authentication codes are used between two parties that share a secret key in order to validate information transmitted between these parties.

[0153] Keyed Hashing for Message Authentication Codes (HMAC), is a mechanism for message authentication using cryptographic hash functions. HMAC is used with any iterative cryptographic hash function, e.g., MD5, SHA-1, SHA-512, etc. in combination with a secret shared key. The cryptographic strength of HMAC depends on the properties of the underlying hash function. The IETF RFC-2101, entitled “HMAC: Keyed-Hashing for Message Authentication” is incorporated here by reference.

[0154] An Electronic Code Book (ECB) is a mode of operation for a “block cipher,” with the characteristic that each possible block of plaintext has a defined corresponding cipher text value and vice versa. In other words, the same plaintext value will always result in the same cipher text

value. Electronic Code Book is used when a volume of plaintext is separated into several blocks of data, each of which is then encrypted independently of other blocks. The Electronic Code Book has the ability to support a separate encryption key for each block type.

[0155] Diffie and Hellman (DH) describe several different group methods for two parties to agree upon a shared secret in such a way that the secret will be unavailable to eavesdroppers. This secret is then converted into various types of cryptographic keys. A large number of the variants of the DH method exist including ANSI X9.42. The IETF RFC-2631, entitled “Diffie-Hellman Key Agreement Method” is incorporated here by reference.

[0156] The HyperText Transport Protocol (HTTP) Secure (HTTPS), is a standard for encrypted communications on the World Wide Web. HTTPS is actually just HTTP over a Secure Sockets Layer (SSL). For more information on HTTP, see IETF RFC-2616 incorporated herein by reference.

[0157] The SSL protocol is a protocol layer which may be placed between a reliable connection-oriented network layer protocol (e.g. TCP/IP) and the application protocol layer (e.g. HTTP). SSL provides for secure communication between a source and destination by allowing mutual authentication, the use of digital signatures for integrity, and encryption for privacy.

[0158] The SSL protocol is designed to support a range of choices for specific security methods used for cryptography, message digests, and digital signatures. The security methods are negotiated between the source and destination at the start of establishing a protocol session. The SSL 2.0 protocol specification, by Kipp E. B. Hickman, 1995 is incorporated herein by reference. More information on SSL is available at the domain name See “netscape.com/eng/security/SSL_2.html.”

[0159] Transport Layer Security (TLS) provides communications privacy over the Internet. The protocol allows client/server applications to communicate over a transport layer (e.g., TCP) in a way that is designed to prevent eavesdropping, tampering, or message forgery. For more information on TLS see IETF RFC-2246, incorporated herein by reference.

[0160] In one embodiment, the security functionality includes Cisco Compatible EXtensions (CCX). CCX includes security specifications for makers of 802.11xx

wireless LAN chips for ensuring compliance with Cisco’s proprietary wireless security LAN protocols. As is known in the art, Cisco Systems, Inc. of San Jose, California is supplier of networking hardware and software, including router and security products.

[0161] However, the present invention is not limited to such security and encryption methods described herein and more, fewer and/or other types of security and encryption methods can be used to practice the invention. The security and encryption methods described herein can also be used in various combinations and/or in different layers of the protocol stack **38** with each other.

Cloud Computing Networks

[0162] FIG. 4 is a block diagram **60** illustrating an exemplary cloud computing network **18**. The cloud computing network **18** is also referred to as a “cloud communications network” **18**. However, the present invention is not limited to this cloud computing model and other cloud computing models can also be used to practice the invention. The exemplary cloud communications network includes both wired and/or wireless components of public and private networks.

[0163] In one embodiment, the cloud computing network **18** includes a cloud communications network **18** comprising plural different cloud component networks **72, 74, 76, 78**. “Cloud computing” is a model for enabling, on-demand network access to a shared pool of configurable computing resources (e.g., public and private networks, servers, storage, applications, and services) that are shared, rapidly provisioned and released with minimal management effort or service provider interaction.

[0164] This exemplary cloud computing model for electronic information retrieval promotes availability for shared resources and comprises: (1) cloud computing essential characteristics; (2) cloud computing service models; and (3) cloud computing deployment models. However, the present invention is not limited to this cloud computing model and other cloud computing models can also be used to practice the invention.

[0165] Exemplary cloud computing essential characteristics appear in Table 1. However, the present invention is not limited to these essential characteristics and more, fewer or other characteristics can also be used to practice the invention.

TABLE 1

1. On-demand energy audit, energy retrofit and energy electrification project services.	Automatic energy audit, energy retrofit and energy electrification projects services can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each network server on the cloud communications network 18 .
2. Broadband network access.	Automatic energy audit, energy retrofit and energy electrification projects services capabilities are available over plural broadband communications networks and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, smart phones 14 , tablet computers 12 , laptops, PDAs, etc.). The broadband network access includes high speed network access such as 4G and 5G wireless and/or wired and broadband and/or ultra-broad band (e.g., WiMAX, etc.) network access.
3. Resource pooling.	Automatic energy audit, energy retrofit and energy electrification projects resources are pooled to serve multiple requesters using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to demand. There is location independence in that a requester of services has no control and/or knowledge over the exact location of the provided by the energy retrofit and energy

TABLE 1-continued

	electrification projects service resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or data center). Examples of pooled resources include storage, processing, memory, network bandwidth, virtual server network device and virtual target network devices.
4. Rapid elasticity.	Capabilities can be rapidly and elastically provisioned, in some cases automatically, to quickly scale out and rapidly released to quickly scale for Energy Audit service collaboration. For automatic energy audit, energy retrofit and energy electrification projects services, multi-media collaboration converters, the automatic energy audit, energy retrofit and energy electrification projects services collaboration and analytic conversion capabilities available for provisioning appear to be unlimited and can be used in any quantity at any time.
5. Measured Services.	Cloud computing systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of automatic energy audit, energy retrofit and energy electrification projects services (e.g., storage, processing, bandwidth, custom electronic content retrieval applications, etc.). Electronic v interoperability services collaboration conversion usage is monitored, controlled, and reported providing transparency for both the automatic energy audit, energy retrofit and energy electrification projects services provider and the v service requester of the utilized electronic content storage retrieval service.

[0166] Exemplary cloud computing service models illustrated in FIG. 4 appear in Table 2. However, the present invention is not limited to these service models and more, fewer or other service models can also be used to practice the invention.

TABLE 2

1. Cloud Computing Software Applications 62 for Electrocutation Project, Installation, Tracking, Inspection and Verification Services (CCSA, SaaS 64, 64').	The capability to use the provider's applications 30, 30a, 30b, 30c, 30d, 30e, running on a cloud infrastructure 66. The cloud computing applications 62, are accessible from the server network device 20 from various client devices 12, 14, 16 through a thin client interface such as a web browser, etc. The user does not manage or control the underlying cloud infrastructure 66 including network, servers, operating systems, storage, or even individual application 30, 30a, 30b, 30c, 30d, 30e capabilities, with the possible exception of limited user-specific application configuration settings. The SaaS 64 includes a network device component for interacting with the SaaS 64 provided by a cloud server network device 20, 22, 24, 26.
2. Cloud Computing Infrastructure 66 for Electrocutation Project, Installation, Tracking, Inspection and Verification Services (CCI 68).	The capability provided to the user is to provision processing, storage and retrieval, networks 18, 72, 74, 76, 78 and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications 30, 30a, 30b, 30c, 30d 30e, 30f. The user does not manage or control the underlying cloud infrastructure 66 but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components (e.g., application layers, host firewalls, etc.).
3. Cloud Computing Platform 70 for Electrocutation Project, Installation, Tracking, Inspection and Verification Services (CCP 71).	The capability provided to the user to deploy onto the cloud infrastructure 66 created or acquired applications created using programming languages and tools supported servers 20, 22, 24, 26, etc.. The user not manage or control the underlying cloud infrastructure 66 including network, servers, operating systems, or storage, but has control over the deployed applications 30a, 30b, 30c, 30d, 30e, 30f and possibly application hosting environment configurations.

[0167] Exemplary cloud computing deployment models appear in Table 3. However, the present invention is not limited to these deployment models and more, fewer or other deployment models can also be used to practice the invention.

[0172] The cloud storage object **82** includes an envelope portion **84**, with a header portion **86**, and a body portion **88**. However, the present invention is not limited to such a cloud storage object **82** and other cloud storage objects and other cloud storage objects with more, fewer or other portions can also be used to practice the invention.

TABLE 3

1. Private cloud network 72.	The cloud network infrastructure is operated solely for Electrocution Project Project, Installation, Tracking, Inspection and Verification. It may be managed by the electronic content retrieval or a third party and may exist on premise or off premise.
2. Community cloud network 74.	The cloud network infrastructure is shared by several different organizations and supports a specific electronic content storage and retrieval community that has shared concerns (e.g., mission, security requirements, policy, compliance considerations, etc.). It may be managed by the different organizations or a third party and may exist on premise or off premise.
3. Public cloud network 76.	The cloud network infrastructure such as the Internet, PSTN, SATV, CATV, Internet TV, etc. is made available to the general public or a large industry group and is owned by one or more organizations selling cloud services.
4. Hybrid cloud network 78.	The cloud network infrastructure 66 is a composition of two and/or more cloud networks 18 (e.g., private 72, community 74, and/or public 76, etc.) and/or other types of public and/or private networks (e.g., intranets, etc.) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds, etc.)

[0168] Cloud software **64**, **64'** for electronic content retrieval takes full advantage of the cloud paradigm by being service oriented with a focus on statelessness, low coupling, modularity, and semantic interoperability for electronic content retrieval. However, cloud software services **64**, **64'** can include various states.

[0169] Cloud storage of desired electronic content on a cloud computing network includes agility, scalability, elasticity and multi-tenancy. Although a storage foundation may be comprised of block storage or file storage such as that exists on conventional networks, cloud storage is typically exposed to requesters of desired electronic content as cloud objects.

[0170] In one exemplary embodiment, the cloud application **30**, **30a**, **30b**, **30c**, **30d**, **30e**, **30'**, **30a'**, **30b'**, **30c'**, **30d'**, **30e'** offers cloud services for Energy Audits. The application **30**, **30a**, **30b**, **30c**, **30d**, **30e**, **30'**, **30a'**, **30b'**, **30c'**, **30d'**, **30e'** offers the cloud computing Infrastructure **66**, **68** as a Service **62** (IaaS), including a cloud software infrastructure service **62**, the cloud Platform **70**, **71** as a Service **62** (PaaS) including a cloud software platform service **62** and/or offers Specific cloud software services Software as a Service **64** (SaaS) including a specific cloud software service **64**, **64'** for automatic energy audit, energy retrofit and energy electrification project services. The IaaS, PaaS and SaaS include one or more of cloud services **62** comprising networking, storage, server network device, virtualization, operating system, middleware, run-time, data and/or application services, or plural combinations thereof, on the cloud communications network **18**.

[0171] FIG. 5 is a block diagram **80** illustrating an exemplary cloud storage object **82**. One or more server network devices (e.g., **20**, **22**, **24**, **26**, etc.) store portions **13'**, **15'** of the electronic content **13**, **15**, **138**, **146**, **146'** as cloud storage objects **82** (FIG. 5) **13'/82**, **15'/82** as is described herein.

[0173] The envelope portion **84** uses unique namespace Uniform Resource Identifiers (URIs) and/or Uniform Resource Names (URNs), and/or Uniform Resource Locators (URLs) unique across the cloud communications network **18** to uniquely specify, location and version information and encoding rules used by the cloud storage object **82** across the whole cloud communications network **18**. For more information, see IETF RFC-3305, Uniform Resource Identifiers (URIs), URLs, and Uniform Resource Names (URNs), the contents of which are incorporated by reference.

[0174] The envelope portion **84** of the cloud storage object **82** is followed by a header portion **86**. The header portion **86** includes extended information about the cloud storage objects such as authorization and/or transaction information, etc.

[0175] The body portion **88** includes methods **90** (i.e., a sequence of instructions, etc.) for using embedded application-specific data in data elements **92**. The body portion **88** typically includes only one portion of plural portions of application-specific data **92** and independent data **94** so the cloud storage object **82** can provide distributed, redundant fault tolerant, security and privacy features described herein.

[0176] Cloud storage objects **82** have proven experimentally to be a highly scalable, available and reliable layer of abstraction that also minimizes the limitations of common file systems. Cloud storage objects **82** also provide low latency and low storage and transmission costs.

[0177] Cloud storage objects **82** are comprised of many distributed resources, but function as a single storage object, are highly fault tolerant through redundancy and provide distribution of desired electronic content across public communication networks **76**, and one or more private networks **72**, community networks **74** and hybrid networks **78** of the cloud communications network **18**. Cloud storage objects **82** are also highly durable because of creation of copies of

portions of desired electronic content across such networks **72, 74, 76, 78** of the cloud communications network **18**. Cloud storage objects **82** includes one or more portions of desired electronic content and can be stored on any of the **72, 74, 76, 78** networks of the cloud communications network **18**. Cloud storage objects **82** are transparent to a requester of desired electronic content and are managed by cloud applications **30, 30a, 30b, 30c, 30d, 30e, 30e**.

[0178] In one embodiment, cloud storage objects **82** are configurable arbitrary objects with a size up to hundreds of terabytes, each accompanied by with a few kilobytes of metadata. Cloud objects are organized into and identified by a unique identifier unique across the whole cloud communications network **18**. However, the present invention is not limited to the cloud storage objects described, and more fewer and other types of cloud storage objects can be used to practice the invention.

[0179] Cloud storage objects **82** present a single unified namespace or object-space and manages desired electronic content by user or administrator-defined policies storage and retrieval policies. Cloud storage objects include Representational state transfer (REST), Simple Object Access Protocol (SOAP), Lightweight Directory Access Protocol (LDAP) and/or Application Programming Interface (API) objects and/or other types of cloud storage objects. However, the present invention is not limited to the cloud storage objects described, and more fewer and other types of cloud storage objects can be used to practice the invention.

[0180] REST is a protocol specification that characterizes and constrains macro-interactions storage objects of the four components of a cloud communications network **18**, namely origin servers, gateways, proxies and clients, without imposing limitations on the individual participants.

[0181] SOAP is a protocol specification for exchanging structured information in the implementation of cloud services with storage objects. SOAP has at least three major characteristics: (1) Extensibility (including security/encryption, routing, etc.); (2) Neutrality (SOAP can be used over any transport protocol such as HTTP, SMTP or even TCP, etc.), and (3) Independence (SOAP allows for almost any programming model to be used, etc.)

[0182] LDAP is a software protocol for enabling storage and retrieval of electronic content and other resources such as files and devices on the cloud communications network **18**. LDAP is a “lightweight” version of Directory Access Protocol (DAP), which is part of X.500, a standard for directory services in a network. LDAP may be used with X.509 security and other security methods for secure storage and retrieval. X.509 is public key digital certificate standard developed as part of the X.500 directory specification. X.509 is used for secure management and distribution of digitally signed certificates across networks.

[0183] An API is a particular set of rules and specifications that software programs can follow to communicate with each other. It serves as an interface between different software programs and facilitates their interaction and provides access to automatic energy audit, energy retrofit and energy electrification project services in a cloud or non-cloud environment. In one embodiment, the API for energy audit, energy retrofit and energy electrification project services is available to network devices **12, 14, 16, 20, 22, 24, 26, 31, 98-104** and networks **18, 18'**. However, the present invention is not limited to such an embodiment and other embodiments can be used to practice the invention.

Wearable Devices

[0184] Wearable technology” and/or “wearable devices” are clothing and accessories incorporating computer and advanced electronic technologies. Wearable network devices provide several advantages including, but not limited to: (1) Quicker access to notifications. Important and/or summary notifications are sent to alert a user to view the whole message. (2) Heads-up information. Digital eye wear allows users to display relevant information like directions without having to constantly glance down; (3) Always-on Searches. Wearable devices provide always-on, hands-free searches; and (4) Recorded data and feedback. Wearable devices take telemetric data recordings and providing useful feedback for users for exercise, health, fitness, etc. activities.

[0185] FIG. 6 is a block diagram with **96** illustrating wearable devices. The wearable devices include one or more processors and include, but are not limited to, wearable digital glasses **98**, Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR) headsets **99**, clothing **100**, jewelry **102** (e.g., smart rings, smart earrings, etc.) and/or watches **104**. However, the present invention is not limited to such embodiments and more, fewer and other types of wearable devices can also be used to practice the invention.

[0186] In one specific embodiment, the application **30, 30a, 30b, 30c, 30d, 30e** interacts with wearable devices **98-104** automatic energy audit, energy retrofit and energy electrification project services the methods described herein. However, the present invention is not limited this embodiment and other embodiments can also be used to practice the invention.

Artificial Intelligence (AI) and Big Data

[0187] “Artificial intelligence” (AI), also known as machine intelligence (MI), is intelligence demonstrated by machines, in contrast to the natural intelligence (NI) displayed by humans and other animals. AI research is defined as the study of “intelligent agents.” Intelligent agents are any software application or hardware device that perceives its environment and takes actions that maximize its chance of successfully achieving its goals. Colloquially, the term “artificial intelligence” is applied when a machine mimics “cognitive” functions that humans associate with human brains, such as learning, problem solving and comparing large number of data points.

[0188] In one embodiment, the present invention uses one or more AI methods including, but are not limited to, AI knowledge-based methods **30c** for energy audit, energy retrofit and energy electrification project services. However, the present invention is not limited to such an embodiment and more, fewer and/or other AI methods can be used to practice the invention.

[0189] In one embodiment, SaaS **64, 64'** includes and AI application **30c** with the AI methods described herein. In another embodiment, the AI application **30c** is a standalone application. However, the present invention is not limited to such an embodiment, and the AI application **30c** can be provided in other than the SaaS **64, 64'**.

[0190] “Big Data” refers to the use of predictive analytic methods that extract value from data, and to a particular size of data set. The quantities of data used are very large, at least 100,000 data points and more typically 500,000 to 1 Million+ data points. Analysis of Big Data sets are used to find

new correlations and to spot trends. In one embodiment, SaaS **64**, **64'** includes and Big Data application **30c'** with the Big Data described herein.

[0191] In one embodiment, the AI methods described herein collect data information to create and store (e.g., in cloud storage object **82**, etc.) a Big Data that is used to analyze trends find new correlations and to spot trends. However, the present invention is not limited to such an embodiment and the AI methods described herein can be used without Big Data sets.

[0192] In one embodiment, the AI methods include predictive analysis methods. "Predictive analysis" is an AI method that uses data mining, statistics, and modelling to make predictions. This AI method mines and analyses historical data patterns to predict future outcomes by extracting information from data sets including Big Data sets to determine patterns and trends. For example, energy consuming component-X manufactured by company-Y consumes 10% more energy than energy consuming component-A manufactured by company-B, energy consuming component-C burns out 15% faster than energy consuming component-D, etc. However, the present invention is not limited to such embodiments and other AI methods can be used to practice the invention.

[0193] In another embodiment, the AI methods include anomaly detection. "Anomaly detection" AI methods analyze data, including Big Data and pinpoints anything out of usual operations or usual expectations. For example, energy consuming component-E manufactured by company-F in batch-XYZ, fails 60% of the time within one month after installation, etc. However, the present invention is not limited to such embodiments and other AI methods can be used to practice the invention.

[0194] In another embodiment, the AI methods include clustering analysis. "Clustering analysis" AI methods include dividing a population or data points or Big Data points into a number of groups such that data points in the same groups are more similar to other data points in the same group and dissimilar to the data points in other groups. It includes a collection of objects on a basis of similarity and dissimilarity between them. For example, energy consuming component-LED-1 manufactured by companies G, H and I, consume 5% less energy than energy consuming component LED-2 manufactures by companies J, K and L, etc. However, the present invention is not limited to such embodiments and other AI methods can be used to practice the invention.

[0195] In another embodiment, the AI methods include predictive analysis, anomaly detection and/or clustering analysis and/or a combination thereof. However, the present invention is not limited to such embodiments and other AI methods can be used to practice the invention.

Short Message Service (SMS) Messaging

[0196] Short Message Service (SMS) is an electronic text messaging service component of phone, Web, or mobile communication systems. It uses standardized communications protocols to allow fixed line or mobile phone devices to exchange short text messages.

[0197] SMS messages were defined in 1985 as part of the Global System for Mobile Communications (GSM) series of standards as a means of sending messages of up to 160 characters to and from GSM mobile handsets. Though most SMS messages are mobile-to-mobile text messages, support

for the service has expanded to include other mobile technologies as well as satellite and landline networks.

[0198] The SMS Internet Engineering Task Force (IETF) Request for Comments (RFC) 5724, ISSN: 2070-1721, 2010, is incorporated herein by reference.

Rich Communication Suite (RCS)

[0199] Rich Communications Suite/Rich Communications System (RCS) is a communication protocol between mobile telephone carriers, between phones and carriers, and between individual devices aiming at replacing SMS messages with a message system that is richer, provides phone-book polling (e.g., for service discovery, etc.), and can transmit in-call multimedia. It is also marketed under the names of Advanced Messaging, Advanced Communications, Chat, joyn, Message+ and SMS+. RCS is also a communication protocol available for device-to-device (D2D) exchanges without using a telecommunications carrier for devices that are in close physical proximity (e.g., between two IoT devices, smart phones, smart phone and electronic tablet, etc.)

[0200] One advantage RCS Messaging has over SMS is that RCS enables users to send rich, verified messages including photos, videos and audio messages, group messages, read receipts, indicators to show other users are typing a message, carousel messages, suggested chips, chat bots, barcodes, location integration, calendar integration, dialer integration, and other RCS messaging features. RCS messaging includes person-to-person (P2P), application-to-person (A2P), application-to-application (A2A), application-to-device (A2D) and/or device-to-device (D2D) messaging.

[0201] The RCS Interworking Guidelines Version 14.0, 13 Oct. 2017, GSM Association, Rich Communication Suite RCS API Detailed Requirements, version 3.0, Oct. 19, 2017, Rich Communication Suite 8.0 Advanced Communications Services and Client Specification Version 9.0, 16 May 2018, RCS Universal Profile Service Definition Document Version 2.2, 16 May 2018, and Rich Communication Suite Endorsement of OMA CPM 2.2 Conversation Functions Version 9.0, 16 Oct. 2019, are all incorporated herein by reference.

[0202] The Rich Communication Suite-Enhanced (RCS-e) includes methods of providing first stage interoperability among Mobile Network Operators (MNOs). RCS-e is a later version of RCS which enables mobile phone end users to use instant messaging (IM), live video sharing and file transfer across any device on any MNO.

[0203] The RCS functionality of the present invention includes, but is not limited to, one and two way, rich, verified, multimedia messages including photos, videos and audio messages, group messages, read receipts, indicators to show other users are typing a message, predefined quick-reply suggestions, rich cards, carousels, action buttons, maps, click-to-call, calendar integration, geo-location, etc. The RCS functionality also includes RCS emulators and/or thin RCS applications that provide full and/or selected features of available RCS functionality.

Energy Audit Services

[0204] FIGS. 7A, 7B and 7C are a flow diagram illustrating a Method **106** for energy audit services. In FIG. 7A at Step **108**, an electronic floor plan of a building for conducting an energy audit of energy consuming devices is provided

on an energy audit application on network device with one or more processors, a non-transitory computer readable medium and a communications interface for connecting the network device to a communications network. At Step 110, a configurable electronic template is provided on energy audit application the network device for conducting the energy audit of the energy consuming devices, the configurable electronic template dynamically configurable during the energy audit. At Step 112, the network device is moved physically into the building to plural different physical locations within the building, the network device is not connected to the communication network within the building and a selected physical location of the network device in the building corresponds to a selected virtual location on the electronic floor plan for the building. At Step 114, a creation selection input is received on the energy audit application on the network device to create a graphical marker for a selected energy consuming component at a specific location on the electronic floor plan of the building. At Step 116, a graphical marker is displayed in a first color on the energy audit application on the network component at the specific location on the electronic floor plan of the building for the selected energy consuming component. In FIG. 7B at Step 118, plural collection selection inputs are received on the energy audit application on network device including: (a) audio information from an audio component on the network device (b) visual information from a camera component on the network device, and (c) electronic information from an electronic keyboard component on the network device, the audio, visual and electronic information including information collected for the selected energy consuming component at the specific location on the electronic floor plan of the building. At Step 120, the received audio, visual and electronic information is stored in the configurable electronic template on the energy audit application on network device. At Step 122, steps 114 through 120 are repeated for plural of other selected energy consuming components at plural other specific locations on the electronic floor plan of the building. At Step 124, an energy audit summary report is created on the energy audit application on the network device with the stored audio, visual and electronic information. The energy audit summary report is dynamically updated on the energy audit application on the network device as energy consuming components are identified in the building. In FIG. 7C at Step 126, the network device is moved to a new physical location inside or outside of the building that allows the network device to connect to the communications network via the communications interface on the network device. At Step 128, the energy audit application on the network device connects via the communications interface on the network device to a server energy audit application on a server network device with one or more processor via the communications network. At Step 130, the data stored on the energy audit application on network device is automatically synchronized with the server energy audit application on the server network device, thereby allowing energy audit information for the electronic floor plan of the building to be identical on the network device and the server network device.

[0205] FIG. 8 is a block diagram 132 illustrating components of energy audits. The energy audit components include an electronic floor plan 134 for a structure including building 136, a configurable electronic template 138, energy consuming devices 142, location 144.

[0206] In FIG. 7, Method 106 is illustrated with an illustrative embodiment. However, the present invention is not limited to this illustrative embodiment and other embodiments can be used to practice the invention.

[0207] In such an illustrative embodiment in FIG. 7A at Step 108, an electronic floor plan 134 of a building for conducting an energy audit of energy consuming devices 142 is provided on an energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 (SaaS 64 when online) on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 with one or more processors, a non-transitory computer readable medium and a communications interface for connecting the network device to a communications network 18, 18'.

[0208] In one embodiment, the electronic floor plan of the building is created with a Computer Aided Design (CAD), Building Information Modeling (BIM) program, 3D modeling program and/or other design program. However, the present invention is not limited to such embodiments and other program can be used to create electronic floor plans 134 of a building 136, 136'.

[0209] For example, the BIM and/or 3D modeling programs, include, but are not limited to, AUTODESK REVIT, AUTODESK INVENTOR, AUTOCAD, SKETCHUP, VECTORWORKS, MICROSTATION, ARCHICAD, SOLIDWORKS, PROE, BLUEBEAM, etc. However, the present invention is not limited to such embodiments and other embodiment can be used to practice the invention.

[0210] In one embodiment, the energy consuming components 142 include, but are not limited to, lighting, heating, ventilation, air conditioning, motors, variable frequency drives, transformers, generators and/or combinations thereof. However, the present invention is not limited so such embodiments and any other energy consuming components 142 can be used to practice the invention.

[0211] At Step 110, a configurable electronic template 138 is provided on energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 for conducting the energy audit of the energy consuming components 142, the configurable electronic template 138 dynamically configurable during the energy audit.

[0212] In one embodiment, the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 includes a template wizard to assist in creating plural data fields in configurable electronic template 138. In such an embodiment, the template wizard provides assistance by presenting questions (e.g., "what type of lighting are you auditing? (1) incandescent, (2) fluorescent, (3) LED?", etc.) and then automatically creating appropriate data fields in the configurable electronic template 138. However, the present invention is not limited to such an embodiment and the invention can be practiced with and/out a template wizard.

[0213] For example, the configurable electronic template 138 may be configured to collect information on fluorescent lights with data fields including light bulb length, brand, ballast, starter, diffusers, cover types, etc. as a building 136 may include a majority of fluorescent lights. During the energy audit, it is decided to collect lumen values for all the fluorescent lights, so a lumen data field is dynamically added to the configurable electronic template 138. Then during an energy audit several store rooms with incandescent lights are encountered. The configurable electronic template 138 is dynamically configurable to add additional fields for the

incandescent lights. However, the present invention is not limited to such an embodiment and other embodiment can be used to practice the invention.

[0214] At Step 112, the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 is moved physically into the building 136 to plural different physical locations within the building, the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 is not connected to the communication network 18, 18' within the building 136 (i.e., is offline, etc.) and a selected physical location 144 of the network device 12 in the building 136 corresponds to a selected virtual location 144' on the electronic floor plan 13, 134 for the building 136.

[0215] In one embodiment, the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 must work offline because the building 136 includes a large amount of concrete, steel or other building materials that interfere or prevent the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 device from ever successfully connecting to a communications network 18, 18'. In addition, the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 may be physically located in a service closet, an attic, interior spaces of building, etc. that prevent successful connections to the communications network 18, 18'. The network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 is able to work offline and complete all its desired tasks and automatically synchronize all energy audit data it has collected at a later time. Working offline also provides an additional advantage of not running down a battery on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 that may be constantly trying to connect with the communications network 18, 18' without ever being successful because of its physical location inside the building 136.

[0216] At Step 114, a creation selection input is received on the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 to create a graphical marker 140 for a selected energy consuming component 142 at a specific location 144 on the electronic floor plan 134 of the building 136.

[0217] In one embodiment, the graphical marker includes a specific graphical marker representing and graphically illustrating a specific type of energy consuming component 142 and/or specific type of energy producing component. However, the present invention is not limited to such embodiments and other embodiments can be used to practice the invention.

[0218] In one embodiment, energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 includes a graphical component fixture wizard with plural selectable graphical objects that visually identify a plural different energy consuming components 142 and energy producing components to ensure each energy consuming component 142 and energy producing component 145 is fully defined and consistently identified on the electronic floor plan 134 of the building 136. However, the present invention is not limited to such embodiments and other embodiments can be used to practice the invention.

[0219] FIG. 8 illustrates the graphical marker 140 as a graphical light bulb for an incandescent lighting component for the selected energy consuming component. Unique graphical markers are also used for all other energy consuming components 142 and energy producing components 145. However, the present invention is not limited to this embodiment and other graphical markers can be used for lighting and for the selected energy consuming component.

[0220] Returning to FIG. 7A at Step 116, the graphical marker 140 is displayed in a first color (e.g., yellow, etc.) on the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network component 12 at the specific location 144 on the electronic floor plan 134 of the building 136 for the selected energy consuming component 140.

[0221] In FIG. 7B at Step 118, plural collection selection inputs are received on the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 including: (a) audio information from an audio component on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 (b) visual information from a camera component on the network device, and (c) electronic information from an electronic keyboard component on the network device 12, 14, 16, 31, 98-104, the audio, visual and electronic information including information collected for the selected energy consuming component 140 at the specific location 144 on the electronic floor plan 134 of the building 136.

[0222] In one embodiment, the audio information captured on the camera component on the network device includes, but is not limited to, voice recordings. However, the present invention is not limited to such an embodiment and other embodiments can be used to practice the invention.

[0223] In one embodiment, the visual information captured on the camera component on the network device includes, but is not limited to, still pictures, video and 360 view visual information. However, the present invention is not limited to such an embodiment and other embodiments can be used to practice the invention.

[0224] In one embodiment, the visual information includes still pictures and/or videos that include additional annotated and/or markup information for individual energy consuming components. For example, "this florescent light has the oldest starter in the building and needs to be replaced first," etc. However, the present invention is not limited to this embodiment and other embodiments can be used to practice the invention.

[0225] In another embodiment Step 118 further includes collecting audio, visual and electronic information from energy producing components on an energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on network device 12, 14, 16, 31, 98-104. However, the present invention is not limited to this embodiment and other embodiments can be used to practice the invention.

[0226] In one embodiment, the energy producing components 145 include, but are not limited to, solar (e.g., solar panels, photovoltaic systems, etc.), wind (e.g., turbines, etc.), water (e.g., hydroelectric, etc.), biomass (e.g., algae, garbage, etc.), geothermal (e.g., heat from the earth's crust and earth's water, etc.), and/or batteries (e.g., alkaline, lithium, metal hydride, etc.), energy producing components 145 and/or combinations thereof. However, the present invention is not limited so such embodiments and any other energy producing components can be used to practice the invention.

[0227] In one embodiment, Step 118 further includes applying automatically one or more Artificial Intelligence (AI) methods to the audio, visual and electronic information collected for the selected energy consuming component 140 to provide additional information about the selected energy consuming component 140. In one embodiment, the AI methods include predictive analysis, anomaly detection and/or clustering analysis and/or a combination thereof. The AI

methods are used with and/or without Big Data. However, the present invention is not limited to such embodiments and more, fewer and other AI methods can be used to practice the invention. In addition, the invention can be practice without using AI methods.

[0228] In one embodiment, the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 further includes one or more additional sensing components to measure: distance, location, energy output, light output, airflow, electrical input and output, water flow, pressure or temperature of the energy consuming components.

[0229] For example, a laser meter is used to measure distance, a Global Positioning System (GPS) component is used to measure location, electric meters to measure energy output and electrical input and output, a lumen meter to measure light output, an airflow meter to measure airflow, a water flow meter to measure water flow, a pressure meter to measure pressure, and/or a thermometer to measure temperature, etc. However, the present invention is not limited to such embodiments and other embodiment can be used to practice the invention.

[0230] The sensing components, include but are not limited to, additional components that are added to the network device and/or applications that using existing hardware on the network device 12, 14, 16, 31, 98-104. However, the present invention is not limited to such embodiments and other embodiment can be used to practice the invention.

[0231] The additional sensing components are used to collect additional information about the energy consuming components for the energy audits. If an energy audit is being conducting on lighting components, a lumen meter is used to collect data from existing lighting components. A “lumen” is a unit of measurement for the brightness of light. A lumen is defined as “a unit of luminous flux equal to the amount of light given out through a solid angle of one steradian by a point source of one candela intensity radiating uniformly in all directions.” Different types of lighting components provide light with different amounts of lumens while consuming different amounts of energy.

[0232] For example, a 10-watt Light Emitting Diode (LED) bulb emits about 800 lumens, cost about \$1.34 per year to operate and has an average life span of about 25,000 hours. In stark contrast, a 10-watt florescent light bulb emits about 450 lumens, cost about \$1.73 per year to operate and has an average life span of about 8000 hours. If a room has two florescent bulbs, they would provide about 900 lumens and cost about \$3.46 per year to operate. Thus, the two florescent bulbs could be replaced with one LED bulb that generates a similar amount of lumens (800 vs. 900) and at a yearly cost savings of about \$2.12 (\$3.46-\$1.34), etc. Collecting such additional information is very important if the energy audit is being using to replace and/or retrofit existing energy consuming components in a commercial or industrial building with a large amount of energy consuming components.

[0233] At Step 120, the received audio, visual and electronic information is stored in the configurable electronic template 138 on the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on network device 12, 14, 16, 31, 98-104.

[0234] In one embodiment, the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 stores the received audio, visual and electronic information in one or more cloud storage objects 82 on a cloud communications network 18 when the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 later

connects to the communications network 18' including a cloud communications network 18.

[0235] At Step 122, steps 114 through 120 are repeated for plural of other selected energy consuming components 140a-140e at plural other specific locations on the electronic floor plan 134 of the building 136, 136'.

[0236] One or more graphical markers are changed from the first color 133 (e.g., yellow, etc.) to a second color 135, 135' (e.g., green, etc.) on the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 at selected specific locations 144' on the electronic floor plan 134 of the building 136 to visually indicate where information collection for selected energy consuming components has been completed.

[0237] A third color 137 (e.g., red, etc.) is used on the on the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 at selected specific locations 144' on the electronic floor plan 134 of the building 136 to visually indicate where no information collection for selected energy consuming components has been completed and/or there is some other type of issue or problem that has occurred with the energy consumer and/or energy producing device and needs to be addressed (e.g., non-working order, missing parts, missing device, wrong description, etc.).

[0238] In another embodiment, the third color (e.g., red, etc.), the first color (e.g., yellow, etc.), and the second color (e.g., green, etc.) are used to indicate a level of completion 139 of the energy audit (and/or energy retrofit project, energy electrification project, etc.) For example, a red color indicates a beginning stage, a yellow color indicates an intermediate stage and a green color indicates a completed stage. However, the present invention is not limited to such an embodiment and other embodiments with other colors used for other indications can be used to practice the invention.

[0239] At Step 124, an energy audit summary report 146 is created on the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 with the stored audio, visual and electronic information. The energy audit summary report 146 is dynamically updated on the energy audit application 30, 30a, 30b, 30c, 30d, 30e on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 as energy consuming components 140a-140e are identified in the building 136.

[0240] In FIG. 7C at Step 126, the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 is moved to a new physical location inside or outside of the building that allows the network device to connect to the communications network 18, 18' via the communications interface on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 either with a wireless or wired connection.

[0241] At Step 128, the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 connects via the communications interface on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 to a server energy audit application 30', 30a', 30b', 30c', 30d', 30e', 64' on a server network device 20, 22, 24, 26 with one or more processor via the communications network 18, 18'.

[0242] At Step 130, the data stored 13, 15, 13'/82, 15'/82, 138, 146 on the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 is automatically synchronized with the server

energy audit application 30', 30a', 30b', 30c', 30d, 30e', 64' on the server network device 12, 14, 16, 31, 98-104, thereby allowing energy audit information for the electronic floor plan 134 of the building 136 to be identical on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 and the server network device 20, 22, 24, 26.

[0243] In one embodiment, energy audit information is collected on multiple network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 and at Step 130 the energy audit information from all the multiple network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 is automatically synchronized whenever the multiple network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 connect to the communications network 18, 18' and are in a connect (i.e., on-line) state.

[0244] In one embodiment, the server energy audit application 30', 30a', 30b', 30c', 30d, 30e', 64' on the server network device 20, 22, 24, 26 automatically applies one or more Artificial Intelligence (AI) methods to the audio, visual and electronic information collected for the selected energy consuming component 140 and the selected other plural energy consuming components 140a-140e, 142 to provide additional information the about plural selected energy consuming components 140, 140a-140e, 142. The AI methods are used with and/or without Big Data. However, the present invention is not limited to such embodiments and more, fewer and other AI methods can be used to practice the invention. In addition, the invention can be practice without using AI methods.

[0245] In one embodiment, the additional information about the plural selected energy consuming components 140a-140e, 142 includes one or more electronic links to one or more information sources including existing electronic product and price information about the existing plural selected energy consuming components 140a-140e, 142 and new electronic information and new price information for new energy consuming components that can replace the existing plural selected energy consuming components 140a-140e, 142 in the building 136.

[0246] For example, Table 4 illustrates exemplary additional information that is available via the one or more electronic links (e.g., underlined in Table 4, etc.) for lighting components. Table 4 is exemplary only. The present invention is not limited to such embodiments and other embodiments can be used to practice the invention.

TABLE 4

LED Bulbs	Compact Florescent Lamp (CFL) Bulbs	Incandescent Bulbs
Brightness: 10 W/800 Lumens	Brightness: 10 W/450 Lumens	Brightness: 10 W/80 Lumens
Operating	Operating	Operating
Cost: \$1.34 per year	Cost: \$1.73 per year	Cost: \$3.05 per year
Product Data Sheets	Product Data Sheets	Product Data Sheets
Energy Efficiency	Energy Efficiency	Energy Efficiency
Data	Data	Data
Manufacturer:	Manufacturer:	Manufacturer:
PHILLIPS	PHILLIPS	PHILLIPS
CREE	CREE	CREE
GENERAL	GENERAL	GENERAL
ELECTRIC	ELECTRIC	ELECTRIC
Purchase:	Purchase:	Purchase:
AMAZON.com	AMAZON.com	AMAZON.com
LOWES.com	LOWES.com	LOWES.com
HOME DEPOT.com	HOME DEPOT.com	HOME DEPOT.com

[0247] FIG. 9 is a flow diagram illustrating a Method 150 for providing energy audits. At Step 152, data stored on an energy audit application on a network device is automatically and continuously synchronized with a server energy audit application on a server network device via a communications network every time the network device connects to the communications network, thereby allowing energy audit information for an electronic floor plan of a building to be identical on the network device and the server network device until all energy consuming components in the building have been identified and recorded. At Step 154, a final energy audit report is produced on the server energy audit application on the server network device, the final energy audit report including one or more electronic links to one or more information sources including existing electronic product and price information about the existing plural selected energy consuming components and new electronic information and new price information for new energy consuming components that can replace the existing plural selected energy consuming components in the building. At Step 156, the final energy audit report is sent from the server energy audit application on the server network device to the energy audit application on the network device via the communications network for use in the building.

[0248] Method 150 is illustrated with an illustrative embodiment. However, the present invention is not limited to this illustrative embodiment and other embodiments can be used to practice the invention.

[0249] In such an illustrative embodiment in FIG. 9 at Step 152, data stored on an energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on a network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 is automatically and continuously synchronized with a server energy audit application 30', 30a', 30b', 30c', 30d, 30e', 64' on a server network device 20, 22, 24, 26 via a communications network 18, 18' every time the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 connects to the communications network 18, 18', thereby allowing energy audit information for the electronic floor plan 134 of the building 136 to be identical on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 and the server network device 20, 22, 24, 26 until all energy consuming components in the building 136 have been identified and recorded.

[0250] At Step 154, a final energy audit report 146' is produced on the server energy audit application 30', 30a', 30b', 30c', 30d, 30e', 64' on the server network device 20, 22, 24, 26, the final energy audit report including one or more electronic links (e.g., Table 4, etc.) to one or more information sources including existing electronic product and price information about the existing plural selected energy consuming components and new electronic information and new price information for new energy consuming components that can replace the existing plural selected energy consuming components in the building 136.

[0251] At Step 156, the final energy audit report 146' is sent from the server energy audit application 30', 30a', 30b', 30c', 30d, 30e', 64' on the server network device 20, 22, 24, 26, to the energy audit application 30, 30a, 30b, 30c, 30d, 30e, 64 on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 via the communications network 18, 18' for use on the network device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 in the building 136.

[0252] A method and system for providing energy audits is presented herein. The method and system allow an energy audit to be conducted offline on an energy audit application

on network device for energy consuming components (e.g., lighting, heating, ventilation, air condition, etc.) and energy producing components (e.g., solar, wind, water, etc.) and automatically synchronized to a server network device when the network device is able to connect to a communications network. Energy audit templates are provided to ensure all energy consuming components and energy producing components are fully defined and consistently identified on an electronic floor plan for a commercial or industrial building. Energy Audits and Energy Retrofits with Plural Auditors

[0253] When data collection for an energy audit and/or energy retrofit is being conducted, it is desirable that the audit be completed at different times and different dates by the same people and/or different people from a same company and/or a different company and continuously synchronized in one place at a later time.

[0254] When data collection for an energy audit and/or energy retrofit is being conducted, it is desirable the data collection and energy audit data is automatically synchronized among devices and/or parties so errors are eliminated.

[0255] A complete final synchronized energy audit summary report on a server energy audit application on a server network device completed by plural energy audit applications on plural network devices at different dates and/or times and/or by different auditors and/or the same auditor is presented herein.

[0256] An “energy audit” project includes plural acts comprising, but not limited to: (1) determining what energy components (i.e., energy consuming components and/or energy producing components, etc.) and/or energy accessories (i.e., spare parts, add on components, etc.) exist for an energy system residing in and/or on, and/or outside a structure; (2) determining what new, more efficient energy components and/or energy accessories could be added to the energy system for the structure and the costs thereof; (3) creating plural work orders and/or Request For Proposals (RFP) with associated costs to complete separate portions of an energy audit of the energy components and/or energy accessories in the structure for review by plural auditors; and (4) accepting proposals from plural auditors for the plural work orders and/or RFPs to complete portions of the energy audit. However, the present invention is not limited to such an embodiment and other embodiments with more, fewer and/or other steps can be used to practice the invention.

[0257] An “energy retrofit” project includes, plural acts comprising, but not limited to: (1) determining what energy components (i.e., energy consuming components and/or energy producing components, etc.) and/or energy accessories (i.e., spare parts, add on components, etc.) exist for an energy system residing in and/or on, a structure with an energy audit; (2) determining what new, more efficient energy components and/or energy accessories should be added to the energy system for the structure and the costs thereof; (3) creating one or more work orders creating plural work orders and/or Request For Proposals (RFPs) with associated costs to complete a retrofit of the energy components and/or energy accessories for plural auditors; (4) accepting work orders and/or RFPs from plural auditors for removing some and/or all existing energy components and/or energy accessories for the energy system in and/or on the structure and installing new, more efficient energy components and/or energy accessories to the structure to replace the energy components and/or energy accessories did not exist and/or were not available when the energy system was

installed in the structure to replace the existing energy components and/or energy accessories for the energy system that were removed from the structure. However, the present invention is not limited to such an embodiment and other embodiments with more, fewer and/or other steps can be used to practice the invention.

[0258] In one embodiment, the structure includes, but is not limited to, personal buildings (e.g., homes, etc.), commercial, industrial, federal, state, municipal, university, college, school and/or hospital, buildings. However, the present invention is not limited to such an embodiment and more, fewer and/or other structures can be used to practice the invention.

[0259] FIG. 10A-10D are a flow diagram illustrating a Method 158 for providing energy audits with multiple auditors.

[0260] FIG. 11 is a block diagram 182 illustrating providing energy audits with multiple auditors.

[0261] In FIG. 10A-10D, Method 158 is illustrated with an illustrative embodiment. However, the present invention is not limited to this illustrative embodiment and other embodiments can be used to practice the invention.

[0262] In such an illustrative embodiment, in FIG. 10A at Step 160, providing plural different portions of an electronic floor plan of a building for conducting an energy audit of energy consuming components or energy producing components from a server energy audit application on a server network device with one or more processors via a communications network, to plural energy audit applications on plural network devices each with one or more processors, a non-transitory computer readable medium and a communications interface for connecting the plural network devices to the communications network. At Step 162, providing via the plural energy audit applications a creation component to create and change colors for plural graphical markers for energy consuming components and/or energy producing components at specific locations on the electronic floor plan of the building, the plural graphical markers including progress indicators comprising: (1) a first color for a selected energy consuming component and/or energy producing components indicating energy audit information has not been collected for the selected energy consuming component or energy producing component, (2) a second color indicating the energy audit information has been collected for the selected energy consuming component and/or energy producing component, and (3) a third color indicating a problem exists with the selected energy consuming component and/or energy producing components that requires additional action. In FIG. 10B at Step 164, (1) moving the plural network devices physically in the building to plural first different physical locations within the building based on the plural different portions of the electronic floor plan of the building. At Step 166, (2) displaying the plural graphical markers in the first color on the plural energy audit applications on the plural network devices at the plural different physical locations on the electronic floor plan of the building as a selected energy consuming component or energy producing component is being accessed. At Step 168, (3) receiving plural selection inputs on the plural energy audit applications on the plural network devices including: (a) audio information from an audio component on the plural network devices, (b) visual information from a camera component on the plural network devices, (c) visual markup information with plural different shapes and colors on infor-

mation collected from the camera component or provided directly on the electronic floor plan of the building, and (d) electronic information from an electronic keyboard component on the plural network devices, the audio, visual and electronic information including information collected for the selected energy consuming component or energy producing component at the plural different physical locations on the electronic floor plan of the building. In FIG. 10C at Step 170, (4) displaying the plural graphical markers in (a) the second color on the plural energy audit applications on the plural network devices at the plural different physical locations for the selected energy consuming components and/or energy producing components visually indicating energy audit information for the selected energy consuming components and/or energy producing components has been successfully completed or (b) displaying one or more graphical markers in the third color on the plural energy audit applications on the plural network devices at the plural different physical locations on the electronic floor plan of the building for the selected energy consuming components and/or energy producing components visually indicating a problem exists with the selected energy consuming components and/or energy producing components that require additional actions. At Step 172, (5) storing the audio, visual and electronic information in the plural graphical markers displayed in one of the three colors as energy audit data on the on the plural energy audit applications on the plural network devices. At Step 174, repeating steps (1)-(5) for plural other selected energy consuming components and energy producing components at plural other specific locations from the plural different portions of the electronic floor plan of the building. In FIG. 10D at Step 176, automatically and continuously synchronizing on the server energy audit application on the server network device via the communications network from the stored energy audit data on the plural energy audit applications on plural network devices, from the plural different portions of the electronic floor plan of the building. At Step 178, creating continuously a synchronized energy audit summary report on the server energy audit application on the server network device with the synchronized energy audit data collected on the plural energy audit applications on the plural network devices including the energy audit data for energy consuming components or energy producing components from the plural different portions of the electronic floor plan of the building. At Step 180, providing a complete final synchronized energy audit summary report for the building on the server energy audit application on the server network device completed by the plural energy audit applications on the plural network devices at different dates and times and by different auditors.

[0263] In such an exemplary embodiment, in FIG. 10A at Step 160, providing plural different portions 183, 184 (only two of which are illustrated for simplicity) of an electronic floor plan 134 of a building 136 for conducting an energy audit 146 of energy consuming components 140, 140' and/or energy producing components 141, 141' from a server energy audit application 30', 30a', 30b', 30c', 30d, 30e' on a server network device 20, 22, 24, 26 with one or more processors via a communications network 18, 18', to plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on plural network devices device 12, 14, 16, 25, 27, 29, 31, 33, 98-104 each with one or more processors, a non-transitory computer readable medium and a communications interface

for connecting the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 to the communications network 18, 18'.

[0264] In one embodiment, the electronic floor plan 134 of the building 136 is created with a Computer Aided Design (CAD), Building Information Modeling (BIM) or other three-dimensional (3D) modeling program. For example, the BIM and/or 3D modeling programs, include, but are not limited to, AUTODESK REVIT, AUTODESK INVENTOR, AUTOCAD, SKETCHUP, VECTORWORKS, MICROSTATION, ARCHICAD, SOLIDWORKS, PROE, BLUEBEAM, etc. However, the present invention is not limited to such embodiments and other embodiment can be used to practice the invention.

[0265] In one embodiment, the energy consuming components 142 include, but are not limited to, lighting (e.g., incandescent, light emitting diode (LED), fluorescent, etc.), heating, ventilation, air conditioning (HVAC), air quality, motors, variable frequency drives, transformers, generators, electrical controllers, water supply valves, water pumps, ultra violet (UV) lighting, UV disinfection systems, compressed air systems, thermostats, steam boilers, chillers, air movers, dampers, hoods, building automation systems, power factor equipment, and/or combinations thereof. However, the present invention is not limited to such embodiments and more, fewer and/or other types of energy consuming components can be used to practice the invention.

[0266] The power factor equipment, includes but is not limited to, alternating current (AC) power system devices. A power factor is defined as a ratio of real power absorbed by a load to an apparent power flowing in a circuit. Real power is the average of an instantaneous product of voltage and current and represents a capacity of electricity for performing work. Apparent power is a product of root mean square (RMS) current and voltage. Due to energy stored in a load and returned to a source, or due to a non-linear load that distorts a wave shape of the current drawn from a source, apparent power may be greater than the real power, so more current flows in the circuit than would be required to transfer real power alone.

[0267] In one embodiment, the energy producing components 145 include, include, but are not limited to, solar, wind, water, biomass, geothermal, batteries, Electronic Vehicle Supply Equipment (EVSE), micro-grids, fuel cells, and/or combinations thereof. However, the present invention is not limited to such embodiments and more, fewer and/or other types of energy producing components can be used to practice the invention.

[0268] A micro-grid is a localized electrical grid that can operate independently of a larger utility grid, or in conjunction with it. It is self-contained power system, utilizing distributed energy resources like solar panels, wind turbines, and/or fuel cells, and may also include energy storage. Micro-grids are designed to improve reliability, reduce costs, and integrate renewable energy sources. A fuel cell uses chemical energy of hydrogen and/or other fuels to cleanly and efficiently produce electricity.

[0269] At Step 162, providing via the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 a creation component 30d, 30e to create and change colors for plural graphical markers 140, 140' for energy consuming components 142 and/or energy producing components 141, 141' at specific locations on the electronic floor plan 134 of the building 136, the plural graphical markers including progress indicators comprising: (1) a first color 133 for a selected

energy consuming component 140, 140' and/or energy producing component 141, 141' indicating energy audit information has not been collected for the selected energy consuming component 140, 140' and/or energy producing component 141, 141, (2) a second color 135 indicating the energy audit information has been collected for the selected energy consuming component energy consuming component 140, 140' and/or energy producing component 141, 141', and (3) a third color 137 indicating a problem exists with the selected energy consuming component 140, 140' and/or energy producing components 141, 141' that requires additional action.

[0270] In one embodiment, the creation component 30d, 30e includes a graphical drawing component 30d, 30e' including the ability to draw arrows 196, circles and/or other shapes and include text in plural different shapes and colors as markup information on information collected from the camera components on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 and/or provided directly on the electronic floor plan 134 of the building 136 to identify key areas of concern with existing energy consuming components 140, 140' or energy producing components 141, 140' and/or new opportunities to place new existing energy consuming components 140, 140' and/or energy producing components 141, 141' in and/or outside 140", 141" of the building 136. In another embodiment, the graphical drawing component includes a separate application. However, the present invention is not limited to such embodiments and other embodiments can be used to practice the invention.

[0271] In FIG. 10B at Step 164, (1) moving the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 physically in the building to plural different physical locations within the building 136 based on the plural different portions 183, 184 of the electronic floor plan 134 of the building 136.

[0272] In one embodiment, Step 164, further includes, but is not limited to, moving the plural of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 physically around an exterior of the building 136 to collect energy audit data on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 for the one or more energy consuming components 140" and/or one or more energy producing components 141" located on exterior components of the building. However, the present invention is not limited to such embodiments and other embodiments can be used to practice the invention.

[0273] At Step 166, (2) displaying the plural graphical markers 140, 140, 141, 141' in the first color 135 on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 at the plural different physical locations on the electronic floor plan 134 of the building 136 as a selected energy consuming component 140, 140' and/or energy producing component 141, 141' is being accessed.

[0274] At Step 168, (3) receiving plural selection inputs on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 including: (a) audio information from an audio component on the plural network devices, (b) visual information from a camera component on the plural network devices 12, 14, 16, 31, 98-104, (c) visual markup information with plural different shapes and colors on information collected from the camera component or provided directly

on the electronic floor plan 134 of the building 136, and (d) electronic information from an electronic keyboard component on the plural network devices 12, 14, 16, 31, 98-104, the audio, visual and electronic information including information collected for the selected energy consuming component 140, 140' and/or energy producing component 141, 141' at the plural different physical locations on the electronic floor plan 134 of the building 136.

[0275] In one embodiment the collecting audio, visual and/or electronic information is collected digital cameras 29, including 360-degree view cameras 29', Light Detection and Ranging (LIDR) components 39, or a combination thereof, and/or camera components on the plural network devices 12, 14, 16, 25, 27, 31, 98-104. However, the present is not limited to such embodiments and other embodiments can be used to practice the invention.

[0276] In FIG. 10C at Step 170, (4) displaying the plural graphical markers in (a) the second color 135 on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 at the plural different physical locations for the selected energy consuming components 140, 140' and/or energy producing components 141, 141' visually indicating energy audit information for the selected energy consuming components 140, 140' and/or energy producing components 141, 141' has been successfully completed or (b) displaying one or more graphical markers in the third color 137 on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 at the plural different physical locations on the electronic floor plan 134 of the building 136 for the selected energy consuming components 140, 140' and/or energy producing components 141, 141' visually indicating a problem exists with the selected energy consuming components 140, 140' and/or energy producing components 141, 141' that require additional actions.

[0277] At Step 172, (5) storing the audio, visual and electronic information in the plural graphical markers 140, 140', 141, 141' displayed in one of the three colors 133, 135, 137 as energy audit data 186, 188 on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 31, 98-104.

[0278] At Step 174, repeating steps (1)-(5) at Step 164 through 172 for plural other selected energy consuming components 140, 140' and/or energy producing components 141, 141 at plural other specific locations from the plural different portions 183, 184 of the electronic floor plan 134 of the building 136, 136'.

[0279] At Step 175, a test is conducted to determine if all energy consuming components 140, 140' and/or energy producing components 141, 141' on the electronic floor plan 134 of the building 136 have been accessed by the plural auditors using the plural network devices 12, 14, 16, 31, 98-104. If not, steps 164 through 172 are repeated. If so, then Step 176 in FIG. 10D is executed.

[0280] In FIG. 10D at Step 176, automatically and continuously synchronizing on the server energy audit application applications 30', 30a', 30b', 30c', 30d, 30e', 64' on the server network device 20, 22, 24, 26 via the communications network 18, 18' from the stored energy audit data 186, 188 on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 31, 98-104, from the plural different portions 183, 184 of the electronic floor plan 134 of the building 136, 136'.

[0281] At Step 178, creating continuously a synchronized energy audit summary report 146 on the server energy audit application 30', 30a', 30b', 30c', 30d, 30e', 64' on the server network device 20, 22, 24, 26 with the synchronized energy audit data collected on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 including the energy audit data 186, 188 for energy consuming components 140, 140' and/or energy producing components 141, 141' from the plural different portions 183, 184 of the electronic floor plan 134 of the building 136.

[0282] In one embodiment, Step 178, includes but is not limited to applying automatically one or more Artificial Intelligence (AI) methods to the synchronized energy audit data 186, 188 to provide additional information about the plural energy consuming components 140, 140' and/or energy producing components 141, 141'. However, the present invention is not limited to such an embodiment and other embodiments with and/or without AI can be used to practice the invention.

[0283] In one embodiment, the one or more Artificial Intelligence (AI) methods include, but are not limited to, providing one or more electronic links to one or more information sources including existing electronic product and price information about the existing plural selected energy consuming components 140, 140' and/or energy producing components 141, 141' and new electronic information and new price information for new energy consuming component 140, 140' and/or energy producing components 141, 141' that can replace the existing plural selected energy consuming components 140, 140' and/or energy producing components 141, 141' in the building 136. However, the present invention is not limited to such embodiments and other embodiments with and/or without AI, can be used to practice the invention.

[0284] At Step 180, providing a complete final synchronized energy audit summary report 146' for the building 136 on the server energy audit application 30', 30a', 30b', 30c', 30d, 30e', 64' on the server network device 20, 22, 24, 26 completed by the plural energy audit applications 12, 14, 16, 25, 27, 29, 31, 33, 98-104 on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 at different dates and times and by different auditors 190, 192 and/or the same auditors 190 and/or different auditors 190, 192 (only two of which are illustrated for simplicity).

[0285] In one embodiment, the auditors 190, 192 are from the same company. In another embodiment, the auditors 190, 192 are from different companies. In one embodiment, the different auditors are the same auditor 190 completing audits at different dates and times. In another embodiment, the different auditors are different auditors 190, 192 (i.e., different people, etc.). However, the present invention is not limited to such embodiments, and other embodiments can be used to practice the invention.

[0286] In another embodiment, the plural different portions 186, 188 of the electronic floor plan 134 of the building for conducting the energy audit of energy consuming components 140, 140', 140'' and/or energy producing components 141, 141', 141'' are provided by the server energy audit application 30', 30a', 30b', 30c', 30d, 30e', 64' on the server network device 20, 22, 24, 26 to plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 via the communications network 18, 18' in a work order and/or a

Request for Proposal (RFP) and/or audit job bidding process. However, the present invention is not limited to such embodiments, and other embodiments can be used to practice the invention.

[0287] In one embodiment, Step 180 further includes, but is not limited to, creating an environmental impact report 194 for the complete final synchronized energy audit summary report 146' using energy audit data 186, 188 collected on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 including from the plural different portions 183, 184 of the electronic floor plan 134 of the building 136. However, the present invention is not limited to such embodiments and other embodiments with and/or without environmental impact reports can be used to practice the invention.

[0288] FIG. 12 is a flow diagram illustrating a Method 198 for providing energy audits with multiple auditors.

[0289] In FIG. 12 at Step 200, providing a measurement component on the plural energy audit applications on the plural network devices for automatically collecting physical dimensions of plural individual rooms of the building. At Step 202, automatically calculating with the measurement component, physical dimensions of the plural individual rooms in the building by selecting one wall of an individual room in the building and automatically calculating physical distances to plural of other walls in the room with a proper dimensional scale for the electronic floor plan of the building. At Step 204, automatically calculating with the measurement component one or more distances required to purchase materials including electrical wires, pipes, ductwork and/or other physical components required to install new energy consuming components and/or energy producing components in the plural individual room in the building.

[0290] In FIG. 12, Method 198 is illustrated with an illustrative embodiment. However, the present invention is not limited to this illustrative embodiment and other embodiments can be used to practice the invention.

[0291] In such an illustrative embodiment, in FIG. 12 at Step 200, providing a measurement component 30e on the plural energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plural network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 for automatically collecting physical dimensions of plural individual rooms 199 of the building 136.

[0292] In one embodiment, the measurement component 30e includes a laser and/or LIDR measurement component. However, the present invention is not limited to such an embodiment and other types of measurement components and/or other embodiments may be used to practice the invention.

[0293] In one embodiment, the plural individual rooms 199 of the building 136, include, entrance and exit areas, atriums, bathrooms, office rooms, conference rooms, bedrooms, closets, stairwells, and/or utility closets, basements, attics, etc. However, the present invention is not limited to such an embodiment, and other embodiments, can be used to practice the invention.

[0294] In one embodiment, the measurement component 30e, includes, but is not limited to, capturing Global Positioning System (GPS) coordinates including, but not limited to, longitude, latitude, elevation and/or attitude information in the plural individual rooms 199 in the building 136 for determining precise placement of energy consuming components 140, 140' and/or energy producing components 141,

141' in the plural individual rooms **199** in the building **136**. In another embodiment, the energy audit applications **30**, **30a**, **30b**, **30c**, **30d**, **30e**, **64**, capture GPS coordinates including, but not limited to, longitude, latitude, elevation or attitude information in the plural of individual rooms **199** in the building **136** for determining precise placement of energy consuming components **140**, **140'** and/or energy producing components **141**, **141'** in the plural individual rooms **199** in the building **136**. However, the present invention is not limited to such embodiments and other embodiments and/or other applications can be used to practice the invention.

[0295] At Step **202**, automatically calculating with the measurement component **30e**, physical dimensions of the plural individual rooms **199** in the building **136** by selecting one wall of an individual room in the building **136** and automatically calculating physical distances to plural of other walls in the room with a proper dimensional scale for the electronic floor plan **134** of the building **136**.

[0296] At Step **204**, automatically calculating with the measurement component **30e** one or more distances required to purchase materials including electrical wires, pipes, ductwork and/or other physical components required to install new energy consuming components **140**, **140'** and/or energy producing components **141**, **141'** in the plural individual rooms **199** in the building **136**.

[0297] FIG. **13** is a flow diagram illustrating a Method **206** for providing energy audits with multiple auditors.

[0298] In FIG. **13** at Step **208**, positioning the plural energy audit applications on the plural network devices at a plural different physical locations on the electronic floor plan of the building for the selected energy consuming components or energy producing components. At Step **210**, capturing on the plural energy audit applications on the plural network devices Global Positioning System (GPS) coordinates including longitude, latitude, elevation and/or attitude information in plural individual rooms in the building and outside of the building for determining precise placement of energy consuming components or energy producing components in the plural individual rooms in the building and outside of the building.

[0299] In FIG. **13**, Method **206** is illustrated with an illustrative embodiment.

[0300] However, the present invention is not limited to this illustrative embodiment and other embodiments can be used to practice the invention.

[0301] In such an illustrative embodiment at Step **208**, positioning the plural energy audit applications **30**, **30a**, **30b**, **30c**, **30d**, **30e**, **64** on the plural network devices **12**, **14**, **16**, **25**, **27**, **29**, **31**, **33**, **98-104** at a plural different physical locations on the electronic floor plan **134** of the building **136** for the selected energy consuming components **140**, **140'**, **140''** and/or energy producing components **141**, **141'**, **141''**.

[0302] At Step **210**, capturing on the plural energy audit applications **30**, **30a**, **30b**, **30c**, **30d**, **30e**, **64** on the plural network devices **12**, **14**, **16**, **31**, **98-104**, Global Positioning System (GPS) coordinates including longitude, latitude, elevation and/or attitude information in plural individual rooms in the building and outside of the building **136** for determining precise placement of energy consuming components **140**, **140'**, **140''** and/or energy producing components **141**, **141'**, **141''** in the plural individual rooms in the building and outside of the building **136**.

[0303] FIG. **14** is a flow diagram illustrating a Method **212** for providing energy audits with multiple auditors.

[0304] In FIG. **14** at Step **214**, providing plural different portions of an electronic floor plan of a virtual building that does not physically exist for conducting an a virtual energy audit of energy consuming components and/or energy producing components from the server energy audit application on the server network device via the communications network to the plural energy audit applications on the plural network devices each with one or more processors, a non-transitory computer readable medium and a communications interface for connecting the plural network devices to the communications network. At Step **216**, executing the steps of Method **158** of FIGS. **10A-10D** to conduct an energy audit of energy consuming components and/or energy producing components in the virtual building that does not physically exist and produce a complete final synchronized energy audit summary report for the virtual building that does not physically exist.

[0305] In FIG. **14**, Method **212** is illustrated with an illustrative embodiment. However the present invention is not limited to this illustrative embodiment and other embodiments can be used to practice the invention.

[0306] In such an exemplary embodiment, in FIG. **14** at Step **214**, providing plural different portions **186**, **188** of an electronic floor plan **134** of a virtual building **136'** that does not physically exist for conducting an a virtual energy audit of energy consuming components **140**, **140'**, **140''** and/or energy producing components **141**, **141'**, **141''**, from the server energy audit application **30'**, **30a'**, **30b'**, **30c'**, **30d'**, **30e'** on the server network device **20**, **22**, **24**, **26** via the communications network **18**, **18'**, to the plural energy audit applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** on the plural network devices **12**, **14**, **16**, **25**, **27**, **29**, **31**, **33**, **98-104** each with one or more processors, a non-transitory computer readable medium and a communications interface for connecting the plural network devices **12**, **14**, **16**, **25**, **27**, **29**, **31**, **33**, **98-104** to the communications network **18**, **18'**.

[0307] At Step **216**, executing the steps of Method **158** of FIGS. **10A-10D** to conduct an energy audit of energy consuming components and/or energy producing components in the virtual building **136'** that does not physically exist and produce a complete final synchronized energy audit summary report **146'** for the virtual building **136'** that does not physically exist.

[0308] In one embodiment, the electronic floor plan **134** of the virtual building **136'** that does physically exist to a plurality of Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR) network devices **99**, each including or more processors. However, the present invention is not limited to such an embodiment and other embodiments can be used to practice the invention.

[0309] In another embodiments, the methods (FIGS. **10**, **12**, **13** and **14**) and systems (FIGS. **1-6**, **9** and **11**) described herein include creating energy retrofits and providing a complete final synchronized energy retrofit summary report **146''**. However, the present invention is not limited to such an embodiment and other embodiments can be used to practice the invention.

Energy Audit, Energy Retrofit and Energy Electrification Project System

[0310] In an exemplary embodiment, an energy audit, energy retrofit and electrification audit/retrofit is completed

via the cloud Software as a Service (SaaS) **64** application with additional components on the server network device **20**, **22**, **24**, **26** for use by multiple lighting, energy and electrical contractors and service providers. The SaaS **64** is used with a mobile energy audit application **30**, **30a**, **30b**, **30c**, **30d**, **30e** on a target network device **12**, **14**, **16**, **25**, **27**, **29**, **31**, **33**, **98-104** and cloud-based **18** software platform that automates an entire project lifecycle of an energy audit, energy retrofit and/or energy electrification project including facility site audit or assessment, solutioning (i.e., finding the best energy efficient replacement equipment), pricing, estimation, proposal generation, project mobilization, material purchase order, labor work order, project tracking and/or project close out, etc.

[0311] In an exemplary embodiment, the SaaS application **64** on the cloud server network devices **20**, **22**, **24**, **26**, includes an energy product and service hub where suppliers to the contractor and engineering and other user users of the SaaS application **64** can find and source needed products and materials. The manufacturers upload their products to the product hub the SaaS application **64** on the server network devices **20**, **22**, **24**, **26** and facilitate and monitor pricing requests and orders through the system.

[0312] In one embodiment, the original functionality of SaaS application **64** described herein for FIGS. **1-14** is extended to include the additional functionality included in FIGS. **15-16**. However, the present invention is not limited to such an embodiment and other embodiments may be used to practice the invention.

[0313] FIG. **15** is a block diagram **218** illustrating additional components of the cloud Software as a Service (SaaS) **64** application.

[0314] In FIG. **15**, the SaaS **64**, includes, new additional components, but is not limited to, a real-time inventory availability component **220**, for displaying current product availability for energy consuming components and energy producing components, an obsolete inventory management component **222**, for facilitating exchange of outdated manufacturing inventory for energy consuming components and energy producing components, an artificial intelligence (AI) replacement product recommendation component **224**, for automatically recommending optimal product replacements for energy consuming components and energy producing components, a manufacturer sponsorship component **226**, for sponsoring and prioritizing manufacturer energy consuming and energy producing products in a cloud database system, and a multi-party collaboration component **228** for facilitating project collaboration among service providers providing services for energy audit projects **231**, energy retrofit projects **233** and electrification projects **235**. However, the present invention is not limited to such an embodiment and other embodiments, with more, fewer and/or other additional components added to the SaaS **64** can be used to practice the invention.

[0315] In FIG. **15**, the real-time inventory availability component **220** in the SaaS application **64**, for displaying current product availability includes, but is not limited to: (1) a data integration component connecting to manufacturers' enterprise resource planning (ERP) systems, (2) a real-time data synchronization protocol (e.g., TIP/IP, UDP/IP, etc.) that maintains current inventory levels, (3) a visual indicator component within a specification interface displaying product availability status, (4) a decision-support component enabling contractors and engineers to incorporate

availability considerations into specification decisions, and (5) a prioritization component that sorts potential replacement products by availability timeframes. However, the present invention is not limited to such an embodiment and more, fewer and/or other components and protocols can be used to practice the invention.

[0316] The obsolete inventory management component **222**, for facilitating exchange of outdated manufacturing inventory for energy consuming components and energy producing components, includes but is not limited to, (1) a dedicated digital marketplace component for manufacturers to list obsolescent inventory, (2) a database **20'**, **22'**, **24'**, **26'** component distinguishing standard inventory from obsolescent inventory with multiple metadata fields including reasons for obsolescence (e.g., replaced, retired, recalled, etc.), (3) a search and filter component enabling contractors to identify and procure obsolescent inventory at preferential pricing, (4) a transaction processing component enabling manufacturers to recover partial costs while divesting excess inventory, and (5) a segregated user interface component designated for obsolescent inventory transactions. However, the present invention is not limited to such an embodiment and more, fewer and/or other components and protocols can be used to practice the invention.

[0317] The manufacturer sponsorship component **224**, for sponsoring and prioritizing manufacturer energy consuming and energy producing products in a cloud database **20'**, **22'**, **24'**, **26'** system, includes but is not limited to: (1) a sponsorship component for manufacturers to establish sponsorship relationships with contractors through a rebate method; (2) a data structure with multiple fields for storing manufacturer-defined rebate parameters (e.g., price, purchase date, number of items purchased, etc.) based on product ordering volume; (3) a visual designation component within a user interface for identifying sponsored manufacturers and their associated rebate percentages, (4) a calculation component that determines a percentage of a project that includes sponsored products and products with rebates, and (5) an accounting component that tracks, calculates, and displays year-to-date accumulated rebates for each contractor within a defined fiscal period. However, the present invention is not limited to such an embodiment and more, fewer and/or other components and other data structures can be used to practice the invention.

[0318] The artificial intelligence (AI) replacement product recommendation component **226**, for automatically recommending optimal product replacements for energy consuming components and energy producing components, includes but is not limited to: (1) an AI machine learning component using Big Data trained on specification data from no fewer than 450,000 historical projects; (2) an AI pattern recognition component that identifies correlations between legacy equipment and optimal replacement product solutions based on efficiency metrics; (3) an AI automated recommendation component that suggests appropriate energy-efficient equipment replacements based on contractor and engineer historical specification patterns; (4) an AI computational component that evaluates multiple potential replacement options and ranks them according to energy efficiency, cost effectiveness, and compatibility parameters; and (5) an AI time-saving component that presents optimal time-based performance and value propositions to end customers without requiring manual analysis. However, the present invention is

not limited to such an embodiment and more, fewer and/or other components and protocols can be used to practice the invention.

[0319] In one embodiment, the artificial intelligence (AI) replacement product recommendation component 226 is replaced with another component 226 that includes all replacement product recommendation component functionality included in the AI component that does not use any AI. However, the present invention is not limited to such an embodiment and other embodiments can be used to practice the invention.

[0320] The multi-party collaboration component 228, for facilitating project collaboration among service providers, including, but not limited to: (1) a digital workspace component enabling multiple parties to contribute to energy efficiency projects; (2) a bid submission component allowing subcontractors to provide material, labor, and solution specifications to prime contractors; (3) an evaluation component for comparing partner bids based on cost, fit, and other relevant parameters; (4) a selection and notification component for prime contractors to designate chosen partners and subcontractors; and (5) a project acceleration component that streamlines development timelines, delivery schedules, and increases deliverable accuracy through structured collaboration protocols. However, the present invention is not limited to such an embodiment and more, fewer and/or other components and protocols can be used to practice the invention.

[0321] A plurality of existing SaaS components 230 for: designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services, for conducting and managing the energy audit projects 231, the energy retrofit projects 233 and the electrification projects 235 for the energy consuming components 142 and the energy producing components 145 for the structure 136, 136'. However, the present invention is not limited to such an embodiment and more, fewer and/or other components and protocols can be used to practice the invention.

[0322] In one embodiment, the structure 136, 136' includes, but is not limited to, personal buildings (e.g., homes, sheds, garages, etc.), commercial, industrial, federal, state, municipal, university, college, school and/or hospital buildings. However, the present invention is not limited to such an embodiment and more, fewer and/or other structures can be used to practice the invention.

[0323] The SaaS 64 from the cloud server network device 20, 22, 24, 26 provides via the cloud communications network 18 to a plurality of energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on a plurality of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 each with one or more processors for conducting and managing the energy audit projects 231, the energy retrofit projects 232 and/or the electrification projects 235 for the energy consuming components 142 and the energy producing components 145 for the structure 136, 136'.

[0324] The SaaS application 64 on the cloud server network device executes cloud services for designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials labor and other services 13, 15, 13'/82, 15'/82, 138, 146 to the plurality of energy audit applications 30 on a plurality of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 via the cloud communications network 18 for conducting and managing the energy audit projects 231, the

energy retrofit projects 233 and the electrification projects 235 for the energy consuming components 142 and/or the energy producing components 145 for the structure 136, 136', including real-time inventory availability with the real-time inventory availability component 220, obsolete inventory management with the obsolete inventory management component 222, artificial intelligence (AI) replacement product recommendations with the artificial intelligence (AI) replacement product recommendation component 224, manufacturer sponsoring with the manufacturer sponsorship component 226 for energy consuming and the energy producing products in the cloud database system and multi-party collaboration with the multi-party collaboration component 228 providing collaboration among service providers providing services for the energy audit projects 231, the energy retrofit projects 233 and the electrification projects 235 for the structure 136, 136'.

[0325] The SaaS application 64 on the cloud server network device 20, 22, 24, 26 sends complete, final, synchronized summary reports 146/s for the energy audit projects 231, the energy retrofit projects 233 and the electrification projects 235 for the energy consuming components 142 and/or the energy producing components 145 for the structure, to the plurality of energy audit applications 30 on the plurality of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 via the cloud communications network 18.

[0326] In one embodiment, energy audit projects include an energy audit for the energy consuming components 142 and/or the energy producing components 145 of the structure 136, 136'. However, the present invention is not limited to such an embodiment and more, fewer and/or other types of energy audit projects 231 can be used to practice the invention.

[0327] In one embodiment, the energy retrofit projects include identifying and replacing obsolete or inefficient energy consuming components 142 and/or energy producing components 145 of the structure 136, 136' with new energy consuming components 142 and/or energy producing components 145. However, the present invention is not limited to such an embodiment and more, fewer and/or other types of energy retrofit projects 233 can be used to practice the invention.

[0328] In one embodiment, the electrification projects include identifying and replacing energy consuming components 142 of the structure powered by fossil fuels with energy consuming components 142 powered by electricity created from stored power devices 145 and/or other electricity sources. However, the present invention is not limited to such an embodiment and more, fewer and/or other types of energy electrification projects 235 can be used to practice the invention.

[0329] In one embodiment, the stored power devices 145, include, but are not limited to, batteries, capacitors, long-duration energy storage (LDES) devices, thermal energy storage (TES) devices, hydro energy storage (HES) devices, flywheel energy storage devices, compressed air storage devices, cryogenic energy storage (CES) devices, superconducting magnetic energy storage (SMES) devices, pumped storage hydraulic energy storage (PSHES) devices, Electric Vehicle (EV) charging station devices, EV infrastructure devices or combinations thereof. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0330] Long-duration energy storage (LDES) devices include, but are not limited to, electric storage systems that can provide at least **10** hours or more of stored energy. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0331] Thermal energy storage (TES) devices, include but are not limited to, devices that temporarily store energy by heating or cooling various storage mediums for later reuse. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0332] Hydro energy storage (HES) devices, include but are not limited to, hydroelectric energy storage devices used by electric power systems. The devices store energy in the form of gravitational potential energy of water, pumped from a lower elevation reservoir to a higher elevation. Low-cost surplus off-peak electric power is typically used to run the pumps. During periods of high electrical demand, the stored water is released through turbines to produce electric power. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0333] Flywheel energy storage devices, include but are not limited to, devices that use electric energy input which is stored in the form of kinetic energy. Kinetic energy can be described as “energy of motion.” Flywheel energy storage devices use motion of a spinning mass, called a rotor. The rotor spins in a nearly frictionless enclosure. When power is required, the inertia allows the rotor to spin and the resulting kinetic energy is converted to electricity. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0334] Compressed air storage devices, include but are not limited to, devices that allow air to be compressed and expanded. Compression of air creates heat. The air is warmer after compression. Expansion removes heat. If no extra heat is added, the air will be much colder after expansion. If the heat generated during compression can be stored and used during expansion, the efficiency of the storage improves considerably. Compressed air storage devices include adiabatic, diabatic, isothermal, and/or near-isothermal compressed air storage devices. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0335] Cryogenic energy storage (CES) devices, include but are not limited to, the use of low temperature (cryogenic) liquids such as liquid air or liquid nitrogen to store energy. CES devices are sustainable, low-carbon, energy storage systems. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0336] Superconducting magnetic energy storage (SMES) devices, include but are not limited to, devices that store energy in a magnetic field created by the flow of direct current in a superconducting coil which has been cryogenically cooled to a temperature below its superconducting critical temperature. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0337] Pumped hydraulic energy storage (PHES) devices, include, but are not limited to, devices that produce kinetic

energy in the form of flow and potential energy in the form of pressure. PHES devices include, impoundment, diversion, and pumped storage devices. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0338] Electric Vehicle (EV) charging station devices, include, but are not limited to, vehicles that use a battery pack to store electrical energy that powers a motor in the electronic vehicle. The batteries are charged by plugging the vehicle in to an electric power source. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0339] EV infrastructure devices, include, but are not limited to, structures, machinery, and equipment necessary and integral to support an EV, including battery charging stations, rapid charging stations, and/or battery exchange stations. A battery charging station is defined as an electrical component assembly or cluster of component assemblies designed specifically to charge batteries within an EV. A rapid charging station is defined as an industrial grade electrical outlet that allows for faster recharging of EV batteries through higher power levels. A battery exchange station is defined as a fully automated facility that will enable an EV with a swappable battery to enter a drive lane and exchange the depleted battery with a fully charged battery through a fully automated process. However, the present invention is not limited to such embodiments and other embodiments can also be used to practice the invention.

[0340] In one embodiment, the energy consuming components **142** include one or more of: lighting, heating, ventilation, air conditioning, air quality, motors, variable frequency drives, transformers, generators, electrical controllers, water supply valves, water pumps, or ultra violet (UV) lighting and disinfection systems, compressed air systems, thermostats, steam boilers, chillers, air movers, dampers, hoods, building automation systems, or combinations thereof. However, the present invention is not limited to such an embodiment and more, fewer and/or other type of energy consuming components can be used to practice the invention.

[0341] In one embodiment, the energy producing components **145** include one or more of: solar, wind, water, biomass, batteries, Electric Vehicle Supply Equipment (EVSE), micro-grids fuel cells, and/or combinations thereof. However, the present invention is not limited to such an embodiment and more, fewer and/or other type of energy producing components can be used to practice the invention.

[0342] In one embodiment, the structure **136'**, **136** includes, both physical and virtual personal buildings including personal homes and personal sheds, federal, state, municipal, university, college, school and/or hospital buildings and/or other types of buildings and other structures (e.g., bridges, pipelines, light poles, communications towers, etc.) the including energy consuming devices and/or energy producing devices. However, the present invention is not limited to such an embodiment and more, fewer and/or other types of structures can be used to practice the invention.

[0343] In one embodiment, the SaaS application **64** is included within a Computer Aided Design (CAD), Building Information Modeling (BIM) or other three-dimensional (3D) modeling program. However, the present invention is not limited to such an embodiment and other embodiments may be used to practice the invention.

[0344] In one embodiment, the structure **136** is a virtual structure **136'** included within a Computer Aided Design (CAD), Building Information Modeling (BIM) or other three-dimensional (3D) modeling program. However, the present invention is not limited to such an embodiment and other embodiments may be used to practice the invention.

[0345] In one embodiment, the virtual structure **136'** is viewed using Virtual Reality (VR), Augmented Reality (AR) or Mixed Reality (MR) network devices **99**. However, the present invention is not limited to such an embodiment and other embodiments may be used to practice the invention.

[0346] In one embodiment, the SaaS **64** application is replaced with one or more non-cloud applications **30a'**, **30b'**, **30c'**, **30d'**, **30e'** on the server network devices **20**, **22**, **24**, **26** and are used on a non-cloud communications network **18'** to communicate with plurality of energy audit applications **30**, **30a**, **30b**, **30c**, **30d**, **30e** on the plurality of network devices **12**, **14**, **16**, **25**, **27**, **29**, **31**, **33**, **98-104**. However, the present invention is not limited to such an embodiment and other embodiments can be used to practice the invention.

[0347] FIGS. **16A**, **16B** and **16C** a flow diagram illustrating a Method **232** for conducting and managing energy audits, energy retrofits and electrification projects.

[0348] In FIG. **16A** at Step **234**, creating a Software as a Service (SaaS) application on a cloud server network device with one or more processors and a cloud database system on the cloud communications network for conducting and managing energy audit projects, energy retrofit projects and electrification projects for energy consuming components and energy producing components for a structure including a plurality of existing components and a plurality of new components, comprising: a real-time inventory availability component, for displaying current product availability for energy consuming components and energy producing components, an obsolete inventory management component, for facilitating exchange of outdated manufacturing inventory for energy consuming components and energy producing components, an artificial intelligence (AI) replacement product recommendation component, for automatically recommending optimal product replacements for energy consuming components and energy producing components, a manufacturer sponsorship component in the SaaS application, for sponsoring and prioritizing manufacturer energy consuming and energy producing products in a cloud database system, a multi-party collaboration component for facilitating project collaboration among service providers providing services for energy audit projects, energy retrofit projects and electrification projects, and the plurality of existing SaaS components for: designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other services, for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure. In FIG. **16B** at Step **236**, providing the Software as a System (SaaS) application with the plurality of new components and the plurality of existing components from the cloud server network device via the cloud communications network to a plurality of energy audit applications on a plurality of network devices each with one or more processors for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure. At Step

238, conducting on selected ones of the plurality of energy audit applications on selected ones of the plurality of network devices with the SaaS application on the cloud server network device via the cloud communications network, the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components or the energy producing components for the structure, including designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services, and project service providers, with one or more auditors, the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components or the energy producing components for the structure. In FIG. **16C** at Step **240**, creating with the SaaS application on the cloud server network device, complete, final, synchronized summary reports for the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components and the energy producing components for the structure. At Step **242**, sending with SaaS application on the cloud server network device complete via the cloud communications network to the plurality of energy audit applications on the plurality of network devices the complete, final, synchronized summary reports for the energy audit projects, the energy retrofit projects and/or the electrification projects for the energy consuming components and the energy producing components for the structure. At Step **244**, displaying from SaaS application on the cloud server network device via the cloud communications network, the complete, final, synchronized summary reports for the energy audit projects, the energy retrofit projects and/or the electrification projects for the energy consuming components and the energy producing components for the structure on the plurality of energy audit applications on the plurality of network devices

[0349] Method **232** is illustrated with an illustrative embodiment. However, the present invention is not limited to this illustrative embodiment and other embodiments can be used to practice the invention.

[0350] In such an illustrative embodiment, in FIG. **16A** at Step **234**, creating a Software as a Service (SaaS) application **64** on a cloud server network device **20**, **22**, **24**, **26** with one or more processors and a cloud database system **20'**, **22'**, **24'**, **26'**, **82** on the cloud communications network **18** for conducting and managing energy audit projects, energy retrofit projects and electrification projects for energy consuming components and energy producing components for a structure **136**, **136'**, including a plurality of existing components **230** and a plurality of existing new components, comprising: a real-time inventory availability component **220**, for displaying current product availability for energy consuming components and energy producing components, an obsolete inventory management component **222**, for facilitating exchange of outdated manufacturing inventory for energy consuming components and energy producing components, an artificial intelligence (AI) replacement product recommendation component **224**, for automatically recommending optimal product replacements for energy consuming components and energy producing components, a manufacturer sponsorship component **226** in the SaaS application, for sponsoring and prioritizing manufacturer energy consuming and energy producing products in a cloud database system **20'**, **22'**, **24'**, **26'**, **82**, a multi-party collaboration component **228** for facilitating project collaboration among service providers providing services for energy audit proj-

ects, energy retrofit projects and electrification projects, and a plurality of existing SaaS 64 components 230 for: designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services (e.g., collaboration, sub-contractor, rebate, etc.) for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components 142 and the energy producing components 145 for the structure 136, 136'.

[0351] In FIG. 16B at Step 236, providing the Software as a System (SaaS) application 64 with the plurality of new components 220-228 and plurality of existing components 230 (FIGS. 1-14 and related text) from the cloud server network device 20, 22, 24, 26 via the cloud communications network 18 to a plurality of energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on a plurality of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 each with one or more processors for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components 142 and the energy producing components 145 for the structure 136, 136'.

[0352] At Step 238, conducting on selected ones of the plurality of energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on selected ones of the plurality of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 with the SaaS application 64 on the cloud server network device 20, 22, 24, 26 via the cloud communications network 18, the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components 142 and/or the energy producing components 145 for the structure 136, 136', including designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services, and project service providers, with one or more auditors, the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components 142 or the energy producing components 145 for the structure 136, 136'.

[0353] In FIG. 16C at Step 240, creating with the SaaS application 64 on the cloud server network device 20, 22, 24, 26 a complete, final, synchronized summary report 146/s for the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components 142 and the energy producing components 145 for the structure 136, 136'.

[0354] At Step 242, sending with SaaS application 64 on the cloud server network device 20, 22, 24, 26 via the cloud communications network 18 to the plurality of energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plurality of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104 the final, synchronized summary reports 146/s for the energy audit projects, the energy retrofit projects and/or the electrification projects for the energy consuming components 142 and the energy producing components 145 for the structure, 136, 136'.

[0355] In one embodiment, the final, synchronized summary reports 146/s for the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components 142 and the energy producing components 145 for the structure, 136, 136' are synchronized across the plurality of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104. In another embodiment, the final, synchronized summary reports 146/s are synchronized across only selected ones of the plurality of network devices 12, 14, 16,

25, 27, 29, 31, 33, 98-104. However, the present invention is not limited to such embodiments and other embodiments can be used to practice the invention.

[0356] At Step 244, displaying from SaaS application 64 on the cloud server network device 20, 22, 24, 26 via the cloud communications network 18, the complete, final, synchronized summary reports 146/s for the energy audit projects 231, the energy retrofit projects 233 and/or the electrification projects 235 for the energy consuming components 142 and the energy producing components 145 for the structure 136, 136' on the plurality of energy audit applications 30, 30a, 30b, 30c, 30d, 30e, 64 on the plurality of network devices 12, 14, 16, 25, 27, 29, 31, 33, 98-104. However, the present invention is not limited to such an embodiment and other embodiments can be used to practice the invention.

[0357] In one embodiment, the complete, final, synchronized summary reports 146/s, include, but are not limited to, reports referred to as "solutions" which include all energy consuming components 142 and the energy producing components 145 for the structure 136, 136' and/or other required materials and/or labor costs to remove and/or replace and/or install the energy consuming components 142 and the energy producing components 145 for the structure 136, 136' for the energy audit projects 231, the energy retrofit projects 233 and/or the electrification projects 235. However, the present invention is not limited to such an embodiment, and the final reports 146/s can include, more, fewer and/or other items, and/or other embodiments may be used to practice the invention.

[0358] A method and system for providing energy audits and energy retrofits with multiple participants is presented. The method and system include a cloud Software as a Service (SaaS) application with components for designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services and additional components including a real-time inventory availability component, an obsolete inventory management component, an artificial intelligence (AI) replacement product recommendation component, a manufacturer sponsorship component and a multi-party collaboration component, for conducting and managing energy audit projects, energy retrofit projects and electrification projects for energy consuming components and energy producing components for a building or other structures.

[0359] It should be understood that the architecture, programs, processes, methods and systems described herein are not related or limited to any particular type of computer or network system (hardware or software), unless indicated otherwise. Various types of specialized computer systems may be used with or perform operations in accordance with the teachings described herein.

[0360] In view of the wide variety of embodiments to which the principles of the present invention can be applied, it should be understood that the illustrated embodiments are exemplary only, and should not be taken as limiting the scope of the present invention. For example, the steps of the flow diagrams may be taken in sequences other than those described, and more or fewer elements may be used in the block diagrams.

[0361] The claims should not be read as limited to the described order or elements unless stated to that effect. In addition, use of the term "means" in any claim is intended to invoke 35 U.S.C. § 112, paragraph 6, and any claim

without the word “means” is not so intended. Therefore, all embodiments that come within the scope and spirit of the following claims and equivalents thereto are claimed as the invention.

[0362] While various elements of the preferred embodiments have been described as being implemented in software, in other embodiments hardware or firmware implementations may alternatively be used, and vice-versa.

[0363] Therefore, all embodiments that come within the scope and spirit of the proceeding described and equivalents thereto are identified and claimed as the invention.

We claim:

1. A system for conducting and managing energy audits, energy retrofits and electrification projects, comprising in combination:

for creating a Software as a Service (SaaS) application on a cloud server network device with one or more processors and a cloud database system on a cloud communications network for conducting and managing energy audit projects, energy retrofit projects and electrification projects for energy consuming components and energy producing components for a structure, including a plurality of existing components and a plurality of new components, comprising:

a real-time inventory availability component in the SaaS application, for displaying current product availability, comprising:

- (1) a data integration component connecting to manufacturers' enterprise resource planning (ERP) systems,
- (2) a real-time data synchronization protocol that maintains current inventory levels,
- (3) a visual indicator component within a specification interface displaying product availability status,
- (4) a decision-support component enabling contractors and engineers to incorporate availability considerations into specification decisions, and
- (5) a prioritization component that sorts potential replacement products by availability timeframes;

an obsolete inventory management component in the SaaS application, for facilitating exchange of outdated manufacturing inventory comprising:

- (1) a dedicated digital marketplace component for manufacturers to list obsolescent inventory,
- (2) a cloud database component distinguishing standard inventory from obsolescent inventory with metadata fields including reasons for obsolescence,
- (3) a search and filter component enabling contractors to identify and procure obsolescent inventory at preferential pricing,
- (4) a transaction processing component enabling manufacturers to recover partial costs while divesting excess inventory, and
- (5) a segregated user interface component designated for obsolescent inventory transactions;

a manufacturer sponsorship component in the SaaS application, for sponsoring and prioritizing manufacturer energy consuming products and manufacturer energy producing products in the cloud database system, comprising:

- (1) a sponsorship component for manufacturers to establish sponsorship relationships with contractors through a rebate mechanism,

- (2) a data structure with a plurality of fields for storing manufacturer-defined rebate parameters based on product ordering volume,
- (3) a visual designation component within a user interface for identifying sponsored manufacturers and their associated rebate percentages,
- (4) a calculation component that determines a percentage of a project that includes sponsored products and products with rebates, and
- (5) an accounting component that tracks, calculates, and displays year-to-date accumulated rebates for each contractor within a defined fiscal period;

an artificial intelligence (AI) replacement product recommendation component in the SaaS application, for automatically recommending optimal product replacements, comprising:

- (1) an AI machine learning component using Big Data trained on specification data from no fewer than 450,000 historical projects,
- (2) an AI pattern recognition component that identifies correlations between legacy equipment and optimal replacement product solutions based on efficiency metrics,
- (3) an AI automated recommendation component that suggests appropriate energy-efficient equipment replacements based on contractor and engineer historical specification patterns,
- (4) an AI computational component that evaluates multiple potential replacement options and ranks them according to energy efficiency, cost effectiveness, and compatibility parameters, and
- (5) an AI time-saving component that presents optimal time-based performance and value propositions to end customers without requiring manual analysis;

a multi-party collaboration component in the SaaS application for facilitating project collaboration among service providers, comprising:

- (1) a digital workspace component enabling multiple stakeholders to contribute to energy efficiency projects,
- (2) a bid submission component allowing subcontractors to provide material, labor, and solution specifications to prime contractors,
- (3) an evaluation component for comparing partner bids based on cost, fit, and other relevant parameters,
- (4) a selection and notification component for prime contractors to designate chosen partners, and
- (5) a project acceleration component that streamlines development timelines, delivery schedules, and increases deliverable accuracy through structured collaboration protocols; and

including the plurality of existing SaaS components in the SaaS application for:

designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services, for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure;

for providing the SaaS application with plurality of new components and existing components, from the cloud server network device via the cloud communications network to a plurality of energy audit applications on a plurality of network devices each with one or more

processors for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure;

for conducting and managing with SaaS application on the cloud server network device, the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure, providing designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services on the plurality of energy audit applications on plurality of network devices via the cloud communications network; and

for sending with the SaaS application on the cloud server network device, complete, final, synchronized summary reports for the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure, to the plurality of energy audit applications on the plurality of network devices via the cloud communications network.

2. The system of claim 1, wherein the energy audit projects include an energy audit for the energy consuming components or the energy producing components of the structure.

3. The system of claim 1, wherein the energy retrofit projects include identifying and replacing obsolete or inefficient energy consuming components or energy producing components of the structure with new energy consuming components or energy producing components.

4. The system of claim 1, wherein the electrification projects include identifying and energy consuming components of the structure powered by fossil fuels and replacing with energy consuming components powered by electricity created from stored power devices.

5. The system of claim 4, wherein the stored power devices include or more of: stored power components include batteries, capacitors, long-duration energy storage (LDES) technologies, thermal energy storage (TES) devices, hydro energy storage (HES) devices, flywheel energy storage devices, compressed air storage (CAS) devices, cryogenic energy storage devices, superconducting magnetic energy storage (SMES) devices, pumped storage hydraulic energy storage (PSHES) devices, electric vehicle (EV) charging stations, EV infrastructures, or combinations thereof.

6. The system of claim 1, wherein the energy consuming components include one or more of: lighting, heating, ventilation, air conditioning, air quality, motors, variable frequency drives, transformers, generators, electrical controllers, water supply valves, water pumps, or ultra violet (UV) lighting and disinfection systems, compressed air systems, thermostats, steam boilers, chillers, air movers, dampers, hoods, building automation systems, or combinations thereof.

7. The system of claim 1, wherein the energy producing components include one or more of: solar, wind, water, biomass, geothermal, batteries, Electronic Vehicle Supply Equipment (EVSE), micro-grids, fuel cells, or combinations thereof.

8. The system of claim 1, wherein the structure includes, personal buildings, commercial, industrial, federal, state, municipal, university, college, school, hospital buildings, or combinations thereof.

9. The system claim 1, wherein the cloud server network device includes a plurality of cloud applications available on the cloud communications network, the plurality of cloud applications providing energy audit projects, energy retrofit projects and electrification projects management services including: a plurality of applications including: a cloud computing Infrastructure as a Service (IaaS), a cloud computing Platform as a Service (PaaS) and a components for managing energy audit projects, energy retrofit projects and electrification projects for energy consuming components and energy producing components for the structure, as a Software as a Service (SaaS).

10. The system of claim 1, wherein the plurality of network devices include one or more of: desktop computers, laptop computers, tablet computers, mobile phones, smart phones, personal digital/data assistants (PDA), digital cameras, 360-degree cameras, Light Detection and Ranging (LIDAR) devices, unmanned aerial vehicles (UAVs), driverless vehicles, portable game consoles, wearable network devices, Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR) network devices, Internet of Things (IoT) devices, or combinations thereof.

11. The system of claim 1, wherein the plurality of network devices and the cloud server network device include one or more wireless communications interfaces comprising: cellular telephone, 802.11a, 802.11b, 802.11g, 802.11n, 802.15.1 (Bluetooth), 802.15.4 (ZigBee), Wireless Fidelity (Wi-Fi), Wi-Fi Aware, Worldwide Interoperability for Microwave Access (WiMAX), ETSI High Performance Radio Metropolitan Area Network (HIPERMAN), Near Field Communications (NFC), Machine-to-Machine (M2M), or infra data association (IrDA), wireless communication interfaces, or combinations thereof.

12. The system of claim 1, wherein the cloud database system includes one or more cloud storage objects stored in one or more cloud databases.

13. The system of claim 12, wherein the one or more cloud storage objects include one or more of: Representational state transfer (REST), Simple Object Access Protocol (SOAP), Lightweight Directory Access Protocol (LDAP) or Application Programming Interface (API) objects, or a combination thereof.

14. The system of claim 1, wherein the SaaS application is included within a Computer Aided Design (CAD), Building Information Modeling (BIM) or other three-dimensional (3D) modeling program.

15. The system of claim 1, wherein the structure includes a virtual structure included within a Computer Aided Design (CAD), Building Information Modeling (BIM) or other three-dimensional (3D) modeling program.

16. The system of claim 15, wherein the virtual structure is viewed using Virtual Reality (VR), Augmented Reality (AR) or Mixed Reality (MR) network devices.

17. The system of claim 1, wherein the complete, final, synchronized summary reports for the energy audit projects the energy retrofit projects and the electrification projects are synchronized across the plurality of network devices.

18. A method for conducting and managing energy audits, energy retrofits and electrification projects, comprising:

creating a Software as a Service (SaaS) application on a cloud server network device with one or more processors and a cloud database system on the cloud communications network for conducting and managing energy audit projects, energy retrofit projects and electrification projects for energy consuming components and energy producing components for a structure, with a plurality of existing components and a plurality of new components, comprising:

a real-time inventory availability component, for displaying current product availability for the energy consuming components and the energy producing components, an obsolete inventory management component, for facilitating exchange of outdated manufacturing inventory for the energy consuming components and the energy producing components,

an artificial intelligence (AI) replacement product recommendation component, for automatically recommending optimal product replacements for the energy consuming components and the energy producing components,

a manufacturer sponsorship component in the SaaS application, for sponsoring and prioritizing manufacturer energy consuming products and energy producing products in a cloud database system,

a multi-party collaboration component for facilitating project collaboration among service providers providing services for the energy audit projects, the energy retrofit projects and the electrification projects, and

including the plurality of existing SaaS components for: designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services, for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure;

providing the Software as a System (SaaS) application with the plurality of new components and the plurality of existing components from the cloud server network device via the cloud communications network to a plurality of energy audit applications on a plurality of network devices each with one or more processors for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure;

conducting on selected ones of the plurality of energy audit applications on selected ones of the plurality of network devices with the SaaS application on the cloud server network device via the cloud communications network, the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components or the energy producing components for the structure, including designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services, and project service providers, with one or more auditors, the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components or the energy producing components for the structure;

creating with the SaaS application on the cloud server network device complete, final, synchronized summary reports for the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components and the energy producing components for the structure;

sending with SaaS application on the cloud server network device complete via the cloud communications network to the plurality of energy audit applications on the plurality of network devices the final, synchronized summary reports for the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components and the energy producing components for the structure;

displaying from SaaS application on the cloud server network device via the cloud communications network, the complete, final, synchronized summary reports for the energy audit projects, the energy retrofit projects or the electrification projects for the energy consuming components and the energy producing components for the structure on the plurality of energy audit applications on the plurality of network devices.

19. The method of claim **18** further including, a non-transitory computer readable medium having stored therein a plurality of instructions for causing one or more processors on one or more network devices to execute steps of the method.

20. A system for automatically providing energy audits, comprising in combination:

a plurality of network devices each with one or more processors and a non-transitory computer readable medium;

one or more cloud server network devices each with one or more processors and a non-transitory computer readable medium;

a cloud communications network;

for providing a Software as a Service (SaaS) application on a cloud server network device with one or more processors and a cloud database system on the cloud communications network for conducting and managing energy audit projects, energy retrofit projects and electrification projects for energy consuming components and energy producing components for a structure, with a plurality of existing components and a plurality of new components comprising:

a real-time inventory availability component, for displaying current product availability for energy consuming components and energy producing components, an obsolete inventory management component, for facilitating exchange of outdated manufacturing inventory for energy consuming components and energy producing components,

an artificial intelligence (AI) replacement product recommendation component, for automatically recommending optimal product replacements for energy consuming components and energy producing components,

a manufacturer sponsorship component in the SaaS application, for sponsoring and prioritizing manufacturer energy consuming and energy producing products in a cloud database system,

a multi-party collaboration component for facilitating project collaboration among service providers providing services for energy audit projects, energy retrofit projects and electrification projects, and

including the plurality of existing SaaS components for: designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services, for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure;

for providing the SaaS application with plurality of new components and the plurality of existing components, from the cloud server network device via the cloud communications network to a plurality of energy audit applications on a plurality of network devices each with one or more processors for conducting and managing the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure;

for conducting and managing with SaaS application on the cloud server network device, the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure,

providing designing, specifying, searching, pricing, quoting, ordering, and tracking of project materials, labor and other project services on the plurality of energy audit applications on plurality of network devices via the cloud communications network;

for sending with the SaaS application on the cloud server network device, complete, final, synchronized summary reports for the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure, to the plurality of energy audit applications on the plurality of network devices via the cloud communications network; and

for displaying from SaaS application on the cloud server network device via the cloud communications network, the complete, final, synchronized summary reports for the energy audit projects, the energy retrofit projects and the electrification projects for the energy consuming components and the energy producing components for the structure on the plurality of energy audit applications on the plurality of network devices.

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